Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	30-03-2020	29-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	30-03-2020	29-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

ATTACHMENT 3 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR GROUND PORTAL**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 28-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART GROUND PORTAL FOUNDATIONS
JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0.125 -0.55; 2 -6.15 0.125 -0.55; 3 -12.55 0.125 -0.55; 4 0 0.41 -0.55;
5 -6.15 0.743 -0.55; 6 -12.55 1.091 -0.55; 7 6.73533e-016 0.125 -4.505;
8 -6.15 0.125 -4.505; 9 -12.55 0.125 -4.505; 13 0 0.41 -0.795;
14 2.28389e-016 0.41 -1.865; 15 4.45144e-016 0.41 -3.435;
16 5.76177e-016 0.41 -4.505; 17 -6.15 0.743 -0.795; 18 -12.55 1.091 -0.795;
24 -6.15 0.743 -1.865; 25 -12.55 1.091 -1.865; 27 -6.15 0.743 -3.435;
28 -12.55 1.091 -3.435; 30 -6.15 0.743 -4.505; 31 -12.55 1.091 -4.505;
33 3.025 -0.0449997 -0.795; 34 3.025 -0.0449997 -1.865;
35 3.025 -0.0449997 -3.435; 36 3.025 -0.0449997 -4.505; 37 -18.622 -0.15 0.1;
38 -18.622 1.515 0.1; 39 -18.622 -0.15 -5.4; 40 -18.622 1.515 -5.4;
41 -18.622 1.515 -0.795; 42 -18.622 1.515 -1.865; 43 -18.622 1.515 -3.435;
44 -18.622 1.515 -4.505; 45 -24.622 -0.15 0.1; 46 -24.622 1.935 0.1;
47 -24.622 -0.15 -5.4; 48 -24.622 1.935 -5.4; 49 -24.622 1.935 -0.795;
50 -24.622 1.935 -1.865; 51 -24.622 1.935 -3.435; 52 -24.622 1.935 -4.505;
53 -30.747 -0.15 0.1; 54 -30.747 2.363 0.1; 55 -30.747 -0.15 -5.4;
56 -30.747 2.363 -5.4; 57 -30.747 2.363 -0.795; 58 -30.747 2.363 -1.865;
59 -30.747 2.363 -3.435; 60 -30.747 2.363 -4.505; 61 -35.128 -0.15 0.1;
62 -35.128 2.953 0.1; 63 -35.128 -0.15 -5.4; 64 -35.128 2.953 -5.4;
65 -35.128 2.953 -0.795; 66 -35.128 2.953 -1.865; 67 -35.128 2.953 -3.435;
68 -35.128 2.953 -4.505; 69 -39.385 -0.15 0.1; 70 -39.385 3.526 0.1;
71 -39.385 -0.15 -5.4; 72 -39.385 3.526 -5.4; 73 -39.385 3.526 -0.795;
74 -39.385 3.526 -1.865; 75 -39.385 3.526 -3.435; 76 -39.385 3.526 -4.505;
77 -45.385 -0.15 0.1; 78 -45.385 4.333 0.1; 79 -45.385 -0.15 -5.4;
80 -45.385 4.333 -5.4; 81 -45.385 4.333 -0.795; 82 -45.385 4.333 -1.865;
83 -45.385 4.333 -3.435; 84 -45.385 4.333 -4.505; 85 -49.514 2.8 -1.65;
87 -49.514 2.8 -3.65; 89 -49.514 4.889 -0.795; 90 -49.514 4.889 -1.865;
91 -49.514 4.889 -3.435; 92 -49.514 4.889 -4.505; 93 -49.514 4.889 -1.65;
94 -49.514 4.889 -3.65; 95 -30.747 2.25 0.1; 96 -30.747 2.25 -5.4;
97 -30.747 2.25 -2.65; 98 -21.622 1.725 -0.795; 99 -27.6845 2.149 -0.795;
100 -32.9375 2.658 -0.795; 101 -37.2565 3.2395 -0.795;
102 -42.385 3.9295 -0.795; 103 -42.385 3.9295 0.1; 104 -37.2565 3.2395 0.1;
105 -32.9375 2.658 0.1; 106 -27.6845 2.149 0.1; 107 -21.622 1.725 0.1;
108 -47.4495 4.611 -3.435; 109 -47.4495 4.611 -1.865; 110 -41.385 3.795 -3.435;
111 -43.385 4.064 -3.435; 112 -41.385 3.795 -1.865; 113 -43.385 4.064 -1.865;
114 -37.2565 3.2395 -3.435; 115 -37.2565 3.2395 -1.865;
116 -32.9375 2.658 -3.435; 117 -32.9375 2.658 -1.865;
118 -26.6637 2.07767 -3.435; 119 -28.7053 2.22033 -3.435;
120 -26.6637 2.07767 -1.865; 121 -28.7053 2.22033 -1.865;
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122 -20.622 1.655 -3.435; 123 -22.622 1.795 -3.435; 124 -20.622 1.655 -1.865;
125 -22.622 1.795 -1.865; 126 -14.574 1.23233 -3.435;
127 -16.598 1.37367 -3.435; 128 -14.574 1.23233 -1.865;
129 -16.598 1.37367 -1.865; 130 -8.28333 0.859333 -1.865;
131 -10.4167 0.974667 -1.865; 132 -8.28333 0.859333 -3.435;
133 -10.4167 0.974667 -3.435; 134 -2.05 0.520667 -1.865;
135 -4.1 0.632333 -1.865; 136 -2.05 0.520667 -3.435; 137 -4.1 0.632333 -3.435;
138 1.5125 0.1825 -1.865; 139 1.5125 0.1825 -3.435; 140 1.5125 0.1825 -0.795;
141 1.5125 0.1825 -4.505; 142 -32.9375 2.658 -4.505;
143 -37.2565 3.2395 -4.505; 144 -47.4495 4.611 -4.505;
145 -47.4495 4.611 -0.795; 146 -2.05 0.520667 -0.795; 147 -4.1 0.632333 -0.795;
148 -8.28333 0.859333 -0.795; 149 -10.4167 0.974667 -0.795;
150 -2.05 0.520667 -4.505; 151 -4.1 0.632333 -4.505;
152 -8.28333 0.859333 -4.505; 153 -10.4167 0.974667 -4.505;
154 -14.574 1.23233 -4.505; 155 -16.598 1.37367 -4.505;
156 -14.574 1.23233 -0.795; 157 -16.598 1.37367 -0.795;
158 -20.622 1.655 -4.505; 159 -22.622 1.795 -4.505;
160 -26.6637 2.07767 -4.505; 161 -28.7053 2.22033 -4.505;
162 -41.385 3.795 -4.505; 163 -43.385 4.064 -4.505; 164 -20.622 1.655 -0.795;
165 -22.622 1.795 -0.795; 166 -26.6637 2.07767 -0.795;
167 -28.7053 2.22033 -0.795; 168 -41.385 3.795 -0.795;
169 -43.385 4.064 -0.795; 170 -37.2565 3.2395 -5.4; 171 -18.622 1.515 -2.65;
172 -24.622 1.935 -2.65; 173 -21.622 1.725 -5.4; 201 -45.385 -1.5 -5.4;
202 -45.385 -1.5 0.1; 203 -39.385 -1.5 -5.4; 204 -39.385 -1.5 0.1;
205 -35.128 -1.5 -5.4; 206 -35.128 -1.5 0.1; 207 -30.747 -1.5 -5.4;
208 -30.747 -1.5 0.1; 209 -24.622 -1.5 -5.4; 210 -24.622 -1.5 0.1;
211 -18.622 -2.5 -5.4; 212 -18.622 -2.5 0.1;
MEMBER INCIDENCES
1 1 4; 2 2 5; 3 3 6; 4 7 16; 5 8 30; 6 9 31; 7 4 13; 8 5 17; 9 6 18; 10 13 14;
11 14 15; 12 15 16; 14 17 24; 15 13 146; 16 18 25; 17 17 148; 23 24 27;
24 14 134; 25 25 28; 26 24 130; 29 27 30; 30 15 136; 31 28 31; 32 27 132;
36 16 150; 38 30 152; 41 33 140; 42 34 138; 43 35 139; 44 36 141; 45 37 38;
46 39 40; 47 38 41; 48 41 42; 49 42 171; 50 43 44; 51 44 40; 52 31 154;
53 28 126; 54 25 128; 55 18 156; 56 45 46; 57 47 48; 58 46 49; 59 49 50;
60 50 172; 61 51 52; 62 52 48; 63 44 158; 64 43 122; 65 42 124; 66 41 164;
67 53 95; 68 55 96; 69 54 57; 70 57 58; 71 58 59; 72 59 60; 73 60 56; 74 61 62;
75 63 64; 76 62 65; 77 65 66; 78 66 67; 79 67 68; 80 68 64; 81 69 70; 82 71 72;
83 70 73; 84 73 74; 85 74 75; 86 75 76; 87 76 72; 88 77 78; 89 79 80; 90 78 81;
91 81 82; 92 82 83; 93 83 84; 94 84 80; 95 85 93; 96 87 94; 98 89 93; 99 90 91;
100 91 94; 102 52 160; 103 60 142; 104 68 143; 105 76 162; 106 84 144;
107 51 118; 108 59 116; 109 67 114; 110 75 110; 111 83 108; 112 50 120;
113 58 117; 114 66 115; 115 74 112; 116 82 109; 117 49 166; 118 57 100;
119 65 101; 120 73 168; 121 81 145; 122 78 103; 123 70 104; 124 62 105;
125 54 106; 126 46 107; 127 80 72; 128 72 170; 129 64 56; 130 48 56;
131 48 173; 132 69 62; 133 61 70; 134 71 64; 135 63 72; 136 93 90; 137 94 92;
138 87 93; 139 85 94; 140 87 92; 141 85 89; 142 95 54; 143 96 56; 144 96 97;
145 97 95; 146 55 97; 147 97 53; 148 98 165; 149 99 167; 150 100 65;
151 101 73; 152 102 169; 153 103 70; 154 104 62; 155 105 54; 156 106 46;
157 107 38; 158 103 102; 159 104 101; 160 105 100; 161 106 99; 162 107 98;
163 108 91; 164 109 90; 165 110 111; 166 111 83; 167 112 113; 168 113 82;
169 114 75; 170 115 74; 171 116 67; 172 117 66; 173 118 119; 174 119 59;
175 120 121; 176 121 58; 177 122 123; 178 123 51; 179 124 125; 180 125 50;
181 126 127; 182 127 43; 183 128 129; 184 129 42; 185 130 131; 186 131 25;
187 132 133; 188 133 28; 189 134 135; 190 135 24; 191 136 137; 192 137 27;

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193 138 14; 194 139 15; 195 108 109; 196 111 113; 197 110 112; 198 114 115;
199 116 117; 200 119 121; 201 118 120; 202 123 125; 203 122 124; 204 127 129;
205 126 128; 206 133 131; 207 132 130; 208 137 135; 209 136 134; 210 139 138;
211 90 108; 212 108 82; 213 82 111; 214 111 112; 215 112 75; 216 75 115;
217 115 67; 218 67 117; 219 117 59; 220 59 121; 221 121 118; 222 118 50;
223 50 123; 224 123 124; 225 124 43; 226 43 129; 227 129 126; 228 126 25;
229 25 133; 230 133 130; 231 130 27; 232 27 135; 233 135 136; 234 136 14;
235 14 139; 236 139 34; 237 140 13; 238 141 16; 239 142 68; 240 143 76;
241 144 92; 242 145 89; 243 146 147; 244 147 17; 245 148 149; 246 149 18;
247 150 151; 248 151 30; 249 152 153; 250 153 31; 251 154 155; 252 155 44;
253 156 157; 254 157 41; 255 158 159; 256 159 52; 257 160 161; 258 161 60;
259 162 163; 260 163 84; 261 144 108; 262 109 145; 263 163 111; 264 162 110;
265 143 114; 266 142 116; 267 161 119; 268 160 118; 269 159 123; 270 158 122;
271 155 127; 272 154 126; 274 153 133; 275 150 136; 276 151 137; 277 152 132;
278 134 146; 279 135 147; 280 130 148; 281 131 149; 282 128 156; 283 129 157;
284 141 139; 285 138 140; 286 164 98; 287 124 164; 288 165 49; 289 125 165;
290 166 99; 291 120 166; 292 167 57; 293 121 167; 294 117 100; 295 115 101;
296 168 102; 297 112 168; 298 169 81; 299 113 169; 300 70 101; 301 101 62;
302 170 64; 303 170 143; 304 72 143; 305 143 64; 306 79 78; 307 77 80;
308 71 70; 309 69 72; 310 61 64; 311 63 62; 312 171 43; 313 172 51; 318 45 107;
319 107 37; 320 173 40; 321 47 173; 322 173 39; 323 201 79; 324 202 77;
325 203 71; 326 204 69; 327 205 63; 328 206 61; 329 207 55; 330 208 53;
331 209 47; 332 210 45; 333 211 39; 334 212 37; 349 39 37; 350 47 45;
351 55 53; 352 63 61; 353 71 69; 354 79 77; 355 71 63; 356 69 61;
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
67 68 74 75 81 82 88 89 142 143 TABLE ST IPE240
95 96 98 TO 100 136 137 TABLE ST IPE200
1 TO 12 14 16 23 25 29 31 TAPERED 0.2 0.008 0.2 0.12 0.012 0.12 0.012
45 TO 51 56 TO 62 69 TO 73 76 TO 80 83 TO 87 90 TO 94 312 -
313 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
15 17 36 38 41 44 52 55 63 66 102 TO 106 117 TO 121 148 TO 152 237 TO 260 -
286 288 290 292 296 298 TABLE ST UPN140
24 26 30 32 42 43 53 54 64 65 107 TO 116 163 TO 194 TABLE ST UPN200
122 TO 126 153 TO 162 TABLE ST UPN200
306 307 TABLE LD L100X100X8 SP 0.01
308 TO 311 TABLE LD L90X90X8 SP 0.01
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132 TO 135 144 TO 147 TABLE LD L75X75X8 SP 0.01
318 319 321 322 TABLE LD L65X65X6 SP 0.01
127 TO 131 302 320 TABLE SA L75X75X8
300 301 304 305 TABLE ST L90X90X8
198 211 TO 236 265 295 TABLE ST L65X65X6
138 TO 141 TABLE ST L75X75X8
195 TO 197 199 TO 210 261 TO 264 266 TO 272 274 TO 285 287 289 291 293 294 -
297 299 303 TABLE ST L50X50X6
MEMBER PROPERTY EUROPEAN
349 TO 354 PRIS YD 0.5 ZD 0.3
355 356 PRIS YD 0.6 ZD 0.3
323 TO 334 PRIS YD 0.6 ZD 0.6
CONSTANTS
BETA 90 MEMB 1 TO 6 45 46 56 57 67 68 74 75 81 82 88 89 95 96 142 143
MATERIAL STEEL MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 -
98 TO 100 102 TO 272 274 TO 313 318 TO 322
MATERIAL CONCRETE MEMB 323 TO 334 349 TO 356
SUPPORTS
1 TO 3 7 TO 9 33 TO 36 85 87 PINNED
201 TO 212 FIXED
MEMBER TRUSS
127 129 132 TO 135 138 TO 141 195 TO 236 261 TO 272 274 TO 285 287 289 291 -
293 TO 295 297 299 TO 301 303 TO 311 318 319 321 322
MEMBER RELEASE
15 24 30 36 45 46 56 57 67 TO 69 74 TO 76 81 TO 83 88 TO 90 103 108 113 118 -
122 TO 124 126 128 130 131 144 158 TO 162 START MP 0.99
73 80 87 94 145 153 TO 162 193 194 237 238 302 320 END MP 0.99
130 174 176 258 292 END FX MY MZ
125 START FX MY MZ
MEMBER OFFSET
349 TO 354 START 0 -0.25 0
349 TO 354 END 0 -0.25 0
355 356 START 0 -0.3 0
355 356 END 0 -0.3 0
LOAD 1 DL
SELFWEIGHT Y -1.1 LIST 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 -
99 TO 100 102 TO 272 274 TO 313 318 TO 322
SELFWEIGHT Y -1 LIST 323 TO 334 349 TO 356
*WALKWAY LOAD @50KG/SQ. M Including Seal Plate & H/R
MEMBER LOAD
15 17 36 38 41 44 52 55 63 66 102 TO 106 117 TO 121 148 TO 152 237 TO 260 -
286 288 290 292 296 298 UNI GY -0.32
24 26 30 32 42 43 53 54 64 65 107 TO 116 163 TO 194 UNI GY -0.47
*PIPE DEAD LOAD
MEMBER LOAD
47 58 69 76 83 90 CON GY -2.03
51 62 73 80 87 94 CON GY -3.57
*CONVEYOR DEAD LOAD
MEMBER LOAD
42 43 CON GY -5 0.7
24 30 CON GY -5 0.67
191 CON GY -5 1.61857
189 CON GY -5 1.61857
26 32 CON GY -5 0.5
```

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187 CON GY -5 1.36138
185 CON GY -5 1.36138
53 54 CON GY -5 0.023
183 CON GY -5 0.994071
181 CON GY -5 0.994071
184 CON GY -5 1.96514
182 CON GY -5 1.96514
179 CON GY -5 0.619106
177 CON GY -5 0.619106
180 CON GY -5 1.61421
178 CON GY -5 1.61421
175 CON GY -5 0.563355
173 CON GY -5 0.563355
176 CON GY -5 1.51671
174 CON GY -5 1.51671
108 113 CON GY -5 1.03
172 CON GY -5 0.319725
171 CON GY -5 0.319725
109 114 CON GY -5 1.11
170 CON GY -5 1.9623
169 CON GY -5 1.9623
167 CON GY -5 0.796991
165 CON GY -5 0.796991
168 CON GY -5 1.77898
166 CON GY -5 1.77898
164 CON GY -5 0.676864
163 CON GY -5 0.676864
164 CON GY -5 1.87686
163 CON GY -5 1.87686
*
MEMBER LOAD
42 43 CON GX 1.5 0.7
24 30 CON GX 1.5 0.67
191 CON GX 1.5 1.61857
189 CON GX 1.5 1.61857
26 32 CON GX 1.5 0.5
187 CON GX 1.5 1.36138
185 CON GX 1.5 1.36138
53 54 CON GX 1.5 0.023
183 CON GX 1.5 0.994071
181 CON GX 1.5 0.994071
184 CON GX 1.5 1.96514
182 CON GX 1.5 1.96514
179 CON GX 1.5 0.619106
177 CON GX 1.5 0.619106
180 CON GX 1.5 1.61421
178 CON GX 1.5 1.61421
175 CON GX 1.5 0.563355
173 CON GX 1.5 0.563355
176 CON GX 1.5 1.51671
174 CON GX 1.5 1.51671
108 113 CON GX 1.5 1.03
172 CON GX 1.5 0.319725
171 CON GX 1.5 0.319725
```

```
109 114 CON GX 1.5 1.11
170 CON GX 1.5 1.9623
169 CON GX 1.5 1.9623
167 CON GX 1.5 0.796991
165 CON GX 1.5 0.796991
168 CON GX 1.5 1.77898
166 CON GX 1.5 1.77898
164 CON GX 1.5 0.676864
163 CON GX 1.5 0.676864
164 CON GX 1.5 1.87686
163 CON GX 1.5 1.87686
*
MEMBER LOAD
42 43 CON GZ 0.8 0.7
24 30 CON GZ 0.8 0.67
191 CON GZ 0.8 1.61857
189 CON GZ 0.8 1.61857
26 32 CON GZ 0.8 0.5
187 CON GZ 0.8 1.36138
185 CON GZ 0.8 1.36138
53 54 CON GZ 0.8 0.023
183 CON GZ 0.8 0.994071
181 CON GZ 0.8 0.994071
184 CON GZ 0.8 1.96514
182 CON GZ 0.8 1.96514
179 CON GZ 0.8 0.619106
177 CON GZ 0.8 0.619106
180 CON GZ 0.8 1.61421
178 CON GZ 0.8 1.61421
175 CON GZ 0.8 0.563355
173 CON GZ 0.8 0.563355
176 CON GZ 0.8 1.51671
174 CON GZ 0.8 1.51671
108 113 CON GZ 0.8 1.03
172 CON GZ 0.8 0.319725
171 CON GZ 0.8 0.319725
109 114 CON GZ 0.8 1.11
170 CON GZ 0.8 1.9623
169 CON GZ 0.8 1.9623
167 CON GZ 0.8 0.796991
165 CON GZ 0.8 0.796991
168 CON GZ 0.8 1.77898
166 CON GZ 0.8 1.77898
164 CON GZ 0.8 0.676864
163 CON GZ 0.8 0.676864
164 CON GZ 0.8 1.87686
163 CON GZ 0.8 1.87686
*CABLE TRAY LOAD
47 58 69 76 83 90 158 TO 162 CON GY -11.76
47 58 69 76 83 90 158 TO 162 CMOM GX 4.7
LOAD 2 LL
*WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
MEMBER LOAD
15 17 24 26 30 32 36 38 41 TO 44 52 TO 55 63 TO 66 102 TO 121 148 TO 152 163 -
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164 TO 194 237 TO 260 286 288 290 292 296 298 UNI GY -2
LOAD 3 ML
*CONVEYOR MATERIAL LOAD
MEMBER LOAD
42 43 CON GY -5.5 0.7
24 30 CON GY -5.5 0.67
191 CON GY -5.5 1.61857
189 CON GY -5.5 1.61857
26 32 CON GY -5.5 0.5
187 CON GY -5.5 1.36138
185 CON GY -5.5 1.36138
53 54 CON GY -5.5 0.023
183 CON GY -5.5 0.994071
181 CON GY -5.5 0.994071
184 CON GY -5.5 1.96514
182 CON GY -5.5 1.96514
179 CON GY -5.5 0.619106
177 CON GY -5.5 0.619106
180 CON GY -5.5 1.61421
178 CON GY -5.5 1.61421
175 CON GY -5.5 0.563355
173 CON GY -5.5 0.563355
176 CON GY -5.5 1.51671
174 CON GY -5.5 1.51671
108 113 CON GY -5.5 1.03
172 CON GY -5.5 0.319725
171 CON GY -5.5 0.319725
109 114 CON GY -5.5 1.11
170 CON GY -5.5 1.9623
169 CON GY -5.5 1.9623
167 CON GY -5.5 0.796991
165 CON GY -5.5 0.796991
168 CON GY -5.5 1.77898
166 CON GY -5.5 1.77898
164 CON GY -5.5 0.676864
163 CON GY -5.5 0.676864
164 CON GY -5.5 1.87686
163 CON GY -5.5 1.87686
** Spillage Load @75kg/ Sq. m on seal plate
24 26 30 32 42 43 53 54 64 65 107 TO 116 163 TO 194 UNI GY -0.375
LOAD 4 TL1
TEMPERATURE LOAD
1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 102 TO 121 123 -
124 127 TO 152 154 155 158 TO 162 TEMP 25
LOAD 5 TL2
TEMPERATURE LOAD
1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 102 TO 121 123 -
124 127 TO 152 154 155 158 TO 162 TEMP -25
LOAD 6 WL +Z
MEMBER LOAD
1 TO 6 45 46 56 57 67 68 74 75 81 82 88 89 95 96 142 143 UNI GZ 0.5
*DUE TO CONVEYOR HOOD
42 43 CON GZ 0.75 0.7
24 30 CON GZ 0.75 0.67
```

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191 CON GZ 0.75 1.61857
189 CON GZ 0.75 1.61857
26 32 CON GZ 0.75 0.5
187 CON GZ 0.75 1.36138
185 CON GZ 0.75 1.36138
53 54 CON GZ 0.75 0.023
183 CON GZ 0.75 0.994071
181 CON GZ 0.75 0.994071
184 CON GZ 0.75 1.96514
182 CON GZ 0.75 1.96514
179 CON GZ 0.75 0.619106
177 CON GZ 0.75 0.619106
180 CON GZ 0.75 1.61421
178 CON GZ 0.75 1.61421
175 CON GZ 0.75 0.563355
173 CON GZ 0.75 0.563355
176 CON GZ 0.75 1.51671
174 CON GZ 0.75 1.51671
108 113 CON GZ 0.75 1.03
172 CON GZ 0.75 0.319725
171 CON GZ 0.75 0.319725
109 114 CON GZ 0.75 1.11
170 CON GZ 0.75 1.9623
169 CON GZ 0.75 1.9623
167 CON GZ 0.75 0.796991
165 CON GZ 0.75 0.796991
168 CON GZ 0.75 1.77898
166 CON GZ 0.75 1.77898
164 CON GZ 0.75 0.676864
163 CON GZ 0.75 0.676864
164 CON GZ 0.75 1.87686
163 CON GZ 0.75 1.87686
*DUE TO CABLE TRAY
47 58 69 76 83 90 158 TO 162 CON GZ 2.38
47 58 69 76 83 90 158 TO 162 CMOM GX 2
LOAD 7 WL -Z
MEMBER LOAD
1 TO 6 45 46 56 57 67 68 74 75 81 82 88 89 95 96 142 143 UNI GZ -0.5
*DUE TO CONVEYOR HOOD
42 43 CON GZ -0.75 0.7
24 30 CON GZ -0.75 0.67
191 CON GZ -0.75 1.61857
189 CON GZ -0.75 1.61857
26 32 CON GZ -0.75 0.5
187 CON GZ -0.75 1.36138
185 CON GZ -0.75 1.36138
53 54 CON GZ -0.75 0.023
183 CON GZ -0.75 0.994071
181 CON GZ -0.75 0.994071
184 CON GZ -0.75 1.96514
182 CON GZ -0.75 1.96514
179 CON GZ -0.75 0.619106
177 CON GZ -0.75 0.619106
180 CON GZ -0.75 1.61421
```

```
178 CON GZ -0.75 1.61421
175 CON GZ -0.75 0.563355
173 CON GZ -0.75 0.563355
176 CON GZ -0.75 1.51671
174 CON GZ -0.75 1.51671
108 113 CON GZ -0.75 1.03
172 CON GZ -0.75 0.319725
171 CON GZ -0.75 0.319725
109 114 CON GZ -0.75 1.11
170 CON GZ -0.75 1.9623
169 CON GZ -0.75 1.9623
167 CON GZ -0.75 0.796991
165 CON GZ -0.75 0.796991
168 CON GZ -0.75 1.77898
166 CON GZ -0.75 1.77898
164 CON GZ -0.75 0.676864
163 CON GZ -0.75 0.676864
164 CON GZ -0.75 1.87686
163 CON GZ -0.75 1.87686
*DUE TO CABLE TRAY
47 58 69 76 83 90 158 TO 162 CON GZ -2.38
47 58 69 76 83 90 158 TO 162 CMOM GX -2
LOAD 11 G
REPEAT LOAD
1 1.0
LOAD 12 G + QN + MN
REPEAT LOAD
1 1.0 2 1.0 3 1.0
****LOAD COMBINATION for structural design
*LOAD COMB 201 1.4 DL
*1 1.4
*LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
*1 1.2 4 1.2 2 1.6 3 1.6
*LOAD COMB 203 1.2 (DL + TL1) + 1.6  (LL+ML)
*1 1.2 5 1.2 2 1.6 3 1.6
*LOAD COMB 204 1.2 DL + 1.6 WL + 0.5 (LL+ML)
*1 1.2 6 1.6 2 0.5 3 0.5
*LOAD COMB 205 1.2 (DL + TL1) + 1.6  (LL+ML)
*1 1.2 7 1.6 2 0.5 3 0.5
*LOAD COMB 206 0.9 DL+ 1.6 WL
*1 0.9 6 1.6
*LOAD COMB 207 0.9 DL+ 1.6 WL
*1 0.9 7 1.6
****LOAD COMBINATION FOR BEARING PRESSURE CHECK
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 4 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
1 1.0 5 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
```

```
LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
1 0.6 6 1.0
LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
1 0.6 7 1.0
************************************************************
****LOAD COMBINATION FOR FOUNDATION DESIGN
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 4 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
1 1.4 5 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
1 0.9 6 1.6
LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
1 0.9 7 1.6
************************************************************
PERFORM ANALYSIS
*STEEL TAKE OFF LIST 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 -
*98 TO 100 102 TO 272 274 TO 313 318 TO 322
*LOAD LIST 201 TO 207
*PARAMETER 1
*CODE LRFD
*FYLD 275000 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 -
*102 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
*296 298 302 312 313 320
*FU 410000 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 -
*102 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
*296 298 302 312 313 320
*TRACK 0 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 102 -
*103 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
*296 298 302 312 313 320
*BEAM 1 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 102 -
*103 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
*296 298 302 312 313 320
*TMAIN 300 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 -
*102 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
*296 298 302 312 313 320
*MAIN 200 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 -
*102 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
*296 298 302 312 313 320
*NSF 0.8 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 102 -
*103 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
*296 298 302 312 313 320
*UNT 1.9 MEMB 7 TO 12 14 16 23 25 29 31 47 TO 51 58 TO 62 69 TO 73 76 TO 80 -
*83 TO 87 90 TO 94 98 TO 100 136 137 312 313
*LY 1.9 MEMB 7 TO 12 14 16 23 25 29 31 47 TO 51 58 TO 62 69 TO 73 76 TO 80 -
*83 TO 87 90 TO 94 98 TO 100 136 137 312 313
*LX 1.9 MEMB 7 TO 12 14 16 23 25 29 31 47 TO 51 58 TO 62 69 TO 73 76 TO 80 -
*83 TO 87 90 TO 94 98 TO 100 136 137 312 313
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*LZ 5.5 MEMB 7 TO 12 14 16 23 25 29 31 47 TO 51 58 TO 62 69 TO 73 76 TO 80 -
*83 TO 87 90 TO 94 98 TO 100 136 137 312 313
*KZ 2 MEMB 41 TO 44 103 104 106 108 109 111 113 114 116 118 119 121 TO 126 -
*150 151 153 TO 157 163 164 169 TO 172 193 194 237 TO 242
*KZ 3 MEMB 15 17 24 26 30 32 36 38 52 TO 55 63 TO 65 102 105 107 110 112 115 -
*165 TO 168 173 TO 192 243 TO 260
*LZ 6.1 MEMB 66 117 120 148 149 152 286 288 290 292 296 298
*KZ 0.85 MEMB 211 TO 236 300 301 304 305
*KZ 0.5 MEMB 132 TO 135 306 TO 311
*CHECK CODE MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 -
*102 TO 126 132 TO 272 274 TO 301 303 TO 313 318 319 321 322
LOAD LIST 401 TO 407
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 323 TO 334 349 TO 356
FYMAIN 420000 MEMB 323 TO 334 349 TO 356
TRACK 0 MEMB 323 TO 334 349 TO 356
CLB 0.05 MEMB 323 TO 334 349 TO 356
CLT 0.05 MEMB 323 TO 334 349 TO 356
CLS 0.05 MEMB 323 TO 334 349 TO 356
MAXMAIN 25 MEMB 323 TO 334 349 TO 356
MINMAIN 16 MEMB 323 TO 334 349 TO 356
MINSEC 8 MEMB 323 TO 334 349 TO 356
REINF 0 MEMB 323 TO 334
DESIGN COLUMN 323 TO 334
DESIGN BEAM 349 TO 356
END CONCRETE DESIGN
FINISH
```

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**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 30, 2020                *
*           Time=    17: 9:50                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
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```
     1. STAAD SPACE
INPUT FILE:  D:\F Drive\Maaden Gold\Conveyor CNV-3130-003\Civil\Documents\Staad\3130-CNV-003 GROUND POR..
     2. START JOB INFORMATION
     3. ENGINEER DATE 28-MAR-20
     4. JOB NAME MMGP-SAUDI ARABIA
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045/900707
     7. JOB REV 0
     8. JOB PART GROUND PORTAL FOUNDATIONS
     9. JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
    10. ENGINEER NAME APL
    11. CHECKER NAME SBY
    12. APPROVED NAME PRI
    13. END JOB INFORMATION
    14. INPUT WIDTH 79
    15. UNIT METER KN
    16. JOINT COORDINATES
    17. 1 0 0.125 -0.55; 2 -6.15 0.125 -0.55; 3 -12.55 0.125 -0.55; 4 0 0.41 -0.55
    18. 5 -6.15 0.743 -0.55; 6 -12.55 1.091 -0.55; 7 6.73533E-016 0.125 -4.505
    19. 8 -6.15 0.125 -4.505; 9 -12.55 0.125 -4.505; 13 0 0.41 -0.795
    20. 14 2.28389E-016 0.41 -1.865; 15 4.45144E-016 0.41 -3.435
    21. 16 5.76177E-016 0.41 -4.505; 17 -6.15 0.743 -0.795; 18 -12.55 1.091 -0.795
    22. 24 -6.15 0.743 -1.865; 25 -12.55 1.091 -1.865; 27 -6.15 0.743 -3.435
    23. 28 -12.55 1.091 -3.435; 30 -6.15 0.743 -4.505; 31 -12.55 1.091 -4.505
    24. 33 3.025 -0.0449997 -0.795; 34 3.025 -0.0449997 -1.865
    25. 35 3.025 -0.0449997 -3.435; 36 3.025 -0.0449997 -4.505; 37 -18.622 -0.15 0.1
    26. 38 -18.622 1.515 0.1; 39 -18.622 -0.15 -5.4; 40 -18.622 1.515 -5.4
    27. 41 -18.622 1.515 -0.795; 42 -18.622 1.515 -1.865; 43 -18.622 1.515 -3.435
    28. 44 -18.622 1.515 -4.505; 45 -24.622 -0.15 0.1; 46 -24.622 1.935 0.1
    29. 47 -24.622 -0.15 -5.4; 48 -24.622 1.935 -5.4; 49 -24.622 1.935 -0.795
    30. 50 -24.622 1.935 -1.865; 51 -24.622 1.935 -3.435; 52 -24.622 1.935 -4.505
    31. 53 -30.747 -0.15 0.1; 54 -30.747 2.363 0.1; 55 -30.747 -0.15 -5.4
    32. 56 -30.747 2.363 -5.4; 57 -30.747 2.363 -0.795; 58 -30.747 2.363 -1.865
    33. 59 -30.747 2.363 -3.435; 60 -30.747 2.363 -4.505; 61 -35.128 -0.15 0.1
    34. 62 -35.128 2.953 0.1; 63 -35.128 -0.15 -5.4; 64 -35.128 2.953 -5.4
    35. 65 -35.128 2.953 -0.795; 66 -35.128 2.953 -1.865; 67 -35.128 2.953 -3.435
    36. 68 -35.128 2.953 -4.505; 69 -39.385 -0.15 0.1; 70 -39.385 3.526 0.1
    37. 71 -39.385 -0.15 -5.4; 72 -39.385 3.526 -5.4; 73 -39.385 3.526 -0.795
    38. 74 -39.385 3.526 -1.865; 75 -39.385 3.526 -3.435; 76 -39.385 3.526 -4.505
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39. 77 -45.385 -0.15 0.1; 78 -45.385 4.333 0.1; 79 -45.385 -0.15 -5.4
40. 80 -45.385 4.333 -5.4; 81 -45.385 4.333 -0.795; 82 -45.385 4.333 -1.865
41. 83 -45.385 4.333 -3.435; 84 -45.385 4.333 -4.505; 85 -49.514 2.8 -1.65
42. 87 -49.514 2.8 -3.65; 89 -49.514 4.889 -0.795; 90 -49.514 4.889 -1.865
43. 91 -49.514 4.889 -3.435; 92 -49.514 4.889 -4.505; 93 -49.514 4.889 -1.65
44. 94 -49.514 4.889 -3.65; 95 -30.747 2.25 0.1; 96 -30.747 2.25 -5.4
45. 97 -30.747 2.25 -2.65; 98 -21.622 1.725 -0.795; 99 -27.6845 2.149 -0.795
46. 100 -32.9375 2.658 -0.795; 101 -37.2565 3.2395 -0.795
47. 102 -42.385 3.9295 -0.795; 103 -42.385 3.9295 0.1; 104 -37.2565 3.2395 0.1
48. 105 -32.9375 2.658 0.1; 106 -27.6845 2.149 0.1; 107 -21.622 1.725 0.1
49. 108 -47.4495 4.611 -3.435; 109 -47.4495 4.611 -1.865; 110 -41.385 3.795 -3.435
50. 111 -43.385 4.064 -3.435; 112 -41.385 3.795 -1.865; 113 -43.385 4.064 -1.865
51. 114 -37.2565 3.2395 -3.435; 115 -37.2565 3.2395 -1.865
52. 116 -32.9375 2.658 -3.435; 117 -32.9375 2.658 -1.865
53. 118 -26.6637 2.07767 -3.435; 119 -28.7053 2.22033 -3.435
54. 120 -26.6637 2.07767 -1.865; 121 -28.7053 2.22033 -1.865
55. 122 -20.622 1.655 -3.435; 123 -22.622 1.795 -3.435; 124 -20.622 1.655 -1.865
56. 125 -22.622 1.795 -1.865; 126 -14.574 1.23233 -3.435
57. 127 -16.598 1.37367 -3.435; 128 -14.574 1.23233 -1.865
58. 129 -16.598 1.37367 -1.865; 130 -8.28333 0.859333 -1.865
59. 131 -10.4167 0.974667 -1.865; 132 -8.28333 0.859333 -3.435
60. 133 -10.4167 0.974667 -3.435; 134 -2.05 0.520667 -1.865
61. 135 -4.1 0.632333 -1.865; 136 -2.05 0.520667 -3.435; 137 -4.1 0.632333 -3.435
62. 138 1.5125 0.1825 -1.865; 139 1.5125 0.1825 -3.435; 140 1.5125 0.1825 -0.795
63. 141 1.5125 0.1825 -4.505; 142 -32.9375 2.658 -4.505
64. 143 -37.2565 3.2395 -4.505; 144 -47.4495 4.611 -4.505
65. 145 -47.4495 4.611 -0.795; 146 -2.05 0.520667 -0.795; 147 -4.1 0.632333 -0.795
66. 148 -8.28333 0.859333 -0.795; 149 -10.4167 0.974667 -0.795
67. 150 -2.05 0.520667 -4.505; 151 -4.1 0.632333 -4.505
68. 152 -8.28333 0.859333 -4.505; 153 -10.4167 0.974667 -4.505
69. 154 -14.574 1.23233 -4.505; 155 -16.598 1.37367 -4.505
70. 156 -14.574 1.23233 -0.795; 157 -16.598 1.37367 -0.795
71. 158 -20.622 1.655 -4.505; 159 -22.622 1.795 -4.505
72. 160 -26.6637 2.07767 -4.505; 161 -28.7053 2.22033 -4.505
73. 162 -41.385 3.795 -4.505; 163 -43.385 4.064 -4.505; 164 -20.622 1.655 -0.795
74. 165 -22.622 1.795 -0.795; 166 -26.6637 2.07767 -0.795
75. 167 -28.7053 2.22033 -0.795; 168 -41.385 3.795 -0.795
76. 169 -43.385 4.064 -0.795; 170 -37.2565 3.2395 -5.4; 171 -18.622 1.515 -2.65
77. 172 -24.622 1.935 -2.65; 173 -21.622 1.725 -5.4; 201 -45.385 -1.5 -5.4
78. 202 -45.385 -1.5 0.1; 203 -39.385 -1.5 -5.4; 204 -39.385 -1.5 0.1
79. 205 -35.128 -1.5 -5.4; 206 -35.128 -1.5 0.1; 207 -30.747 -1.5 -5.4
80. 208 -30.747 -1.5 0.1; 209 -24.622 -1.5 -5.4; 210 -24.622 -1.5 0.1
81. 211 -18.622 -2.5 -5.4; 212 -18.622 -2.5 0.1
82. MEMBER INCIDENCES
83. 1 1 4; 2 2 5; 3 3 6; 4 7 16; 5 8 30; 6 9 31; 7 4 13; 8 5 17; 9 6 18; 10 13 14
84. 11 14 15; 12 15 16; 14 17 24; 15 13 146; 16 18 25; 17 17 148; 23 24 27
85. 24 14 134; 25 25 28; 26 24 130; 29 27 30; 30 15 136; 31 28 31; 32 27 132
86. 36 16 150; 38 30 152; 41 33 140; 42 34 138; 43 35 139; 44 36 141; 45 37 38
87. 46 39 40; 47 38 41; 48 41 42; 49 42 171; 50 43 44; 51 44 40; 52 31 154
88. 53 28 126; 54 25 128; 55 18 156; 56 45 46; 57 47 48; 58 46 49; 59 49 50
89. 60 50 172; 61 51 52; 62 52 48; 63 44 158; 64 43 122; 65 42 124; 66 41 164
90. 67 53 95; 68 55 96; 69 54 57; 70 57 58; 71 58 59; 72 59 60; 73 60 56; 74 61 62
91. 75 63 64; 76 62 65; 77 65 66; 78 66 67; 79 67 68; 80 68 64; 81 69 70; 82 71 72
92. 83 70 73; 84 73 74; 85 74 75; 86 75 76; 87 76 72; 88 77 78; 89 79 80; 90 78 81
93. 91 81 82; 92 82 83; 93 83 84; 94 84 80; 95 85 93; 96 87 94; 98 89 93; 99 90 91
94. 100 91 94; 102 52 160; 103 60 142; 104 68 143; 105 76 162; 106 84 144

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 95. 107 51 118; 108 59 116; 109 67 114; 110 75 110; 111 83 108; 112 50 120
 96. 113 58 117; 114 66 115; 115 74 112; 116 82 109; 117 49 166; 118 57 100
 97. 119 65 101; 120 73 168; 121 81 145; 122 78 103; 123 70 104; 124 62 105
 98. 125 54 106; 126 46 107; 127 80 72; 128 72 170; 129 64 56; 130 48 56
 99. 131 48 173; 132 69 62; 133 61 70; 134 71 64; 135 63 72; 136 93 90; 137 94 92
100. 138 87 93; 139 85 94; 140 87 92; 141 85 89; 142 95 54; 143 96 56; 144 96 97
101. 145 97 95; 146 55 97; 147 97 53; 148 98 165; 149 99 167; 150 100 65
102. 151 101 73; 152 102 169; 153 103 70; 154 104 62; 155 105 54; 156 106 46
103. 157 107 38; 158 103 102; 159 104 101; 160 105 100; 161 106 99; 162 107 98
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106. 175 120 121; 176 121 58; 177 122 123; 178 123 51; 179 124 125; 180 125 50
107. 181 126 127; 182 127 43; 183 128 129; 184 129 42; 185 130 131; 186 131 25
108. 187 132 133; 188 133 28; 189 134 135; 190 135 24; 191 136 137; 192 137 27
109. 193 138 14; 194 139 15; 195 108 109; 196 111 113; 197 110 112; 198 114 115
110. 199 116 117; 200 119 121; 201 118 120; 202 123 125; 203 122 124; 204 127 129
111. 205 126 128; 206 133 131; 207 132 130; 208 137 135; 209 136 134; 210 139 138
112. 211 90 108; 212 108 82; 213 82 111; 214 111 112; 215 112 75; 216 75 115
113. 217 115 67; 218 67 117; 219 117 59; 220 59 121; 221 121 118; 222 118 50
114. 223 50 123; 224 123 124; 225 124 43; 226 43 129; 227 129 126; 228 126 25
115. 229 25 133; 230 133 130; 231 130 27; 232 27 135; 233 135 136; 234 136 14
116. 235 14 139; 236 139 34; 237 140 13; 238 141 16; 239 142 68; 240 143 76
117. 241 144 92; 242 145 89; 243 146 147; 244 147 17; 245 148 149; 246 149 18
118. 247 150 151; 248 151 30; 249 152 153; 250 153 31; 251 154 155; 252 155 44
119. 253 156 157; 254 157 41; 255 158 159; 256 159 52; 257 160 161; 258 161 60
120. 259 162 163; 260 163 84; 261 144 108; 262 109 145; 263 163 111; 264 162 110
121. 265 143 114; 266 142 116; 267 161 119; 268 160 118; 269 159 123; 270 158 122
122. 271 155 127; 272 154 126; 274 153 133; 275 150 136; 276 151 137; 277 152 132
123. 278 134 146; 279 135 147; 280 130 148; 281 131 149; 282 128 156; 283 129 157
124. 284 141 139; 285 138 140; 286 164 98; 287 124 164; 288 165 49; 289 125 165
125. 290 166 99; 291 120 166; 292 167 57; 293 121 167; 294 117 100; 295 115 101
126. 296 168 102; 297 112 168; 298 169 81; 299 113 169; 300 70 101; 301 101 62
127. 302 170 64; 303 170 143; 304 72 143; 305 143 64; 306 79 78; 307 77 80
128. 308 71 70; 309 69 72; 310 61 64; 311 63 62; 312 171 43; 313 172 51; 318 45 107
129. 319 107 37; 320 173 40; 321 47 173; 322 173 39; 323 201 79; 324 202 77
130. 325 203 71; 326 204 69; 327 205 63; 328 206 61; 329 207 55; 330 208 53
131. 331 209 47; 332 210 45; 333 211 39; 334 212 37; 349 39 37; 350 47 45
132. 351 55 53; 352 63 61; 353 71 69; 354 79 77; 355 71 63; 356 69 61
133. DEFINE MATERIAL START
134. ISOTROPIC STEEL
135. E 2.05E+008
136. POISSON 0.3
137. DENSITY 76.8195
138. ALPHA 1.2E-005
139. DAMP 0.03
140. TYPE STEEL
141. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
142. ISOTROPIC CONCRETE
143. E 2.17185E+007
144. POISSON 0.17
145. DENSITY 23.5616
146. ALPHA 1E-005
147. DAMP 0.05
148. TYPE CONCRETE
149. STRENGTH FCU 27579
150. END DEFINE MATERIAL
```

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151. MEMBER PROPERTY EUROPEAN
152. 67 68 74 75 81 82 88 89 142 143 TABLE ST IPE240
153. 95 96 98 TO 100 136 137 TABLE ST IPE200
154. 1 TO 12 14 16 23 25 29 31 TAPERED 0.2 0.008 0.2 0.12 0.012 0.12 0.012
155. 45 TO 51 56 TO 62 69 TO 73 76 TO 80 83 TO 87 90 TO 94 312 -
156. 313 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
157. 15 17 36 38 41 44 52 55 63 66 102 TO 106 117 TO 121 148 TO 152 237 TO 260 -
158. 286 288 290 292 296 298 TABLE ST UPN140
159. 24 26 30 32 42 43 53 54 64 65 107 TO 116 163 TO 194 TABLE ST UPN200
160. 122 TO 126 153 TO 162 TABLE ST UPN200
161. 306 307 TABLE LD L100X100X8 SP 0.01
162. 308 TO 311 TABLE LD L90X90X8 SP 0.01
163. 132 TO 135 144 TO 147 TABLE LD L75X75X8 SP 0.01
164. 318 319 321 322 TABLE LD L65X65X6 SP 0.01
165. 127 TO 131 302 320 TABLE SA L75X75X8
166. 300 301 304 305 TABLE ST L90X90X8
167. 198 211 TO 236 265 295 TABLE ST L65X65X6
168. 138 TO 141 TABLE ST L75X75X8
169. 195 TO 197 199 TO 210 261 TO 264 266 TO 272 274 TO 285 287 289 291 293 294 -
170. 297 299 303 TABLE ST L50X50X6
171. MEMBER PROPERTY EUROPEAN
172. 349 TO 354 PRIS YD 0.5 ZD 0.3
173. 355 356 PRIS YD 0.6 ZD 0.3
174. 323 TO 334 PRIS YD 0.6 ZD 0.6
175. CONSTANTS
176. BETA 90 MEMB 1 TO 6 45 46 56 57 67 68 74 75 81 82 88 89 95 96 142 143
177. MATERIAL STEEL MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 -
178. 98 TO 100 102 TO 272 274 TO 313 318 TO 322
179. MATERIAL CONCRETE MEMB 323 TO 334 349 TO 356
180. SUPPORTS
181. 1 TO 3 7 TO 9 33 TO 36 85 87 PINNED
182. 201 TO 212 FIXED
183. MEMBER TRUSS
184. 127 129 132 TO 135 138 TO 141 195 TO 236 261 TO 272 274 TO 285 287 289 291 -
185. 293 TO 295 297 299 TO 301 303 TO 311 318 319 321 322
186. MEMBER RELEASE
187. 15 24 30 36 45 46 56 57 67 TO 69 74 TO 76 81 TO 83 88 TO 90 103 108 113 118 -
188. 122 TO 124 126 128 130 131 144 158 TO 162 START MP 0.99
189. 73 80 87 94 145 153 TO 162 193 194 237 238 302 320 END MP 0.99
190. 130 174 176 258 292 END FX MY MZ
191. 125 START FX MY MZ
192. MEMBER OFFSET
193. 349 TO 354 START 0 -0.25 0
194. 349 TO 354 END 0 -0.25 0
195. 355 356 START 0 -0.3 0
196. 355 356 END 0 -0.3 0
197. LOAD 1 DL
198. SELFWEIGHT Y -1.1 LIST 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 -
199. 99 TO 100 102 TO 272 274 TO 313 318 TO 322
200. SELFWEIGHT Y -1 LIST 323 TO 334 349 TO 356
201. *WALKWAY LOAD @50KG/SQ. M INCLUDING SEAL PLATE _H/R
202. MEMBER LOAD
203. 15 17 36 38 41 44 52 55 63 66 102 TO 106 117 TO 121 148 TO 152 237 TO 260 -
204. 286 288 290 292 296 298 UNI GY -0.32
205. 24 26 30 32 42 43 53 54 64 65 107 TO 116 163 TO 194 UNI GY -0.47
206. *PIPE DEAD LOAD
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207. MEMBER LOAD
208. 47 58 69 76 83 90 CON GY -2.03
209. 51 62 73 80 87 94 CON GY -3.57
210. *CONVEYOR DEAD LOAD
211. MEMBER LOAD
212. 42 43 CON GY -5 0.7
213. 24 30 CON GY -5 0.67
214. 191 CON GY -5 1.61857
215. 189 CON GY -5 1.61857
216. 26 32 CON GY -5 0.5
217. 187 CON GY -5 1.36138
218. 185 CON GY -5 1.36138
219. 53 54 CON GY -5 0.023
220. 183 CON GY -5 0.994071
221. 181 CON GY -5 0.994071
222. 184 CON GY -5 1.96514
223. 182 CON GY -5 1.96514
224. 179 CON GY -5 0.619106
225. 177 CON GY -5 0.619106
226. 180 CON GY -5 1.61421
227. 178 CON GY -5 1.61421
228. 175 CON GY -5 0.563355
229. 173 CON GY -5 0.563355
230. 176 CON GY -5 1.51671
231. 174 CON GY -5 1.51671
232. 108 113 CON GY -5 1.03
233. 172 CON GY -5 0.319725
234. 171 CON GY -5 0.319725
235. 109 114 CON GY -5 1.11
236. 170 CON GY -5 1.9623
237. 169 CON GY -5 1.9623
238. 167 CON GY -5 0.796991
239. 165 CON GY -5 0.796991
240. 168 CON GY -5 1.77898
241. 166 CON GY -5 1.77898
242. 164 CON GY -5 0.676864
243. 163 CON GY -5 0.676864
244. 164 CON GY -5 1.87686
245. 163 CON GY -5 1.87686
246. *
247. MEMBER LOAD
248. 42 43 CON GX 1.5 0.7
249. 24 30 CON GX 1.5 0.67
250. 191 CON GX 1.5 1.61857
251. 189 CON GX 1.5 1.61857
252. 26 32 CON GX 1.5 0.5
253. 187 CON GX 1.5 1.36138
254. 185 CON GX 1.5 1.36138
255. 53 54 CON GX 1.5 0.023
256. 183 CON GX 1.5 0.994071
257. 181 CON GX 1.5 0.994071
258. 184 CON GX 1.5 1.96514
259. 182 CON GX 1.5 1.96514
260. 179 CON GX 1.5 0.619106
261. 177 CON GX 1.5 0.619106
262. 180 CON GX 1.5 1.61421
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263. 178 CON GX 1.5 1.61421
264. 175 CON GX 1.5 0.563355
265. 173 CON GX 1.5 0.563355
266. 176 CON GX 1.5 1.51671
267. 174 CON GX 1.5 1.51671
268. 108 113 CON GX 1.5 1.03
269. 172 CON GX 1.5 0.319725
270. 171 CON GX 1.5 0.319725
271. 109 114 CON GX 1.5 1.11
272. 170 CON GX 1.5 1.9623
273. 169 CON GX 1.5 1.9623
274. 167 CON GX 1.5 0.796991
275. 165 CON GX 1.5 0.796991
276. 168 CON GX 1.5 1.77898
277. 166 CON GX 1.5 1.77898
278. 164 CON GX 1.5 0.676864
279. 163 CON GX 1.5 0.676864
280. 164 CON GX 1.5 1.87686
281. 163 CON GX 1.5 1.87686
282. *
283. MEMBER LOAD
284. 42 43 CON GZ 0.8 0.7
285. 24 30 CON GZ 0.8 0.67
286. 191 CON GZ 0.8 1.61857
287. 189 CON GZ 0.8 1.61857
288. 26 32 CON GZ 0.8 0.5
289. 187 CON GZ 0.8 1.36138
290. 185 CON GZ 0.8 1.36138
291. 53 54 CON GZ 0.8 0.023
292. 183 CON GZ 0.8 0.994071
293. 181 CON GZ 0.8 0.994071
294. 184 CON GZ 0.8 1.96514
295. 182 CON GZ 0.8 1.96514
296. 179 CON GZ 0.8 0.619106
297. 177 CON GZ 0.8 0.619106
298. 180 CON GZ 0.8 1.61421
299. 178 CON GZ 0.8 1.61421
300. 175 CON GZ 0.8 0.563355
301. 173 CON GZ 0.8 0.563355
302. 176 CON GZ 0.8 1.51671
303. 174 CON GZ 0.8 1.51671
304. 108 113 CON GZ 0.8 1.03
305. 172 CON GZ 0.8 0.319725
306. 171 CON GZ 0.8 0.319725
307. 109 114 CON GZ 0.8 1.11
308. 170 CON GZ 0.8 1.9623
309. 169 CON GZ 0.8 1.9623
310. 167 CON GZ 0.8 0.796991
311. 165 CON GZ 0.8 0.796991
312. 168 CON GZ 0.8 1.77898
313. 166 CON GZ 0.8 1.77898
314. 164 CON GZ 0.8 0.676864
315. 163 CON GZ 0.8 0.676864
316. 164 CON GZ 0.8 1.87686
317. 163 CON GZ 0.8 1.87686
318. *CABLE TRAY LOAD
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319. 47 58 69 76 83 90 158 TO 162 CON GY -11.76
320. 47 58 69 76 83 90 158 TO 162 CMOM GX 4.7
321. LOAD 2 LL
322. *WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
323. MEMBER LOAD
324. 15 17 24 26 30 32 36 38 41 TO 44 52 TO 55 63 TO 66 102 TO 121 148 TO 152 163 -
325. 164 TO 194 237 TO 260 286 288 290 292 296 298 UNI GY -2
326. LOAD 3 ML
327. *CONVEYOR MATERIAL LOAD
328. MEMBER LOAD
329. 42 43 CON GY -5.5 0.7
330. 24 30 CON GY -5.5 0.67
331. 191 CON GY -5.5 1.61857
332. 189 CON GY -5.5 1.61857
333. 26 32 CON GY -5.5 0.5
334. 187 CON GY -5.5 1.36138
335. 185 CON GY -5.5 1.36138
336. 53 54 CON GY -5.5 0.023
337. 183 CON GY -5.5 0.994071
338. 181 CON GY -5.5 0.994071
339. 184 CON GY -5.5 1.96514
340. 182 CON GY -5.5 1.96514
341. 179 CON GY -5.5 0.619106
342. 177 CON GY -5.5 0.619106
343. 180 CON GY -5.5 1.61421
344. 178 CON GY -5.5 1.61421
345. 175 CON GY -5.5 0.563355
346. 173 CON GY -5.5 0.563355
347. 176 CON GY -5.5 1.51671
348. 174 CON GY -5.5 1.51671
349. 108 113 CON GY -5.5 1.03
350. 172 CON GY -5.5 0.319725
351. 171 CON GY -5.5 0.319725
352. 109 114 CON GY -5.5 1.11
353. 170 CON GY -5.5 1.9623
354. 169 CON GY -5.5 1.9623
355. 167 CON GY -5.5 0.796991
356. 165 CON GY -5.5 0.796991
357. 168 CON GY -5.5 1.77898
358. 166 CON GY -5.5 1.77898
359. 164 CON GY -5.5 0.676864
360. 163 CON GY -5.5 0.676864
361. 164 CON GY -5.5 1.87686
362. 163 CON GY -5.5 1.87686
363. ** SPILLAGE LOAD @75KG/ SQ. M ON SEAL PLATE
364. 24 26 30 32 42 43 53 54 64 65 107 TO 116 163 TO 194 UNI GY -0.375
365. LOAD 4 TL1
366. TEMPERATURE LOAD
367. 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 102 TO 121 123 -
368. 124 127 TO 152 154 155 158 TO 162 TEMP 25
369. LOAD 5 TL2
370. TEMPERATURE LOAD
371. 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 102 TO 121 123 -
372. 124 127 TO 152 154 155 158 TO 162 TEMP -25
373. LOAD 6 WL +Z
374. MEMBER LOAD
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375. 1 TO 6 45 46 56 57 67 68 74 75 81 82 88 89 95 96 142 143 UNI GZ 0.5
376. *DUE TO CONVEYOR HOOD
377. 42 43 CON GZ 0.75 0.7
378. 24 30 CON GZ 0.75 0.67
379. 191 CON GZ 0.75 1.61857
380. 189 CON GZ 0.75 1.61857
381. 26 32 CON GZ 0.75 0.5
382. 187 CON GZ 0.75 1.36138
383. 185 CON GZ 0.75 1.36138
384. 53 54 CON GZ 0.75 0.023
385. 183 CON GZ 0.75 0.994071
386. 181 CON GZ 0.75 0.994071
387. 184 CON GZ 0.75 1.96514
388. 182 CON GZ 0.75 1.96514
389. 179 CON GZ 0.75 0.619106
390. 177 CON GZ 0.75 0.619106
391. 180 CON GZ 0.75 1.61421
392. 178 CON GZ 0.75 1.61421
393. 175 CON GZ 0.75 0.563355
394. 173 CON GZ 0.75 0.563355
395. 176 CON GZ 0.75 1.51671
396. 174 CON GZ 0.75 1.51671
397. 108 113 CON GZ 0.75 1.03
398. 172 CON GZ 0.75 0.319725
399. 171 CON GZ 0.75 0.319725
400. 109 114 CON GZ 0.75 1.11
401. 170 CON GZ 0.75 1.9623
402. 169 CON GZ 0.75 1.9623
403. 167 CON GZ 0.75 0.796991
404. 165 CON GZ 0.75 0.796991
405. 168 CON GZ 0.75 1.77898
406. 166 CON GZ 0.75 1.77898
407. 164 CON GZ 0.75 0.676864
408. 163 CON GZ 0.75 0.676864
409. 164 CON GZ 0.75 1.87686
410. 163 CON GZ 0.75 1.87686
411. *DUE TO CABLE TRAY
412. 47 58 69 76 83 90 158 TO 162 CON GZ 2.38
413. 47 58 69 76 83 90 158 TO 162 CMOM GX 2
414. LOAD 7 WL -Z
415. MEMBER LOAD
416. 1 TO 6 45 46 56 57 67 68 74 75 81 82 88 89 95 96 142 143 UNI GZ -0.5
417. *DUE TO CONVEYOR HOOD
418. 42 43 CON GZ -0.75 0.7
419. 24 30 CON GZ -0.75 0.67
420. 191 CON GZ -0.75 1.61857
421. 189 CON GZ -0.75 1.61857
422. 26 32 CON GZ -0.75 0.5
423. 187 CON GZ -0.75 1.36138
424. 185 CON GZ -0.75 1.36138
425. 53 54 CON GZ -0.75 0.023
426. 183 CON GZ -0.75 0.994071
427. 181 CON GZ -0.75 0.994071
428. 184 CON GZ -0.75 1.96514
429. 182 CON GZ -0.75 1.96514
430. 179 CON GZ -0.75 0.619106
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431. 177 CON GZ -0.75 0.619106
432. 180 CON GZ -0.75 1.61421
433. 178 CON GZ -0.75 1.61421
434. 175 CON GZ -0.75 0.563355
435. 173 CON GZ -0.75 0.563355
436. 176 CON GZ -0.75 1.51671
437. 174 CON GZ -0.75 1.51671
438. 108 113 CON GZ -0.75 1.03
439. 172 CON GZ -0.75 0.319725
440. 171 CON GZ -0.75 0.319725
441. 109 114 CON GZ -0.75 1.11
442. 170 CON GZ -0.75 1.9623
443. 169 CON GZ -0.75 1.9623
444. 167 CON GZ -0.75 0.796991
445. 165 CON GZ -0.75 0.796991
446. 168 CON GZ -0.75 1.77898
447. 166 CON GZ -0.75 1.77898
448. 164 CON GZ -0.75 0.676864
449. 163 CON GZ -0.75 0.676864
450. 164 CON GZ -0.75 1.87686
451. 163 CON GZ -0.75 1.87686
452. *DUE TO CABLE TRAY
453. 47 58 69 76 83 90 158 TO 162 CON GZ -2.38
454. 47 58 69 76 83 90 158 TO 162 CMOM GX -2
455. LOAD 11 G
456. REPEAT LOAD
457. 1 1.0
458. LOAD 12 G + QN + MN
459. REPEAT LOAD
460. 1 1.0 2 1.0 3 1.0
461. ****LOAD COMBINATION FOR STRUCTURAL DESIGN
462. *LOAD COMB 201 1.4 DL
463. *1 1.4
464. *LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
465. *1 1.2 4 1.2 2 1.6 3 1.6
466. *LOAD COMB 203 1.2 (DL + TL1) + 1.6  (LL+ML)
467. *1 1.2 5 1.2 2 1.6 3 1.6
468. *LOAD COMB 204 1.2 DL + 1.6 WL + 0.5 (LL+ML)
469. *1 1.2 6 1.6 2 0.5 3 0.5
470. *LOAD COMB 205 1.2 (DL + TL1) + 1.6  (LL+ML)
471. *1 1.2 7 1.6 2 0.5 3 0.5
472. *LOAD COMB 206 0.9 DL+ 1.6 WL
473. *1 0.9 6 1.6
474. *LOAD COMB 207 0.9 DL+ 1.6 WL
475. *1 0.9 7 1.6
476. ****LOAD COMBINATION FOR BEARING PRESSURE CHECK
477. LOAD COMB 301 1.0 DL
478. 1 1.0
479. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
480. 1 1.0 4 1.0 2 1.0 3 1.0
481. LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
482. 1 1.0 5 1.0 2 1.0 3 1.0
483. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
484. 1 1.0 6 0.75 2 0.75 3 0.75
485. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
486. 1 1.0 7 0.75 2 0.75 3 0.75
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487. LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
488. 1 0.6 6 1.0
489. LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
490. 1 0.6 7 1.0
491. ************************************************************
492. ****LOAD COMBINATION FOR FOUNDATION DESIGN
493. LOAD COMB 401 1.4 DL
494. 1 1.4
495. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
496. 1 1.4 4 1.4 2 1.7 3 1.7
497. LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
498. 1 1.4 5 1.4 2 1.7 3 1.7
499. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
500. 1 1.2 6 1.6 2 0.5 3 0.5
501. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
502. 1 1.2 7 1.6 2 0.5 3 0.5
503. LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
504. 1 0.9 6 1.6
505. LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
506. 1 0.9 7 1.6
507. ************************************************************
508. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**172**	**NUMBER OF MEMBERS**	**321**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**24**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 9, TOTAL DEGREES OF FREEDOM = 924
TOTAL LOAD COMBINATION CASES = 14 SO FAR.

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509. *STEEL TAKE OFF LIST 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 -
510. *98 TO 100 102 TO 272 274 TO 313 318 TO 322
511. *LOAD LIST 201 TO 207
512. *PARAMETER 1
513. *CODE LRFD
514. *FYLD 275000 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 -
515. *102 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
516. *296 298 302 312 313 320
517. *FU 410000 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 -
518. *102 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
519. *296 298 302 312 313 320
520. *TRACK 0 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 102 -
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521. *103 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
522. *296 298 302 312 313 320
523. *BEAM 1 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 102 -
524. *103 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
525. *296 298 302 312 313 320
526. *TMAIN 300 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 -
527. *102 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
528. *296 298 302 312 313 320
529. *MAIN 200 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 -
530. *102 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
531. *296 298 302 312 313 320
532. *NSF 0.8 MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 102 -
533. *103 TO 121 123 124 127 TO 152 154 155 158 TO 194 237 TO 260 286 288 290 292 -
534. *296 298 302 312 313 320
535. *UNT 1.9 MEMB 7 TO 12 14 16 23 25 29 31 47 TO 51 58 TO 62 69 TO 73 76 TO 80 -
536. *83 TO 87 90 TO 94 98 TO 100 136 137 312 313
537. *LY 1.9 MEMB 7 TO 12 14 16 23 25 29 31 47 TO 51 58 TO 62 69 TO 73 76 TO 80 -
538. *83 TO 87 90 TO 94 98 TO 100 136 137 312 313
539. *LX 1.9 MEMB 7 TO 12 14 16 23 25 29 31 47 TO 51 58 TO 62 69 TO 73 76 TO 80 -
540. *83 TO 87 90 TO 94 98 TO 100 136 137 312 313
541. *LZ 5.5 MEMB 7 TO 12 14 16 23 25 29 31 47 TO 51 58 TO 62 69 TO 73 76 TO 80 -
542. *83 TO 87 90 TO 94 98 TO 100 136 137 312 313
543. *KZ 2 MEMB 41 TO 44 103 104 106 108 109 111 113 114 116 118 119 121 TO 126 -
544. *150 151 153 TO 157 163 164 169 TO 172 193 194 237 TO 242
545. *KZ 3 MEMB 15 17 24 26 30 32 36 38 52 TO 55 63 TO 65 102 105 107 110 112 115 -
546. *165 TO 168 173 TO 192 243 TO 260
547. *LZ 6.1 MEMB 66 117 120 148 149 152 286 288 290 292 296 298
548. *KZ 0.85 MEMB 211 TO 236 300 301 304 305
549. *KZ 0.5 MEMB 132 TO 135 306 TO 311
550. *CHECK CODE MEMB 1 TO 12 14 TO 17 23 TO 26 29 TO 32 36 38 41 TO 96 98 TO 100 -
551. *102 TO 126 132 TO 272 274 TO 301 303 TO 313 318 319 321 322
552. LOAD LIST 401 TO 407
553. START CONCRETE DESIGN
554. CODE ACI 2008
555. FC 35000 MEMB 323 TO 334 349 TO 356
556. FYMAIN 420000 MEMB 323 TO 334 349 TO 356
557. TRACK 0 MEMB 323 TO 334 349 TO 356
558. CLB 0.05 MEMB 323 TO 334 349 TO 356
559. CLT 0.05 MEMB 323 TO 334 349 TO 356
560. CLS 0.05 MEMB 323 TO 334 349 TO 356
561. MAXMAIN 25 MEMB 323 TO 334 349 TO 356
562. MINMAIN 16 MEMB 323 TO 334 349 TO 356
563. MINSEC 8 MEMB 323 TO 334 349 TO 356
564. REINF 0 MEMB 323 TO 334
565. DESIGN COLUMN 323 TO 334
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COLUMN NO. 323 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

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COLUMN NO. 324 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

==========

COLUMN NO. 325 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI
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12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

==========

COLUMN NO. 326 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI
---

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

==========

COLUMN NO. 327 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI
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12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

==========

COLUMN NO. 328 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

==========

COLUMN NO. 329 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

==========

COLUMN NO. 330 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

==========

COLUMN NO. 331 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

==========

COLUMN NO. 332 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

==========

COLUMN NO. 333 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

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12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

==========

COLUMN NO. 334 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

---

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

**********END OF COLUMN DESIGN RESULTS**********

566. DESIGN BEAM 349 TO 356

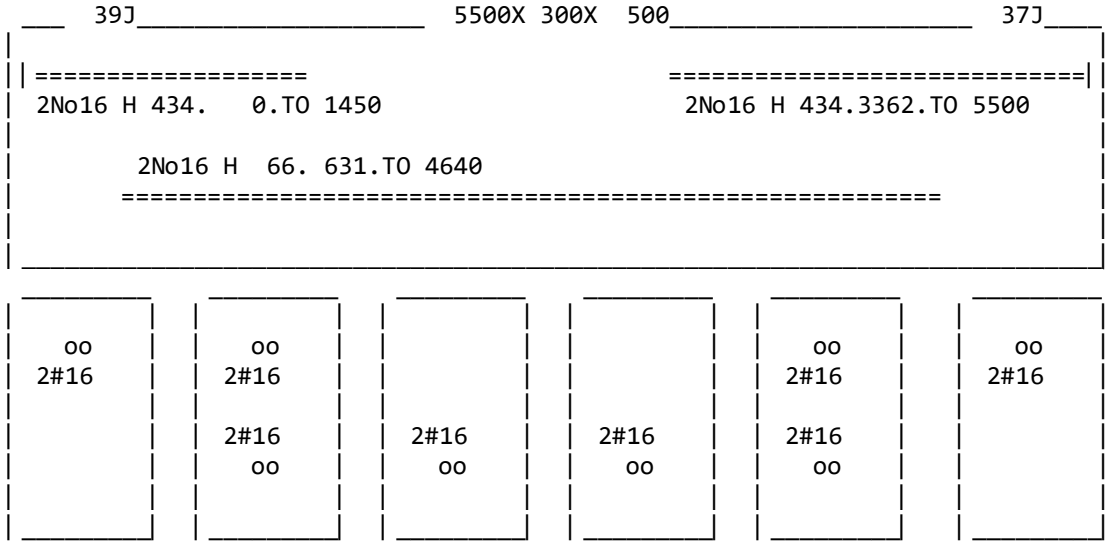
BEAM NO. 349 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	631.	4640.	NO	NO
2	434.	2 - 16MM	0.	1450.	YES	NO
3	434.	2 - 16MM	3362.	5500.	NO	YES

B E A M N O. 349 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.05 KNS Vc= 125.34 KNS Vs= 0.00 KNS
Tu= 0.62 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.87 KNS Vc= 125.34 KNS Vs= 0.00 KNS
Tu= 0.62 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



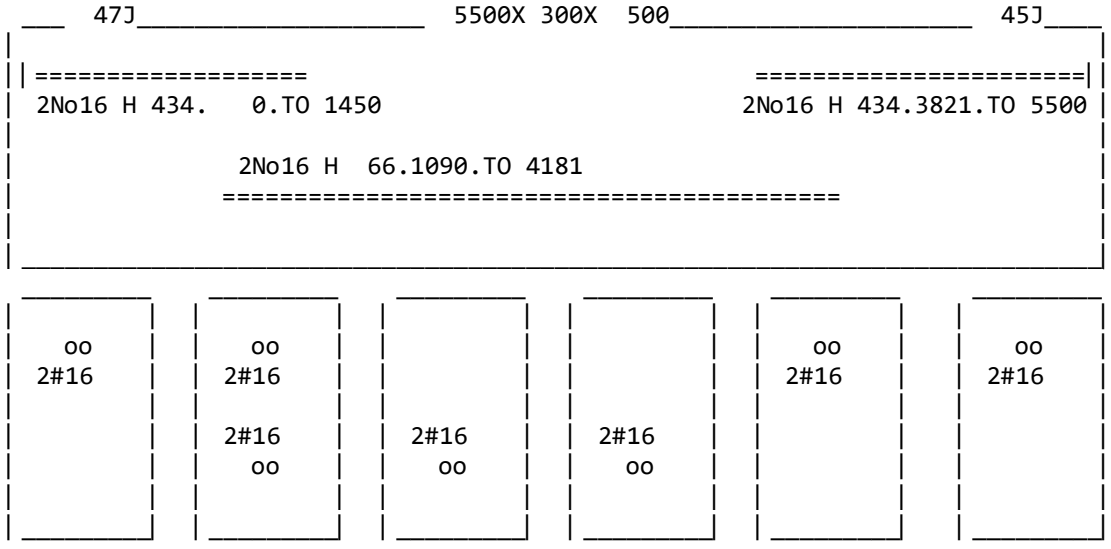
BEAM NO. 350 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	1090.	4181.	NO	NO
2	434.	2 - 16MM	0.	1450.	YES	NO
3	434.	2 - 16MM	3821.	5500.	NO	YES

B E A M N O. 350 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.33 KNS Vc= 126.77 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.59 KNS Vc= 126.77 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



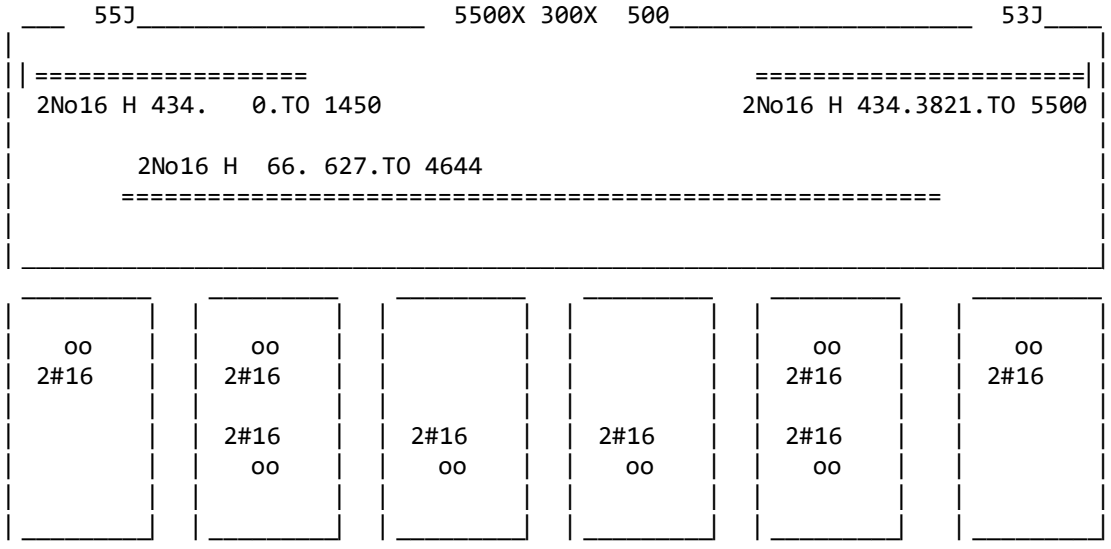
BEAM NO. 351 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	627.	4644.	NO	NO
2	434.	2 - 16MM	0.	1450.	YES	NO
3	434.	2 - 16MM	3821.	5500.	NO	YES

B E A M N O. 351 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.26 KNS Vc= 122.72 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.66 KNS Vc= 122.54 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



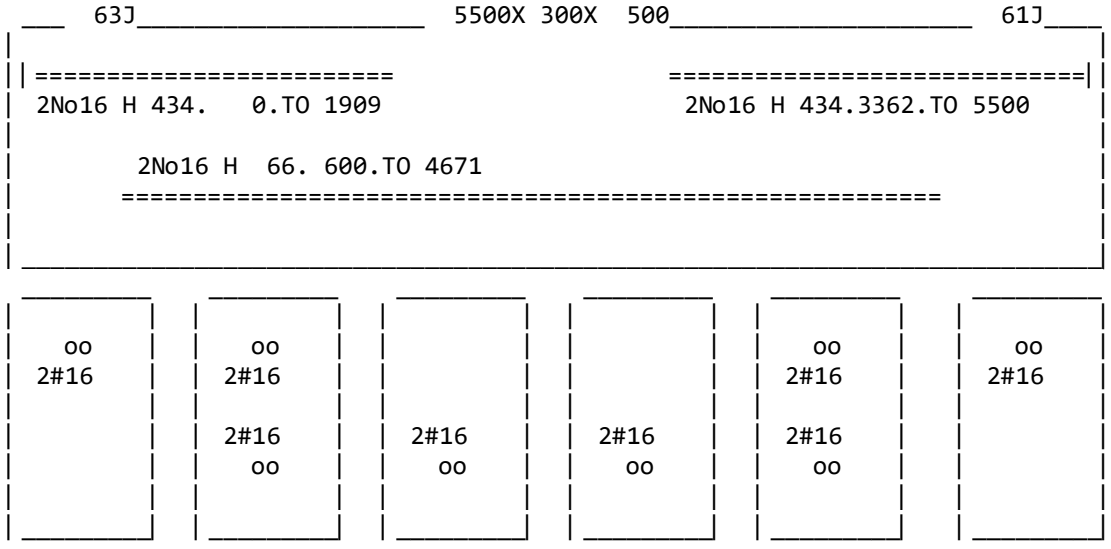
BEAM NO. 352 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	600.	4671.	NO	NO
2	434.	2 - 16MM	0.	1909.	YES	NO
3	434.	2 - 16MM	3362.	5500.	NO	YES

B E A M N O. 352 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.33 KNS Vc= 122.86 KNS Vs= 0.00 KNS
Tu= 0.02 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.59 KNS Vc= 122.76 KNS Vs= 0.00 KNS
Tu= 0.02 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



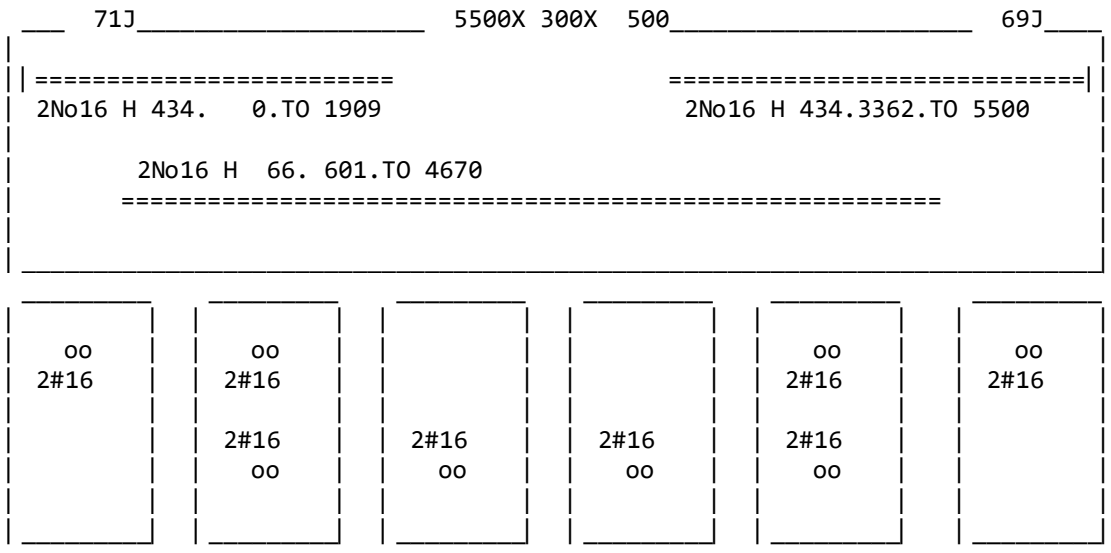
BEAM NO. 353 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	601.	4670.	NO	NO
2	434.	2 - 16MM	0.	1909.	YES	NO
3	434.	2 - 16MM	3362.	5500.	NO	YES

B E A M N O. 353 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.34 KNS Vc= 122.82 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.57 KNS Vc= 122.73 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



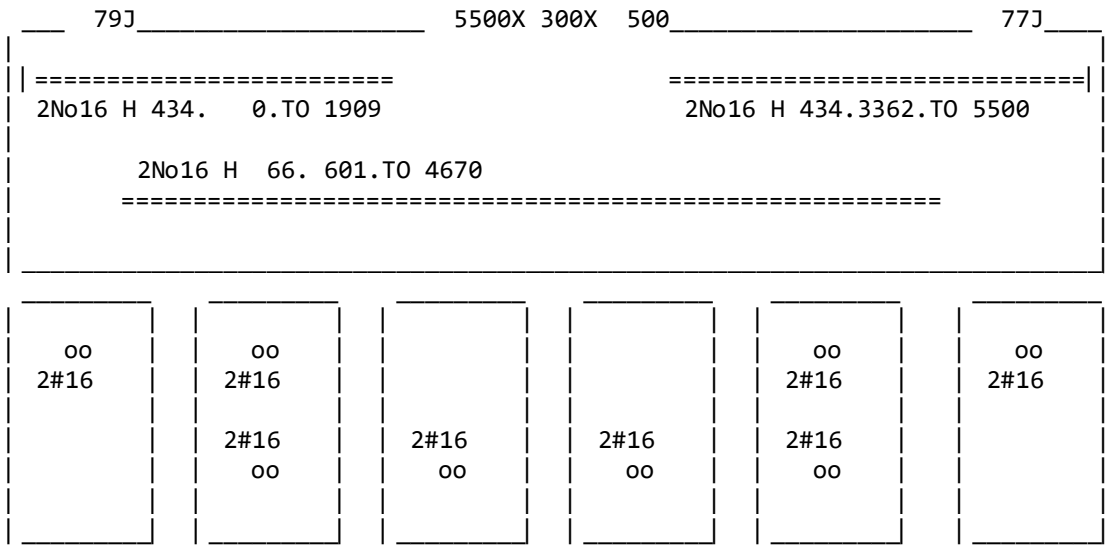
BEAM NO. 354 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	601.	4670.	NO	NO
2	434.	2 - 16MM	0.	1909.	YES	NO
3	434.	2 - 16MM	3362.	5500.	NO	YES

B E A M N O. 354 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.34 KNS Vc= 122.73 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.58 KNS Vc= 122.65 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



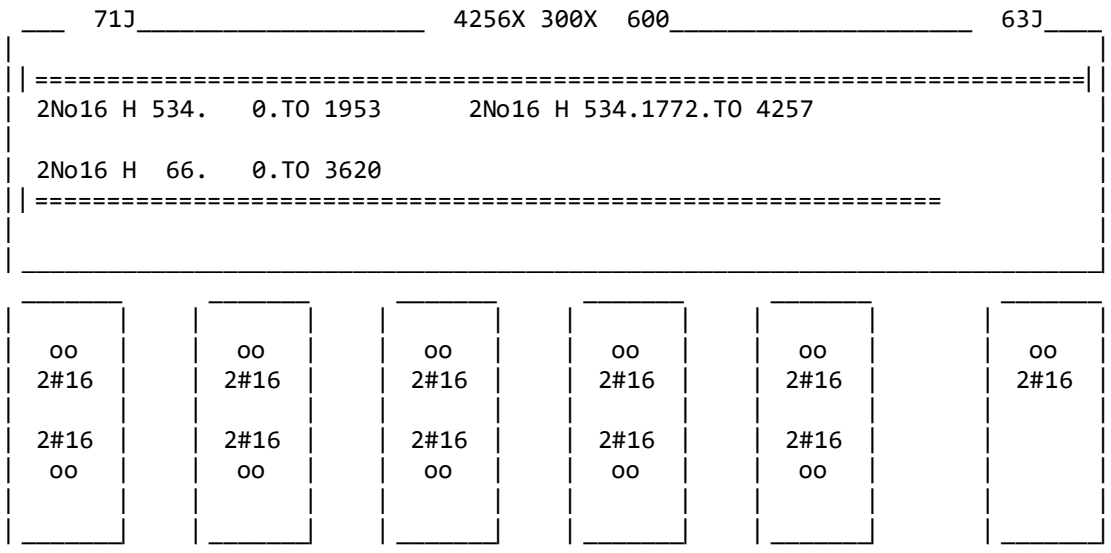
BEAM NO. 355 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4257. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	3620.	YES	NO
2	534.	2 - 16MM	0.	1953.	YES	NO
3	534.	2 - 16MM	1772.	4257.	NO	YES

B E A M N O. 355 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 8.21 KNS Vc= 275.95 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 6.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.72 KNS Vc= 151.68 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 6.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



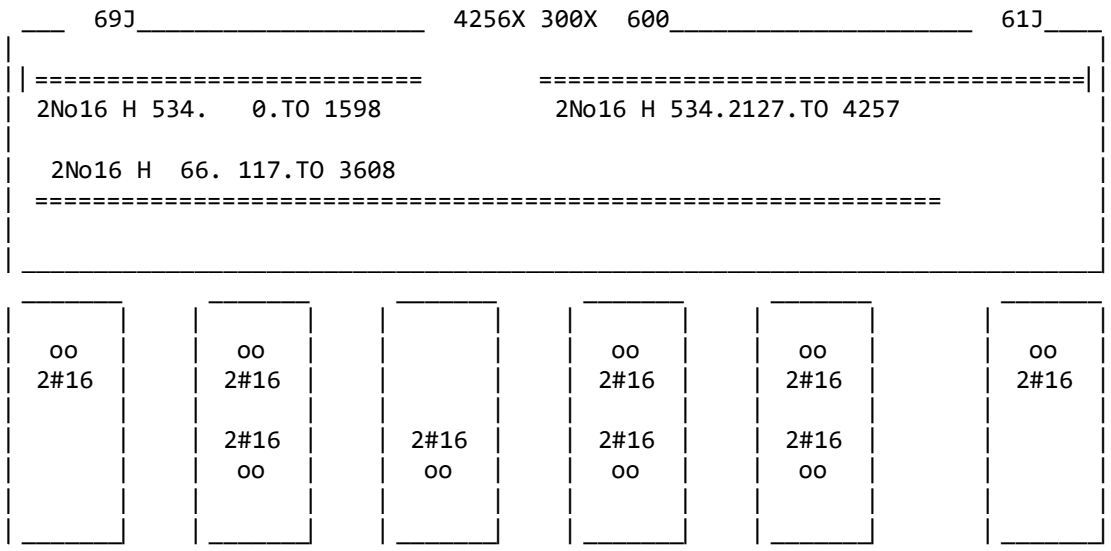
BEAM NO. 356 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4257. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	117.	3608.	NO	NO
2	534.	2 - 16MM	0.	1598.	YES	NO
3	534.	2 - 16MM	2127.	4257.	NO	YES

B E A M N O. 356 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 8.27 KNS Vc= 156.88 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.66 KNS Vc= 156.88 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

567. END CONCRETE DESIGN

568. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 30,2020 TIME= 17: 9:52 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	30-03-2020	29-05-2020	01-11-2023				Date	30-03-2020	29-05-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK/APL	RPK/APL	RPK/APL				Prepared By	RPK/APL	RPK/APL	RPK/APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 4 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-1**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 28-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART TRESTLE TR1 FOUNDATION
JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 2; 3 0 1.49012e-008 0; 4 0 1.49012e-008 2; 5 0 0.35 0;
6 0 0.35 2; 14 -0.75 -3 2.75; 15 -1.501 -3 2.75; 17 -0.5 -3 2.75;
19 -0.25 -3 2.75; 20 0 -3 2.75; 22 0.25 -3 2.75; 24 0.5 -3 2.75;
26 0.75 -3 2.75; 27 1.501 -3 2.75; 28 -0.75 -3 2.5; 29 -1.501 -3 2.5;
30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5; 33 0.25 -3 2.5; 34 0.5 -3 2.5;
35 0.75 -3 2.5; 36 1.501 -3 2.5; 37 -0.75 -3 2.25; 38 -1.501 -3 2.25;
39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25; 42 0.25 -3 2.25;
43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.501 -3 2.25; 46 -0.75 -3 2;
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115 0.75 -3 0.25; 116 1.501 -3 0.25; 117 -0.75 -3 0; 118 -1.501 -3 0;
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148 0.25 -3 -0.75; 149 0.5 -3 -0.75; 150 0.75 -3 -0.75; 151 1.501 -3 -0.75;
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178 -0.75 -3 3; 179 -1.501 -3 3; 181 -0.5 -3 3; 183 -0.25 -3 3; 185 0 -3 3;
187 0.25 -3 3; 189 0.5 -3 3; 191 0.75 -3 3; 193 1.501 -3 3;
212 1.00033 -3 2.75; 213 1.00033 -3 2.5; 214 1.25067 -3 2.75;
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215 1.25067 -3 2.5; 216 1.00033 -3 2.25; 217 1.25067 -3 2.25; 218 1.00033 -3 2;
219 1.25067 -3 2; 220 1.00033 -3 1.75; 221 1.25067 -3 1.75; 222 1.00033 -3 1.5;
223 1.25067 -3 1.5; 224 1.00033 -3 1.25; 225 1.25067 -3 1.25; 226 1.00033 -3 1;
227 1.25067 -3 1; 228 1.00033 -3 0.75; 229 1.25067 -3 0.75; 230 1.00033 -3 0.5;
231 1.25067 -3 0.5; 232 1.00033 -3 0.25; 233 1.25067 -3 0.25; 234 1.00033 -3 0;
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238 1.00033 -3 -0.5; 239 1.25067 -3 -0.5; 240 1.00033 -3 -0.75;
241 1.25067 -3 -0.75; 242 1.00033 -3 -1; 243 1.25067 -3 -1; 247 1.00033 -3 3;
249 1.25067 -3 3; 258 -1.25067 -3 2.75; 259 -1.25067 -3 2.5;
260 -1.00033 -3 2.75; 261 -1.00033 -3 2.5; 262 -1.25067 -3 2.25;
263 -1.00033 -3 2.25; 264 -1.25067 -3 2; 265 -1.00033 -3 2;
266 -1.25067 -3 1.75; 267 -1.00033 -3 1.75; 268 -1.25067 -3 1.5;
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MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
ELEMENT INCIDENCES SHELL
15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33;
20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41;
27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49;
34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56;
41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63;
48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69;
55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77;
60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85;
67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93;
74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99;
80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104;
84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111;
90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117;
96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121;
100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127;
105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131;
111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137;
114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140;
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137 143 145 162 158; 140 145 146 164 162; 143 146 147 166 164;
146 147 148 168 166; 149 148 149 170 168; 152 149 150 172 170;
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175 187 189 24 22; 178 189 191 26 24; 214 26 212 213 35; 215 212 214 215 213;
216 214 27 36 215; 217 35 213 216 44; 218 213 215 217 216; 219 215 36 45 217;
220 44 216 218 52; 221 216 217 219 218; 222 217 45 53 219; 223 52 218 220 61;
224 218 219 221 220; 225 219 53 62 221; 226 61 220 222 70; 227 220 221 223 222;
228 221 62 71 223; 229 70 222 224 79; 230 222 223 225 224; 231 223 71 80 225;
232 79 224 226 88; 233 224 225 227 226; 234 225 80 89 227; 235 88 226 228 97;
236 226 227 229 228; 237 227 89 98 229; 238 97 228 230 106;
239 228 229 231 230; 240 229 98 107 231; 241 106 230 232 115;

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242 230 231 233 232; 243 231 107 116 233; 244 115 232 234 123;
245 232 233 235 234; 246 233 116 124 235; 247 123 234 236 132;
248 234 235 237 236; 249 235 124 133 237; 250 132 236 238 141;
251 236 237 239 238; 252 237 133 142 239; 253 141 238 240 150;
254 238 239 241 240; 255 239 142 151 241; 256 150 240 242 172;
257 240 241 243 242; 258 241 151 174 243; 265 191 247 212 26;
266 247 249 214 212; 267 249 193 27 214; 280 15 258 259 29;
281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38; 284 259 261 263 262;
285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264; 288 263 37 46 265;
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301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275; 304 91 274 276 100;
305 274 275 277 276; 306 275 90 99 277; 307 100 276 278 109;
308 276 277 279 278; 309 277 99 108 279; 310 109 278 280 118;
311 278 279 281 280; 312 279 108 117 281; 313 118 280 282 126;
314 280 281 283 282; 315 281 117 125 283; 316 126 282 284 135;
317 282 283 285 284; 318 283 125 134 285; 319 135 284 286 144;
320 284 285 287 286; 321 285 134 143 287; 322 144 286 288 159;
323 286 287 289 288; 324 287 143 158 289; 331 179 293 258 15;
332 293 295 260 258; 333 295 178 14 260;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 140 149 152 163 166 175 178 -
214 TO 258 265 TO 267 280 TO 324 331 TO 333
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 140 143 146 149 152 163 166 169 172 175 178 214 -
215 TO 219 226 TO 243 250 TO 258 265 TO 267 280 TO 285 292 TO 309 316 TO 324 -
331 TO 333
_SHEAR 63 TO 68 71 TO 76 137 140 143 146 149 152 163 166 169 172 175 178 214 -
215 TO 258 265 TO 267 280 TO 324 331 TO 333
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
149 152 163 166 169 172 175 178 214 TO 258 265 TO 267 280 TO 324 331 TO 332 -
333 THICKNESS 0.45
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
```

```
****************************************
SUPPORTS
1 2 14 15 17 19 20 22 24 26 TO 151 158 159 162 164 166 168 170 172 174 178 -
179 181 183 185 187 189 191 193 212 TO 243 247 249 258 TO 289 293 -
295 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
****************************************
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
5 FX 3.679 FY -50.096 FZ 2.627
6 FX 6.764 FY -99.374 FZ 2.973
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
149 152 163 166 169 172 175 178 214 TO 258 265 TO 267 280 TO 324 331 TO 332 -
333 PR GY -45.9
****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5 FX 3.026 FY -42.937 FZ 0.032
6 FX 2.868 FY -41.463 FZ 0.068
****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
5 FX 1.691 FY -34.5 FZ -0.014
6 FX 1.582 FY -33.4 FZ 0.014
****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
5 FX -2.294 FY 56.373 FZ 20.864
6 FX 2.275 FY -56.573 FZ 20.864
****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5 FX 2.294 FY -56.31 FZ -20.814
6 FX -2.275 FY 56.51 FZ -20.814
****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 0.316 FY 0 FZ 0
6 FX 0.316 FY 0 FZ 0
****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5 FX -0.316 FY 0 FZ 0
6 FX -0.316 FY 0 FZ 0
****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5 FX 0 FY 2.781 FZ 2.131
6 FX 0 FY -2.781 FZ 2.269
****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
```

```
5 FX 0 FY -2.781 FZ -2.131
6 FX 0 FY 2.781 FZ -2.269
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5 FX -0.03 FY 0.3 FZ 0
6 FX -0.03 FY 0.3 FZ 0
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
5 FX -0.129 FY 4.08 FZ 2.199
6 FX 0.129 FY -4.08 FZ 2.199
*****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
```

```
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
```

```
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
****************************************
PERFORM ANALYSIS
****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 30, 2020                *
*           Time=    17:22:58                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE:  D:\F Drive\Maaden Gold\Conveyor CNV-3130-003\Civil\Documents\Staad\3130-CNV-003_TR1 FDN MK..
     2. START JOB INFORMATION
     3. ENGINEER DATE 28-MAR-20
     4. JOB NAME MMGP-SAUDI ARABIA
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045/900707
     7. JOB REV 0
     8. JOB PART TRESTLE TR1 FOUNDATION
     9. JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
    10. ENGINEER NAME APL
    11. CHECKER NAME SBY
    12. APPROVED NAME PRI
    13. END JOB INFORMATION
    14. INPUT WIDTH 79
    15. UNIT METER KN
    16. JOINT COORDINATES
    17. 1 0 -3 0; 2 0 -3 2; 3 0 1.49012E-008 0; 4 0 1.49012E-008 2; 5 0 0.35 0
    18. 6 0 0.35 2; 14 -0.75 -3 2.75; 15 -1.501 -3 2.75; 17 -0.5 -3 2.75
    19. 19 -0.25 -3 2.75; 20 0 -3 2.75; 22 0.25 -3 2.75; 24 0.5 -3 2.75
    20. 26 0.75 -3 2.75; 27 1.501 -3 2.75; 28 -0.75 -3 2.5; 29 -1.501 -3 2.5
    21. 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5; 33 0.25 -3 2.5; 34 0.5 -3 2.5
    22. 35 0.75 -3 2.5; 36 1.501 -3 2.5; 37 -0.75 -3 2.25; 38 -1.501 -3 2.25
    23. 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25; 42 0.25 -3 2.25
    24. 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.501 -3 2.25; 46 -0.75 -3 2
    25. 47 -1.501 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2; 51 0.5 -3 2
    26. 52 0.75 -3 2; 53 1.501 -3 2; 54 -0.75 -3 1.75; 55 -1.501 -3 1.75
    27. 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75
    28. 60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.501 -3 1.75; 63 -0.75 -3 1.5
    29. 64 -1.501 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5
    30. 69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.501 -3 1.5; 72 -0.75 -3 1.25
    31. 73 -1.501 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25
    32. 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.501 -3 1.25
    33. 81 -0.75 -3 1; 82 -1.501 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1
    34. 86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.501 -3 1; 90 -0.75 -3 0.75
    35. 91 -1.501 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75
    36. 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.501 -3 0.75
    37. 99 -0.75 -3 0.5; 100 -1.501 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5
    38. 103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5
```

```
39. 107 1.501 -3 0.5; 108 -0.75 -3 0.25; 109 -1.501 -3 0.25; 110 -0.5 -3 0.25
40. 111 -0.25 -3 0.25; 112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25
41. 115 0.75 -3 0.25; 116 1.501 -3 0.25; 117 -0.75 -3 0; 118 -1.501 -3 0
42. 119 -0.5 -3 0; 120 -0.25 -3 0; 121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0
43. 124 1.501 -3 0; 125 -0.75 -3 -0.25; 126 -1.501 -3 -0.25; 127 -0.5 -3 -0.25
44. 128 -0.25 -3 -0.25; 129 0 -3 -0.25; 130 0.25 -3 -0.25; 131 0.5 -3 -0.25
45. 132 0.75 -3 -0.25; 133 1.501 -3 -0.25; 134 -0.75 -3 -0.5; 135 -1.501 -3 -0.5
46. 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5; 138 0 -3 -0.5; 139 0.25 -3 -0.5
47. 140 0.5 -3 -0.5; 141 0.75 -3 -0.5; 142 1.501 -3 -0.5; 143 -0.75 -3 -0.75
48. 144 -1.501 -3 -0.75; 145 -0.5 -3 -0.75; 146 -0.25 -3 -0.75; 147 0 -3 -0.75
49. 148 0.25 -3 -0.75; 149 0.5 -3 -0.75; 150 0.75 -3 -0.75; 151 1.501 -3 -0.75
50. 158 -0.75 -3 -1; 159 -1.501 -3 -1; 162 -0.5 -3 -1; 164 -0.25 -3 -1.
51. 166 0 -3 -1; 168 0.25 -3 -1; 170 0.5 -3 -1; 172 0.75 -3 -1; 174 1.501 -3 -1.
52. 178 -0.75 -3 3; 179 -1.501 -3 3; 181 -0.5 -3 3; 183 -0.25 -3 3; 185 0 -3 3
53. 187 0.25 -3 3; 189 0.5 -3 3; 191 0.75 -3 3; 193 1.501 -3 3
54. 212 1.00033 -3 2.75; 213 1.00033 -3 2.5; 214 1.25067 -3 2.75
55. 215 1.25067 -3 2.5; 216 1.00033 -3 2.25; 217 1.25067 -3 2.25; 218 1.00033 -3 2
56. 219 1.25067 -3 2; 220 1.00033 -3 1.75; 221 1.25067 -3 1.75; 222 1.00033 -3 1.5
57. 223 1.25067 -3 1.5; 224 1.00033 -3 1.25; 225 1.25067 -3 1.25; 226 1.00033 -3 1
58. 227 1.25067 -3 1; 228 1.00033 -3 0.75; 229 1.25067 -3 0.75; 230 1.00033 -3 0.5
59. 231 1.25067 -3 0.5; 232 1.00033 -3 0.25; 233 1.25067 -3 0.25; 234 1.00033 -3 0
60. 235 1.25067 -3 0; 236 1.00033 -3 -0.25; 237 1.25067 -3 -0.25
61. 238 1.00033 -3 -0.5; 239 1.25067 -3 -0.5; 240 1.00033 -3 -0.75
62. 241 1.25067 -3 -0.75; 242 1.00033 -3 -1; 243 1.25067 -3 -1; 247 1.00033 -3 3
63. 249 1.25067 -3 3; 258 -1.25067 -3 2.75; 259 -1.25067 -3 2.5
64. 260 -1.00033 -3 2.75; 261 -1.00033 -3 2.5; 262 -1.25067 -3 2.25
65. 263 -1.00033 -3 2.25; 264 -1.25067 -3 2; 265 -1.00033 -3 2
66. 266 -1.25067 -3 1.75; 267 -1.00033 -3 1.75; 268 -1.25067 -3 1.5
67. 269 -1.00033 -3 1.5; 270 -1.25067 -3 1.25; 271 -1.00033 -3 1.25
68. 272 -1.25067 -3 1; 273 -1.00033 -3 1; 274 -1.25067 -3 0.75
69. 275 -1.00033 -3 0.75; 276 -1.25067 -3 0.5; 277 -1.00033 -3 0.5
70. 278 -1.25067 -3 0.25; 279 -1.00033 -3 0.25; 280 -1.25067 -3 0
71. 281 -1.00033 -3 0; 282 -1.25067 -3 -0.25; 283 -1.00033 -3 -0.25
72. 284 -1.25067 -3 -0.5; 285 -1.00033 -3 -0.5; 286 -1.25067 -3 -0.75
73. 287 -1.00033 -3 -0.75; 288 -1.25067 -3 -1; 289 -1.00033 -3 -1.
74. 293 -1.25067 -3 3; 295 -1.00033 -3 3
75. MEMBER INCIDENCES
76. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
77. ELEMENT INCIDENCES SHELL
78. 15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33
79. 20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41
80. 27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49
81. 34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56
82. 41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63
83. 48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69
84. 55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77
85. 60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85
86. 67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93
87. 74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99
88. 80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104
89. 84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111
90. 90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117
91. 96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121
92. 100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127
93. 105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131
94. 111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137
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 95. 114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140
 96. 119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146
 97. 122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149
 98. 137 143 145 162 158; 140 145 146 164 162; 143 146 147 166 164
 99. 146 147 148 168 166; 149 148 149 170 168; 152 149 150 172 170
100. 163 178 181 17 14; 166 181 183 19 17; 169 183 185 20 19; 172 185 187 22 20
101. 175 187 189 24 22; 178 189 191 26 24; 214 26 212 213 35; 215 212 214 215 213
102. 216 214 27 36 215; 217 35 213 216 44; 218 213 215 217 216; 219 215 36 45 217
103. 220 44 216 218 52; 221 216 217 219 218; 222 217 45 53 219; 223 52 218 220 61
104. 224 218 219 221 220; 225 219 53 62 221; 226 61 220 222 70; 227 220 221 223 222
105. 228 221 62 71 223; 229 70 222 224 79; 230 222 223 225 224; 231 223 71 80 225
106. 232 79 224 226 88; 233 224 225 227 226; 234 225 80 89 227; 235 88 226 228 97
107. 236 226 227 229 228; 237 227 89 98 229; 238 97 228 230 106
108. 239 228 229 231 230; 240 229 98 107 231; 241 106 230 232 115
109. 242 230 231 233 232; 243 231 107 116 233; 244 115 232 234 123
110. 245 232 233 235 234; 246 233 116 124 235; 247 123 234 236 132
111. 248 234 235 237 236; 249 235 124 133 237; 250 132 236 238 141
112. 251 236 237 239 238; 252 237 133 142 239; 253 141 238 240 150
113. 254 238 239 241 240; 255 239 142 151 241; 256 150 240 242 172
114. 257 240 241 243 242; 258 241 151 174 243; 265 191 247 212 26
115. 266 247 249 214 212; 267 249 193 27 214; 280 15 258 259 29
116. 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38; 284 259 261 263 262
117. 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264; 288 263 37 46 265
118. 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267; 292 55 266 268 64
119. 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73; 296 268 269 271 270
120. 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272; 300 271 72 81 273
121. 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275; 304 91 274 276 100
122. 305 274 275 277 276; 306 275 90 99 277; 307 100 276 278 109
123. 308 276 277 279 278; 309 277 99 108 279; 310 109 278 280 118
124. 311 278 279 281 280; 312 279 108 117 281; 313 118 280 282 126
125. 314 280 281 283 282; 315 281 117 125 283; 316 126 282 284 135
126. 317 282 283 285 284; 318 283 125 134 285; 319 135 284 286 144
127. 320 284 285 287 286; 321 285 134 143 287; 322 144 286 288 159
128. 323 286 287 289 288; 324 287 143 158 289; 331 179 293 258 15
129. 332 293 295 260 258; 333 295 178 14 260
130. START GROUP DEFINITION
131. ELEMENT
132. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
133. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
134. 108 111 112 115 116 119 120 123 124 137 140 149 152 163 166 175 178 -
135. 214 TO 258 265 TO 267 280 TO 324 331 TO 333
136. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
137. 111 TO 116 119 TO 124 137 140 143 146 149 152 163 166 169 172 175 178 214 -
138. 215 TO 219 226 TO 243 250 TO 258 265 TO 267 280 TO 285 292 TO 309 316 TO 324 -
139. 331 TO 333
140. _SHEAR 63 TO 68 71 TO 76 137 140 143 146 149 152 163 166 169 172 175 178 214 -
141. 215 TO 258 265 TO 267 280 TO 324 331 TO 333
142. END GROUP DEFINITION
143. ELEMENT PROPERTY
144. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
145. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
146. 149 152 163 166 169 172 175 178 214 TO 258 265 TO 267 280 TO 324 331 TO 332 -
147. 333 THICKNESS 0.45
148. DEFINE MATERIAL START
149. ISOTROPIC CONCRETE
150. E 2.17185E+007
```

```
151. POISSON 0.17
152. DENSITY 23.5616
153. ALPHA 1E-005
154. DAMP 0.05
155. TYPE CONCRETE
156. STRENGTH FCU 27579
157. END DEFINE MATERIAL
158. MEMBER PROPERTY AMERICAN
159. 1 TO 4 PRIS YD 0.7 ZD 0.5
160. 5 PRIS YD 0.5 ZD 0.3
161. CONSTANTS
162. MATERIAL CONCRETE ALL
163. ****************************************
164. SUPPORTS
165. 1 2 14 15 17 19 20 22 24 26 TO 151 158 159 162 164 166 168 170 172 174 178 -
166. 179 181 183 185 187 189 191 193 212 TO 243 247 249 258 TO 289 293 -
167. 295 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
```

JOINT	INFLUENCE AREA
1	0.062500
2	0.062500
14	0.062541
15	0.031291
17	0.062500
19	0.062500
20	0.062500
22	0.062500
24	0.062500
26	0.062541
27	0.031291
28	0.062541
29	0.031291
30	0.062500
31	0.062500
32	0.062500
33	0.062500
34	0.062500
35	0.062541
36	0.031291
37	0.062541
38	0.031291
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.062541
45	0.031291
46	0.062541
47	0.031291
48	0.062500
49	0.062500
50	0.062500
51	0.062500

52	0.062541
53	0.031291
54	0.062541
55	0.031291
56	0.062500
57	0.062500
58	0.062500
59	0.062500
60	0.062500
61	0.062541
62	0.031291
63	0.062541
64	0.031291
65	0.062500
66	0.062500
67	0.062500
68	0.062500
69	0.062500
70	0.062541
71	0.031291
72	0.062541
73	0.031291
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.062541
80	0.031291
81	0.062541
82	0.031291
83	0.062500
84	0.062500
85	0.062500
86	0.062500
87	0.062500
88	0.062541
89	0.031291
90	0.062541
91	0.031291
92	0.062500
93	0.062500
94	0.062500
95	0.062500
96	0.062500
97	0.062541
98	0.031291
99	0.062541
100	0.031291
101	0.062500
102	0.062500
103	0.062500
104	0.062500
105	0.062500
106	0.062541
107	0.031291

108	0.062541
109	0.031291
110	0.062500
111	0.062500
112	0.062500
113	0.062500
114	0.062500
115	0.062541
116	0.031291
117	0.062541
118	0.031291
119	0.062500
120	0.062500
121	0.062500
122	0.062500
123	0.062541
124	0.031291
125	0.062541
126	0.031291
127	0.062500
128	0.062500
129	0.062500
130	0.062500
131	0.062500
132	0.062541
133	0.031291
134	0.062541
135	0.031291
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.062541
142	0.031291
143	0.062541
144	0.031291
145	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.062541
151	0.031291
158	0.031271
159	0.015646
162	0.031250
164	0.031250
166	0.031250
168	0.031250
170	0.031250
172	0.031271
174	0.015646
178	0.031271
179	0.015646
181	0.031250

183	0.031250
185	0.031250
187	0.031250
189	0.031250
191	0.031271
193	0.015646
212	0.062584
213	0.062584
214	0.062584
215	0.062584
216	0.062584
217	0.062584
218	0.062584
219	0.062584
220	0.062584
221	0.062584
222	0.062584
223	0.062584
224	0.062584
225	0.062584
226	0.062584
227	0.062584
228	0.062584
229	0.062584
230	0.062584
231	0.062584
232	0.062584
233	0.062584
234	0.062584
235	0.062584
236	0.062584
237	0.062584
238	0.062584
239	0.062584
240	0.062584
241	0.062584
242	0.031292
243	0.031292
247	0.031292
249	0.031292
258	0.062584
259	0.062584
260	0.062584
261	0.062584
262	0.062584
263	0.062584
264	0.062584
265	0.062584
266	0.062584
267	0.062584
268	0.062584
269	0.062584
270	0.062584
271	0.062584
272	0.062584
273	0.062584

274	0.062584
275	0.062584
276	0.062584
277	0.062584
278	0.062584
279	0.062584
280	0.062584
281	0.062584
282	0.062584
283	0.062584
284	0.062584
285	0.062584
286	0.062584
287	0.062584
288	0.031292
289	0.031292
293	0.031292
295	0.031292
TOTAL	12.008017

```
168. ****************************************
169. LOAD 1 DEAD LOAD DL
170. SELFWEIGHT Y -1
171. JOINT LOAD
172. 5 FX 3.679 FY -50.096 FZ 2.627
173. 6 FX 6.764 FY -99.374 FZ 2.973
174. ELEMENT LOAD
175. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
176. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
177. 149 152 163 166 169 172 175 178 214 TO 258 265 TO 267 280 TO 324 331 TO 332 -
178. 333 PR GY -45.9
179. ****************************************
180. LOAD 2 LIVE LOAD LL
181. JOINT LOAD
182. 5 FX 3.026 FY -42.937 FZ 0.032
183. 6 FX 2.868 FY -41.463 FZ 0.068
184. ****************************************
185. LOAD 3 MATERIAL LOAD LL
186. JOINT LOAD
187. 5 FX 1.691 FY -34.5 FZ -0.014
188. 6 FX 1.582 FY -33.4 FZ 0.014
189. ****************************************
190. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
191. JOINT LOAD
192. 5 FX -2.294 FY 56.373 FZ 20.864
193. 6 FX 2.275 FY -56.573 FZ 20.864
194. ****************************************
195. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
196. JOINT LOAD
197. 5 FX 2.294 FY -56.31 FZ -20.814
198. 6 FX -2.275 FY 56.51 FZ -20.814
199. ****************************************
200. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
201. JOINT LOAD
202. 5 FX 0.316 FY 0 FZ 0
203. 6 FX 0.316 FY 0 FZ 0
204. ****************************************
```

```
205. LOAD 7 WIND LOAD ACROSS THE CONV.  WL ( -X)
206. JOINT LOAD
207. 5 FX -0.316 FY 0 FZ 0
208. 6 FX -0.316 FY 0 FZ 0
209. ****************************************
210. LOAD 8 TEMP LOAD   TL1
211. JOINT LOAD
212. 5 FX 0 FY 2.781 FZ 2.131
213. 6 FX 0 FY -2.781 FZ 2.269
214. ****************************************
215. LOAD 9 TEMP LOAD   TL2
216. JOINT LOAD
217. 5 FX 0 FY -2.781 FZ -2.131
218. 6 FX 0 FY 2.781 FZ -2.269
219. ****************************************
220. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
221. JOINT LOAD
222. 5 FX -0.03 FY 0.3 FZ 0
223. 6 FX -0.03 FY 0.3 FZ 0
224. ****************************************
225. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
226. JOINT LOAD
227. 5 FX -0.129 FY 4.08 FZ 2.199
228. 6 FX 0.129 FY -4.08 FZ 2.199
229. ****************************************
230. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
231. LOAD 301 1.0 DL
232. REPEAT LOAD
233. 1 1.0
234. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
235. REPEAT LOAD
236. 1 1.0 8 1.0 2 1.0 3 1.0
237. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
238. REPEAT LOAD
239. 1 1.0 9 1.0 2 1.0 3 1.0
240. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
241. REPEAT LOAD
242. 1 1.0 4 0.75 2 0.75 3 0.75
243. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
244. REPEAT LOAD
245. 1 1.0 5 0.75 2 0.75 3 0.75
246. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
247. REPEAT LOAD
248. 1 1.0 6 0.75 2 0.75 3 0.75
249. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
250. REPEAT LOAD
251. 1 1.0 7 0.75 2 0.75 3 0.75
252. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
253. REPEAT LOAD
254. 1 1.0 10 0.525 2 0.75 3 0.75
255. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
256. REPEAT LOAD
257. 1 1.0 10 -0.525 2 0.75 3 0.75
258. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
259. REPEAT LOAD
260. 1 1.0 11 0.525 2 0.75 3 0.75
```

```
261. LOAD 311 1.0 DL + 0.525 EL( -Z) + 0.75  ( LL+ML)
262. REPEAT LOAD
263. 1 1.0 11 -0.525 2 0.75 3 0.75
264. LOAD 312 0.6 DL + 1.0 WL( +Z)
265. REPEAT LOAD
266. 1 0.6 4 1.0
267. LOAD 313 0.6 DL + 1.0 WL( -Z)
268. REPEAT LOAD
269. 1 0.6 5 1.0
270. LOAD 314 0.6 DL + 1.0 WL( +X)
271. REPEAT LOAD
272. 1 0.6 6 1.0
273. LOAD 315 0.6 DL + 1.0 WL( -X)
274. REPEAT LOAD
275. 1 0.6 7 1.0
276. LOAD 316 0.6 DL + 0.7 EL( +X)
277. REPEAT LOAD
278. 1 0.6 10 0.7
279. LOAD 317 0.6 DL + 0.7 EL( -X)
280. REPEAT LOAD
281. 1 0.6 10 -0.7
282. LOAD 318 0.6 DL + 0.7 EL( +Z)
283. REPEAT LOAD
284. 1 0.6 11 0.7
285. LOAD 319 0.6 DL + 0.7 EL( -Z)
286. REPEAT LOAD
287. 1 0.6 11 -0.7
288. ****************************************
289. *****LOAD COMBINATION FOR FOUNDATION DESIGN
290. LOAD 401 1.4 DL
291. REPEAT LOAD
292. 1 1.4
293. LOAD 402 1.4 ( DL + TL1) + 1.7  ( LL+ML)
294. REPEAT LOAD
295. 1 1.4 8 1.4 2 1.7 3 1.7
296. LOAD 403 1.4 ( DL + TL1) + 1.7  ( LL+ML)
297. REPEAT LOAD
298. 1 1.4 9 1.4 2 1.7 3 1.7
299. LOAD 404 1.2 DL+ 1.6 WL( +Z) + 0.5  ( LL+ML)
300. REPEAT LOAD
301. 1 1.2 4 1.6 2 0.5 3 0.5
302. LOAD 405 1.2 DL+ 1.6 WL( -Z) + 0.5  ( LL+ML)
303. REPEAT LOAD
304. 1 1.2 5 1.6 2 0.5 3 0.5
305. LOAD 406 1.2 DL+ 1.6 WL( +X) + 0.5  ( LL+ML)
306. REPEAT LOAD
307. 1 1.2 6 1.6 2 0.5 3 0.5
308. LOAD 407 1.2 DL+ 1.6 WL( -X) + 0.5  ( LL+ML)
309. REPEAT LOAD
310. 1 1.2 7 1.6 2 0.5 3 0.5
311. LOAD 408 1.2 DL + 1.0 EL( +X) + 0.5  ( LL+ML)
312. REPEAT LOAD
313. 1 1.2 10 1.0 2 0.5 3 0.5
314. LOAD 409 1.2 DL + 1.0 EL( -X) + 0.5  ( LL+ML)
315. REPEAT LOAD
316. 1 1.2 10 -1.0 2 0.5 3 0.5
```

```
317. LOAD 410 1.2 DL + 1.0 EL( +Z) + 0.5  ( LL+ML)
318. REPEAT LOAD
319. 1 1.2 11 1.0 2 0.5 3 0.5
320. LOAD 411 1.2 DL + 1.0 EL( -Z) + 0.5  ( LL+ML)
321. REPEAT LOAD
322. 1 1.2 11 -1.0 2 0.5 3 0.5
323. LOAD 412 0.9 DL + 1.6 WL( +Z)
324. REPEAT LOAD
325. 1 0.9 4 1.6
326. LOAD 413 0.9 DL + 1.6 WL( -Z)
327. REPEAT LOAD
328. 1 0.9 5 1.6
329. LOAD 414 0.9 DL + 1.6 WL( +X)
330. REPEAT LOAD
331. 1 0.9 6 1.6
332. LOAD 415 0.9 DL + 1.6 WL( -X)
333. REPEAT LOAD
334. 1 0.9 7 1.6
335. LOAD 416 0.9 DL + 1.0 EL( +X)
336. REPEAT LOAD
337. 1 0.9 10 1.0
338. LOAD 417 0.9 DL + 1.0 EL( -X)
339. REPEAT LOAD
340. 1 0.9 10 -1.0
341. LOAD 418 0.9 DL + 1.0 EL( +Z)
342. REPEAT LOAD
343. 1 0.9 11 1.0
344. LOAD 419 0.9 DL + 1.0 EL( -Z)
345. REPEAT LOAD
346. 1 0.9 11 -1.0
347. ****************************************
348. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**225**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**192**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**221**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 687
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

```
349. ****************************************
350. LOAD LIST 401 TO 419
```

351. START CONCRETE DESIGN

```
352. CODE ACI 2008
353. FC 35000 MEMB 1 TO 5
354. FYMAIN 420000 MEMB 1 TO 5
355. TRACK 1 ALL
356. CLB 0.05 MEMB 1 TO 5
357. CLT 0.05 MEMB 1 TO 5
358. CLS 0.05 ALL
359. MAXMAIN 25 ALL
360. MINMAIN 16 ALL
361. MINSEC 8 ALL
362. REINF 0 MEMB 1 TO 4
363. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	113.62	3.61	0.01059

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	113.62	3.97	0.01165

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    107.90      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    107.90      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 700.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3500.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
---------------------------------------------------------

 12 - 20 MM               1.077      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION:  MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    235.97      4.26    0.00602
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    235.97      8.02    0.01132
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	214.04	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	214.04	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

364. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

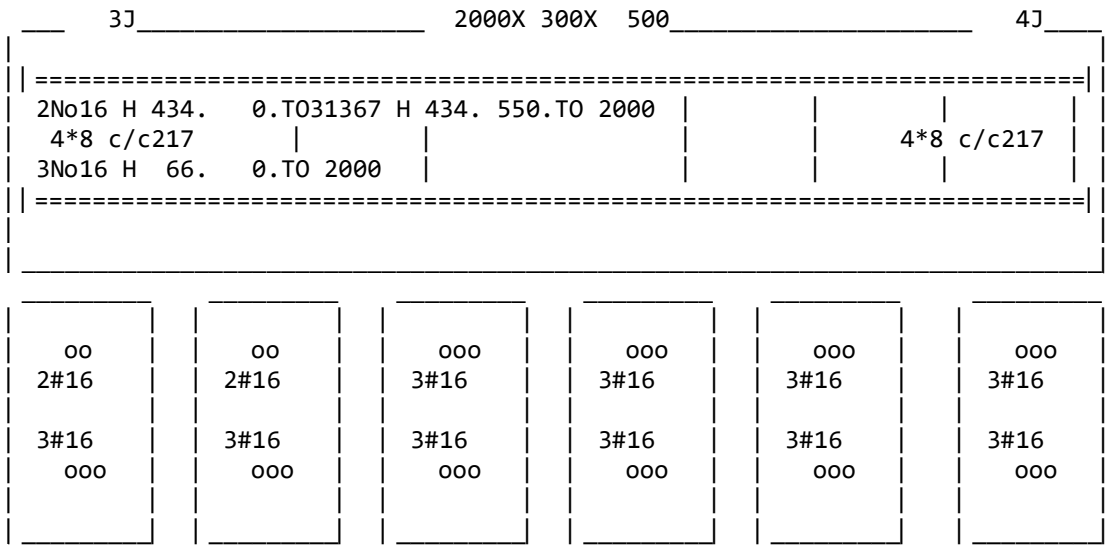
LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	2000.	YES	YES
	CRITICAL POS MOMENT= 53.94 KN-MET AT 0.MM, LOAD 404					
	REQD STEEL= 445.MM2, RHO=0.0034, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
2	434.	2 - 16MM	0.	1367.	YES	NO
	CRITICAL NEG MOMENT= 48.00 KN-MET AT 0.MM, LOAD 413					
	REQD STEEL= 396.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
3	434.	3 - 16MM	550.	2000.	NO	YES
	CRITICAL NEG MOMENT= 56.44 KN-MET AT 2000.MM, LOAD 404					
	REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 52.79 KNS Vc= 127.15 KNS Vs= 0.00 KNS
Tu= 1.37 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM

AT END SUPPORT - Vu= 57.59 KNS Vc= 127.73 KNS Vs= 0.00 KNS
Tu= 1.37 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM



**********END OF BEAM DESIGN**********

365. END CONCRETE DESIGN
366. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 30,2020 TIME= 17: 23: 3 ****

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************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	30-03-2020	29-05-2020	01-11-2023				Date	30-03-2020	29-05-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK/APL	RPK/APL	RPK/APL				Prepared By	RPK/APL	RPK/APL	RPK/APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 5 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-2**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 28-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART GROUND PORTAL FOUNDATIONS
JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
ENGINEER NAME RPK
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 4; 3 0 1.49012e-008 0; 4 0 1.49012e-008 4; 5 0 0.35 0;
6 0 0.35 4;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.6 ZD 0.6
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 2 FIXED
*****************************************
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
JOINT LOAD
5 FX 0 FY -94.1 FZ -22.6
6 FX 0 FY -167.8 FZ 33.8
LOAD 2 LIVE LOAD
JOINT LOAD
5 FX 0 FY -71.9 FZ -15.5
6 FX 0 FY -72.1 FZ 15.6
LOAD 3 MATERIAL LOAD
JOINT LOAD
5 FX 0 FY -50.8 FZ -11
6 FX 0 FY -50.2 FZ 10.9
LOAD 4 WIND LOAD +Z
JOINT LOAD
5 FX 0 FY 145.7 FZ 40.5
```

```
6 FX 0 FY -145.7 FZ 40.6
LOAD 5 WIND LOAD -Z
JOINT LOAD
5 FX -0.04 FY -145.63 FZ -40.53
6 FX 0.04 FY 145.63 FZ -40.62
LOAD 6 WIND LOAD (+X)
JOINT LOAD
5 FX 3.83 FY 0 FZ 0
6 FX 3.83 FY 0 FZ 0
LOAD 7 WIND LOAD (-X)
JOINT LOAD
5 FX -3.83 FY 0 FZ 0
6 FX -3.83 FY 0 FZ 0
LOAD 8 TEMP LOAD 1
JOINT LOAD
5 FX 0 FY -0.16 FZ -22.93
6 FX 0 FY 0.16 FZ 22.83
LOAD 9 TEMP LOAD 2
JOINT LOAD
5 FX 0 FY 0.16 FZ 22.93
6 FX 0 FY -0.16 FZ -22.83
LOAD 10 EARTHQUAKE LOAD +X
JOINT LOAD
5 FX -0.01 FY -0.05 FZ 0
6 FX -0.01 FY 0.05 FZ 0
LOAD 11 EARTHQUAKE LOAD +Z
JOINT LOAD
5 FX -0.01 FY 7.65 FZ 2.08
6 FX 0.01 FY -7.65 FZ 2.09
*****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
```

```
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL( -Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL( +X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL( -X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL( +X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL( -X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL( +Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL( -Z)
1 0.6 11 -0.7
****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 ( DL + TL1) + 1.7  ( LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 ( DL + TL2) + 1.7  ( LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL( +Z) + 0.5  ( LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL( -Z) + 0.5  ( LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL( +X) + 0.5  ( LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL( -X) + 0.5  ( LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL( +X) + 0.5  ( LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL( -X) + 0.5  ( LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL( +Z) + 0.5  ( LL+ML)
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL( -Z) + 0.5  ( LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL( +Z)
1 0.9 4 1.6
LOAD COMB 413 0.9 DL + 1.6 WL( -Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL( +X)
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL( -X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL( +X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL( -X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL( +Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL( -Z)
```

```
1 0.9 11 -1.0
****************************************
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 1 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 30, 2020                *
*           Time=    17:24:55                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Conveyor CNV-3130-003\Civil\Documents\Staad\3130-CNV-003-TR2-Fdn.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 28-MAR-20
     4. JOB NAME MMGP-SAUDI ARABIA
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045/900707
     7. JOB REV 0
     8. JOB PART GROUND PORTAL FOUNDATIONS
     9. JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
    10. ENGINEER NAME RPK
    11. CHECKER NAME SBY
    12. APPROVED NAME PRI
    13. END JOB INFORMATION
    14. INPUT WIDTH 79
    15. UNIT METER KN
    16. JOINT COORDINATES
    17. 1 0 -3 0; 2 0 -3 4; 3 0 1.49012E-008 0; 4 0 1.49012E-008 4; 5 0 0.35 0
    18. 6 0 0.35 4
    19. MEMBER INCIDENCES
    20. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
    21. DEFINE MATERIAL START
    22. ISOTROPIC CONCRETE
    23. E 2.17185E+007
    24. POISSON 0.17
    25. DENSITY 23.5616
    26. ALPHA 1E-005
    27. DAMP 0.05
    28. TYPE CONCRETE
    29. STRENGTH FCU 27579
    30. END DEFINE MATERIAL
    31. MEMBER PROPERTY AMERICAN
    32. 1 TO 4 PRIS YD 0.6 ZD 0.6
    33. 5 PRIS YD 0.5 ZD 0.3
    34. CONSTANTS
    35. MATERIAL CONCRETE ALL
    36. SUPPORTS
    37. 1 2 FIXED
    38. ****************************************
```

```
39. LOAD 1 DEAD LOAD
40. SELFWEIGHT Y -1
41. JOINT LOAD
42. 5 FX 0 FY -94.1 FZ -22.6
43. 6 FX 0 FY -167.8 FZ 33.8
44. LOAD 2 LIVE LOAD
45. JOINT LOAD
46. 5 FX 0 FY -71.9 FZ -15.5
47. 6 FX 0 FY -72.1 FZ 15.6
48. LOAD 3 MATERIAL LOAD
49. JOINT LOAD
50. 5 FX 0 FY -50.8 FZ -11
51. 6 FX 0 FY -50.2 FZ 10.9
52. LOAD 4 WIND LOAD +Z
53. JOINT LOAD
54. 5 FX 0 FY 145.7 FZ 40.5
55. 6 FX 0 FY -145.7 FZ 40.6
56. LOAD 5 WIND LOAD -Z
57. JOINT LOAD
58. 5 FX -0.04 FY -145.63 FZ -40.53
59. 6 FX 0.04 FY 145.63 FZ -40.62
60. LOAD 6 WIND LOAD (+X)
61. JOINT LOAD
62. 5 FX 3.83 FY 0 FZ 0
63. 6 FX 3.83 FY 0 FZ 0
64. LOAD 7 WIND LOAD (-X)
65. JOINT LOAD
66. 5 FX -3.83 FY 0 FZ 0
67. 6 FX -3.83 FY 0 FZ 0
68. LOAD 8 TEMP LOAD 1
69. JOINT LOAD
70. 5 FX 0 FY -0.16 FZ -22.93
71. 6 FX 0 FY 0.16 FZ 22.83
72. LOAD 9 TEMP LOAD 2
73. JOINT LOAD
74. 5 FX 0 FY 0.16 FZ 22.93
75. 6 FX 0 FY -0.16 FZ -22.83
76. LOAD 10 EARTHQUAKE LOAD +X
77. JOINT LOAD
78. 5 FX -0.01 FY -0.05 FZ 0
79. 6 FX -0.01 FY 0.05 FZ 0
80. LOAD 11 EARTHQUAKE LOAD +Z
81. JOINT LOAD
82. 5 FX -0.01 FY 7.65 FZ 2.08
83. 6 FX 0.01 FY -7.65 FZ 2.09
84. ****************************************
85. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
86. LOAD COMB 301 1.0 DL
87. 1 1.0
88. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
89. 1 1.0 8 1.0 2 1.0 3 1.0
90. LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
91. 1 1.0 9 1.0 2 1.0 3 1.0
92. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
93. 1 1.0 4 0.75 2 0.75 3 0.75
94. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
```

```
 95. 1 1.0 5 0.75 2 0.75 3 0.75
 96. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
 97. 1 1.0 6 0.75 2 0.75 3 0.75
 98. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
 99. 1 1.0 7 0.75 2 0.75 3 0.75
100. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
101. 1 1.0 10 0.525 2 0.75 3 0.75
102. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
103. 1 1.0 10 -0.525 2 0.75 3 0.75
104. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
105. 1 1.0 11 0.525 2 0.75 3 0.75
106. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
107. 1 1.0 11 -0.525 2 0.75 3 0.75
108. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
109. 1 0.6 4 1.0
110. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
111. 1 0.6 5 1.0
112. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
113. 1 0.6 6 1.0
114. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
115. 1 0.6 7 1.0
116. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
117. 1 0.6 10 0.7
118. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
119. 1 0.6 10 -0.7
120. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
121. 1 0.6 11 0.7
122. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
123. 1 0.6 11 -0.7
124. ****************************************
125. *****LOAD COMBINATION FOR FOUNDATION DESIGN
126. LOAD COMB 401 1.4 DL
127. 1 1.4
128. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
129. 1 1.4 8 1.4 2 1.7 3 1.7
130. LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
131. 1 1.4 9 1.4 2 1.7 3 1.7
132. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
133. 1 1.2 4 1.6 2 0.5 3 0.5
134. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
135. 1 1.2 5 1.6 2 0.5 3 0.5
136. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
137. 1 1.2 6 1.6 2 0.5 3 0.5
138. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
139. 1 1.2 7 1.6 2 0.5 3 0.5
140. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
141. 1 1.2 10 1.0 2 0.5 3 0.5
142. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
143. 1 1.2 10 -1.0 2 0.5 3 0.5
144. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
145. 1 1.2 11 1.0 2 0.5 3 0.5
146. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
147. 1 1.2 11 -1.0 2 0.5 3 0.5
148. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
149. 1 0.9 4 1.6
150. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
```

```
151. 1 0.9 5 1.6
152. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
153. 1 0.9 6 1.6
154. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
155. 1 0.9 7 1.6
156. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
157. 1 0.9 10 1.0
158. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
159. 1 0.9 10 -1.0
160. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
161. 1 0.9 11 1.0
162. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
163. 1 0.9 11 -1.0
164. ****************************************
165. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**6**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 38 SO FAR.

```
166. LOAD LIST 301 TO 319
167. PRINT SUPPORT REACTION ALL
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	0.00	126.87	-7.25	-13.94	0.00	0.00
	302	0.00	249.76	-13.61	-18.10	0.00	0.00
	303	0.00	249.39	-7.77	-14.37	0.00	0.00
	304	0.00	94.52	-40.25	-87.34	0.00	0.00
	305	0.02	343.23	20.60	56.06	-0.01	-0.08
	306	-2.87	218.90	-9.83	-15.66	0.00	9.62
	307	2.87	218.90	-9.83	-15.66	0.00	-9.62
	308	0.01	218.92	-9.83	-15.66	0.00	-0.02
	309	-0.01	218.87	-9.83	-15.66	0.00	0.02
	310	0.00	214.34	-10.93	-18.24	-0.00	-0.01
	311	-0.00	223.46	-8.74	-13.08	0.00	0.01
	312	0.00	-89.71	-44.91	-103.94	0.00	0.00
	313	0.03	241.90	36.23	87.27	-0.02	-0.10
	314	-3.83	76.12	-4.35	-8.36	0.00	12.83
	315	3.83	76.12	-4.35	-8.36	0.00	-12.83
	316	0.01	76.16	-4.35	-8.36	0.00	-0.02
	317	-0.01	76.09	-4.35	-8.36	0.00	0.02
	318	0.01	70.04	-5.81	-11.81	-0.00	-0.02
	319	-0.01	82.20	-2.89	-4.92	0.00	0.02
2	301	0.00	206.00	-3.95	-12.73	0.00	0.00
	302	0.00	328.11	2.51	-8.34	0.00	0.00
	303	0.00	328.48	-3.53	-12.54	0.00	0.00
	304	0.00	422.10	-31.78	-82.69	0.00	0.00
	305	-0.02	173.39	29.06	60.71	-0.01	0.08
	306	-2.87	297.72	-1.37	-11.01	0.00	9.62
	307	2.87	297.72	-1.37	-11.01	0.00	-9.62
	308	0.01	297.69	-1.37	-11.01	0.00	-0.02
	309	-0.01	297.75	-1.37	-11.01	0.00	0.02
	310	-0.00	302.28	-2.46	-13.59	-0.00	0.01
	311	0.00	293.16	-0.27	-8.43	0.00	-0.01
	312	0.00	289.43	-42.91	-103.21	0.00	0.00
	313	-0.03	-42.18	38.20	87.98	-0.02	0.10
	314	-3.83	123.60	-2.37	-7.64	0.00	12.83
	315	3.83	123.60	-2.37	-7.64	0.00	-12.83
	316	0.01	123.56	-2.37	-7.64	0.00	-0.02
	317	-0.01	123.63	-2.37	-7.64	0.00	0.02
	318	-0.01	129.68	-3.83	-11.08	-0.00	0.02
	319	0.01	117.52	-0.91	-4.20	0.00	-0.02

********** END OF LATEST ANALYSIS RESULT **********

```
168. ****************************************
169. LOAD LIST 401 TO 419
170. START CONCRETE DESIGN
171. CODE ACI 2008
172. FC 35000 ALL
173. FYMAIN 420000 ALL
174. TRACK 1 ALL
175. CLB 0.05 ALL
176. CLT 0.05 ALL
177. CLS 0.05 ALL
178. MAXMAIN 25 ALL
179. MINMAIN 16 ALL
180. MINSEC 8 ALL
181. REINF 0 MEMB 1 TO 4
182. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	218.46	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	218.46	16.82	0.02566

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    202.68      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    202.68      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION:  MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    388.87      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    388.87      1.90    0.00163
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	361.42	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	361.42	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

183. DESIGN BEAM 5

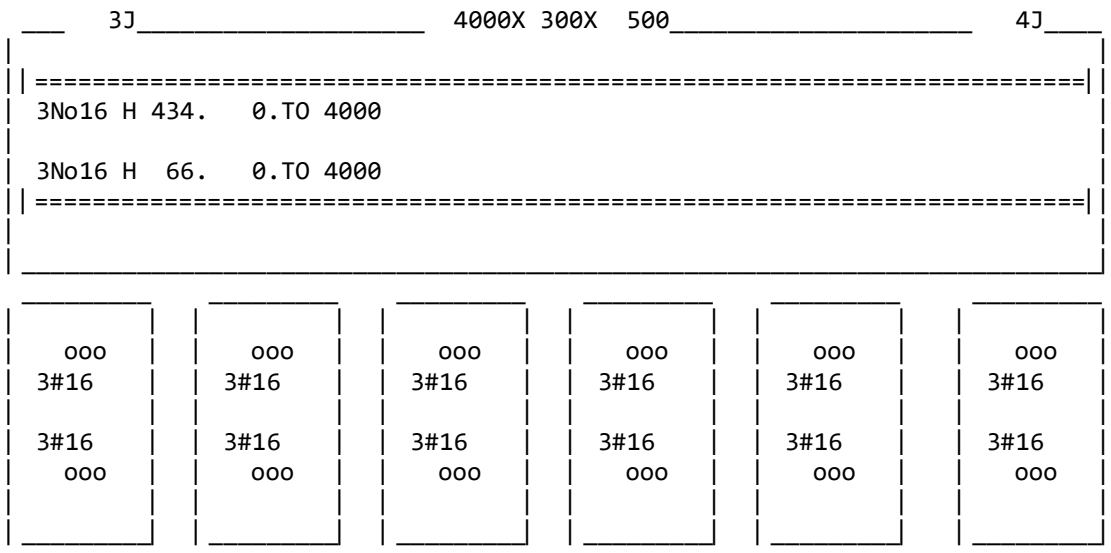
BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL POS MOMENT= 64.34 KN-MET AT 0.MM, LOAD 412					
	REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
2	434.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL NEG MOMENT= 78.20 KN-MET AT 4000.MM, LOAD 404					
	REQD STEEL= 490.MM2, RHO=0.0038, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 3.95 KNS Vc= 117.58 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.54 KNS Vc= 117.58 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

184. END CONCRETE DESIGN
185. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 30,2020 TIME= 17:24:56 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	30-03-2020	29-05-2020	01-11-2023				Date	30-03-2020	29-05-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK/APL	RPK/APL	RPK/APL				Prepared By	RPK/APL	RPK/APL	RPK/APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 6 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-3**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 28-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART GROUND PORTAL FOUNDATIONS
JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
ENGINEER NAME RPK
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 6; 3 0 1.49012e-008 0; 4 0 1.49012e-008 6; 5 0 0.35 0;
6 0 0.35 6;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.8 ZD 0.5
5 PRIS YD 0.6 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 2 FIXED
*****************************************
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
JOINT LOAD
5 FX -0.1 FY -121.3 FZ -20.3
6 FX -0.1 FY -249.8 FZ 35.1
LOAD 2 LIVE LOAD
JOINT LOAD
5 FX 0 FY -89.9 FZ -13.6
6 FX 0 FY -90 FZ 13.6
LOAD 3 MATERIAL LOAD
JOINT LOAD
5 FX 0 FY -61.8 FZ -9.3
6 FX 0 FY -61.7 FZ 9.3
LOAD 4 WIND LOAD +Z
JOINT LOAD
5 FX 0 FY 370.1 FZ 65.1
```

```
6 FX 0 FY -370.1 FZ 65.1
LOAD 5 WIND LOAD -Z
JOINT LOAD
5 FX -0.01 FY -354.02 FZ -62.69
6 FX 0.01 FY 354.02 FZ -62.69
LOAD 6 WIND LOAD (+X)
JOINT LOAD
5 FX 5.63 FY 0 FZ 0
6 FX 5.63 FY 0 FZ 0
LOAD 7 WIND LOAD (-X)
JOINT LOAD
5 FX -5.63 FY 0 FZ 0
6 FX -5.63 FY 0 FZ 0
LOAD 8 TEMP LOAD 1
JOINT LOAD
5 FX 0.04 FY 1.94 FZ -52.27
6 FX -0.04 FY -1.94 FZ 52.17
LOAD 9 TEMP LOAD 2
JOINT LOAD
5 FX -0.04 FY -1.94 FZ 52.27
6 FX 0.04 FY 1.94 FZ -52.17
LOAD 10 EARTHQUAKE LOAD +X
JOINT LOAD
5 FX 0.03 FY 0.03 FZ 0.01
6 FX 0.03 FY 0.07 FZ -0.01
LOAD 11 EARTHQUAKE LOAD +Z
JOINT LOAD
5 FX 0.01 FY 18.61 FZ 2.74
6 FX -0.01 FY -18.61 FZ 2.74
*****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
```

```
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL( -Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL( +X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL( -X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL( +X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL( -X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL( +Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL( -Z)
1 0.6 11 -0.7
****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 ( DL + TL1) + 1.7  ( LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 ( DL + TL2) + 1.7  ( LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL( +Z) + 0.5  ( LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL( -Z) + 0.5  ( LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL( +X) + 0.5  ( LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL( -X) + 0.5  ( LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL( +X) + 0.5  ( LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL( -X) + 0.5  ( LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL( +Z) + 0.5  ( LL+ML)
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL( -Z) + 0.5  ( LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL( +Z)
1 0.9 4 1.6
LOAD COMB 413 0.9 DL + 1.6 WL( -Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL( +X)
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL( -X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL( +X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL( -X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL( +Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL( -Z)
```

```
1 0.9 11 -1.0
****************************************
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 1 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 30, 2020                *
*           Time=    17:26:32                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Conveyor CNV-3130-003\Civil\Documents\Staad\3130-CNV-003-TR3-Fdn.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 28-MAR-20
     4. JOB NAME MMGP-SAUDI ARABIA
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045/900707
     7. JOB REV 0
     8. JOB PART GROUND PORTAL FOUNDATIONS
     9. JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
    10. ENGINEER NAME RPK
    11. CHECKER NAME SBY
    12. APPROVED NAME PRI
    13. END JOB INFORMATION
    14. INPUT WIDTH 79
    15. UNIT METER KN
    16. JOINT COORDINATES
    17. 1 0 -3 0; 2 0 -3 6; 3 0 1.49012E-008 0; 4 0 1.49012E-008 6; 5 0 0.35 0
    18. 6 0 0.35 6
    19. MEMBER INCIDENCES
    20. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
    21. DEFINE MATERIAL START
    22. ISOTROPIC CONCRETE
    23. E 2.17185E+007
    24. POISSON 0.17
    25. DENSITY 23.5616
    26. ALPHA 1E-005
    27. DAMP 0.05
    28. TYPE CONCRETE
    29. STRENGTH FCU 27579
    30. END DEFINE MATERIAL
    31. MEMBER PROPERTY AMERICAN
    32. 1 TO 4 PRIS YD 0.8 ZD 0.5
    33. 5 PRIS YD 0.6 ZD 0.3
    34. CONSTANTS
    35. MATERIAL CONCRETE ALL
    36. SUPPORTS
    37. 1 2 FIXED
    38. ****************************************
```

```
39. LOAD 1 DEAD LOAD
40. SELFWEIGHT Y -1
41. JOINT LOAD
42. 5 FX -0.1 FY -121.3 FZ -20.3
43. 6 FX -0.1 FY -249.8 FZ 35.1
44. LOAD 2 LIVE LOAD
45. JOINT LOAD
46. 5 FX 0 FY -89.9 FZ -13.6
47. 6 FX 0 FY -90 FZ 13.6
48. LOAD 3 MATERIAL LOAD
49. JOINT LOAD
50. 5 FX 0 FY -61.8 FZ -9.3
51. 6 FX 0 FY -61.7 FZ 9.3
52. LOAD 4 WIND LOAD +Z
53. JOINT LOAD
54. 5 FX 0 FY 370.1 FZ 65.1
55. 6 FX 0 FY -370.1 FZ 65.1
56. LOAD 5 WIND LOAD -Z
57. JOINT LOAD
58. 5 FX -0.01 FY -354.02 FZ -62.69
59. 6 FX 0.01 FY 354.02 FZ -62.69
60. LOAD 6 WIND LOAD (+X)
61. JOINT LOAD
62. 5 FX 5.63 FY 0 FZ 0
63. 6 FX 5.63 FY 0 FZ 0
64. LOAD 7 WIND LOAD (-X)
65. JOINT LOAD
66. 5 FX -5.63 FY 0 FZ 0
67. 6 FX -5.63 FY 0 FZ 0
68. LOAD 8 TEMP LOAD 1
69. JOINT LOAD
70. 5 FX 0.04 FY 1.94 FZ -52.27
71. 6 FX -0.04 FY -1.94 FZ 52.17
72. LOAD 9 TEMP LOAD 2
73. JOINT LOAD
74. 5 FX -0.04 FY -1.94 FZ 52.27
75. 6 FX 0.04 FY 1.94 FZ -52.17
76. LOAD 10 EARTHQUAKE LOAD +X
77. JOINT LOAD
78. 5 FX 0.03 FY 0.03 FZ 0.01
79. 6 FX 0.03 FY 0.07 FZ -0.01
80. LOAD 11 EARTHQUAKE LOAD +Z
81. JOINT LOAD
82. 5 FX 0.01 FY 18.61 FZ 2.74
83. 6 FX -0.01 FY -18.61 FZ 2.74
84. ************************************
85. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
86. LOAD COMB 301 1.0 DL
87. 1 1.0
88. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
89. 1 1.0 8 1.0 2 1.0 3 1.0
90. LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
91. 1 1.0 9 1.0 2 1.0 3 1.0
92. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
93. 1 1.0 4 0.75 2 0.75 3 0.75
94. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
```

```
 95. 1 1.0 5 0.75 2 0.75 3 0.75
 96. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
 97. 1 1.0 6 0.75 2 0.75 3 0.75
 98. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
 99. 1 1.0 7 0.75 2 0.75 3 0.75
100. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
101. 1 1.0 10 0.525 2 0.75 3 0.75
102. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
103. 1 1.0 10 -0.525 2 0.75 3 0.75
104. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
105. 1 1.0 11 0.525 2 0.75 3 0.75
106. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
107. 1 1.0 11 -0.525 2 0.75 3 0.75
108. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
109. 1 0.6 4 1.0
110. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
111. 1 0.6 5 1.0
112. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
113. 1 0.6 6 1.0
114. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
115. 1 0.6 7 1.0
116. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
117. 1 0.6 10 0.7
118. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
119. 1 0.6 10 -0.7
120. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
121. 1 0.6 11 0.7
122. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
123. 1 0.6 11 -0.7
124. ****************************************
125. *****LOAD COMBINATION FOR FOUNDATION DESIGN
126. LOAD COMB 401 1.4 DL
127. 1 1.4
128. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
129. 1 1.4 8 1.4 2 1.7 3 1.7
130. LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
131. 1 1.4 9 1.4 2 1.7 3 1.7
132. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
133. 1 1.2 4 1.6 2 0.5 3 0.5
134. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
135. 1 1.2 5 1.6 2 0.5 3 0.5
136. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
137. 1 1.2 6 1.6 2 0.5 3 0.5
138. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
139. 1 1.2 7 1.6 2 0.5 3 0.5
140. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
141. 1 1.2 10 1.0 2 0.5 3 0.5
142. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
143. 1 1.2 10 -1.0 2 0.5 3 0.5
144. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
145. 1 1.2 11 1.0 2 0.5 3 0.5
146. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
147. 1 1.2 11 -1.0 2 0.5 3 0.5
148. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
149. 1 0.9 4 1.6
150. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
```

```
151. 1 0.9 5 1.6
152. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
153. 1 0.9 6 1.6
154. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
155. 1 0.9 7 1.6
156. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
157. 1 0.9 10 1.0
158. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
159. 1 0.9 10 -1.0
160. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
161. 1 0.9 11 1.0
162. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
163. 1 0.9 11 -1.0
164. ****************************************
165. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**6**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 38 SO FAR.

```
166. LOAD LIST 301 TO 319
167. PRINT SUPPORT REACTION ALL
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	0.10	162.71	-5.58	-13.59	0.00	-0.34
	302	0.06	312.50	-14.75	-19.75	0.01	-0.21
	303	0.14	316.33	-2.03	-11.24	-0.01	-0.46
	304	0.10	-20.34	-56.51	-120.84	0.00	-0.34
	305	0.11	560.54	39.33	86.87	-0.00	-0.36
	306	-4.12	276.49	-7.69	-15.02	0.00	13.81
	307	4.32	276.49	-7.69	-15.02	0.00	-14.48
	308	0.08	276.47	-7.69	-15.02	0.00	-0.28
	309	0.12	276.50	-7.69	-15.02	0.00	-0.39
	310	0.10	266.15	-9.13	-18.15	0.00	-0.32
	311	0.10	286.82	-6.25	-11.90	-0.00	-0.35
	312	0.06	-298.14	-68.45	-149.25	0.00	-0.20
	313	0.07	476.36	59.34	127.71	-0.00	-0.23
	314	-5.57	97.63	-3.35	-8.15	0.00	18.66
	315	5.69	97.63	-3.35	-8.15	0.00	-19.06
	316	0.04	97.61	-3.35	-8.15	0.00	-0.13
	317	0.08	97.65	-3.35	-8.15	0.00	-0.27
	318	0.05	83.85	-5.27	-12.32	0.00	-0.18
	319	0.07	111.41	-1.43	-3.99	-0.00	-0.22
2	301	0.10	296.98	-9.22	-18.69	0.00	-0.34
	302	0.14	450.60	0.05	-12.33	0.01	-0.46
	303	0.06	446.76	-12.87	-21.23	-0.01	-0.21
	304	0.10	707.58	-55.94	-123.08	0.00	-0.34
	305	0.09	126.70	39.90	84.64	-0.00	-0.31
	306	-4.12	410.75	-7.11	-17.26	0.00	13.81
	307	4.32	410.75	-7.11	-17.26	0.00	-14.48
	308	0.08	410.72	-7.11	-17.26	0.00	-0.28
	309	0.12	410.79	-7.11	-17.26	0.00	-0.39
	310	0.10	421.09	-8.55	-20.38	0.00	-0.35
	311	0.10	400.42	-5.67	-14.13	-0.00	-0.32
	312	0.06	573.95	-70.63	-152.31	0.00	-0.20
	313	0.05	-200.55	57.16	124.65	-0.00	-0.17
	314	-5.57	178.19	-5.53	-11.21	0.00	18.66
	315	5.69	178.19	-5.53	-11.21	0.00	-19.06
	316	0.04	178.14	-5.53	-11.21	0.00	-0.13
	317	0.08	178.24	-5.53	-11.21	0.00	-0.27
	318	0.07	191.97	-7.45	-15.38	0.00	-0.22
	319	0.05	164.41	-3.61	-7.05	-0.00	-0.18

********** END OF LATEST ANALYSIS RESULT **********

```
168. ****************************************
169. LOAD LIST 401 TO 419
170. START CONCRETE DESIGN
171. CODE ACI 2008
172. FC 35000 ALL
173. FYMAIN 420000 ALL
174. TRACK 1 ALL
175. CLB 0.05 ALL
176. CLT 0.05 ALL
177. CLS 0.05 ALL
178. MAXMAIN 25 ALL
179. MINMAIN 16 ALL
180. MINSEC 8 ALL
181. REINF 0 MEMB 1 TO 4
182. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 800.0 MMS, TIED
AREA OF STEEL REQUIRED = 4140.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 16 MM	1.206	412	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13783.17	11026.53	5191.57	1627.23	313.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
715.55	-2026.73	-540.87	0.34	-0.00021

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13783.17	11026.53	4894.76	976.39	199.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
429.11	-2026.73	-540.87	264.42	-0.16296

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 800.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
13469.31  10775.44   5189.38   1552.22     299.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  602.05  -1688.94    261.26      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
13469.31  10775.44   4903.45    934.93     190.7
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  363.40  -1688.94    261.26      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 800.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  4000.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 20 - 16 MM               1.005      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 256.00 MM

    COLUMN INTERACTION:  MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
13469.31  10775.44   5189.38   1552.22     299.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  602.05  -1688.94    578.75      0.08    0.00004
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
13469.31  10775.44   4903.45    934.93     190.7
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  363.40  -1688.94    578.75     19.32    0.01113
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 800.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13469.31	10775.44	5189.38	1552.22	299.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
602.05	-1688.94	538.03	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13469.31	10775.44	4903.45	934.93	190.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
363.40	-1688.94	538.03	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

183. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

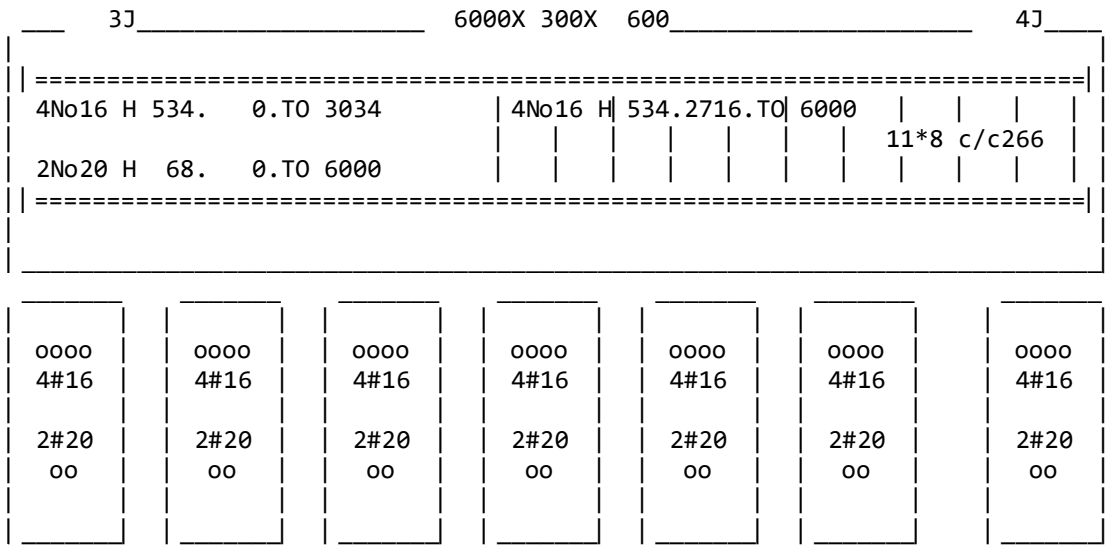
LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	2 - 20MM	0.	6000.	YES	YES
	CRITICAL POS MOMENT= 119.63 KN-MET AT 0.MM, LOAD 412					
	REQD STEEL= 611.MM2, RHO=0.0038, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 45./ 164. MMS					
	REQD. DEVELOPMENT LENGTH = 513. MMS					
	Cracked Moment of Inertia Iz at above location = 93187.1 cm^4					
2	534.	4 - 16MM	0.	3034.	YES	NO
	CRITICAL NEG MOMENT= 124.14 KN-MET AT 0.MM, LOAD 405					
	REQD STEEL= 633.MM2, RHO=0.0039, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 586. MMS					
	Cracked Moment of Inertia Iz at above location = 115622.7 cm^4					
3	534.	4 - 16MM	2716.	6000.	NO	YES
	CRITICAL NEG MOMENT= 149.44 KN-MET AT 6000.MM, LOAD 404					
	REQD STEEL= 766.MM2, RHO=0.0048, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 586. MMS					
	Cracked Moment of Inertia Iz at above location = 115622.7 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.60 KNS Vc= 147.65 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.

AT END SUPPORT - Vu= 57.07 KNS Vc= 145.71 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 6.2 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 266. MM C/C FOR 2466. MM



**********END OF BEAM DESIGN**********

184. END CONCRETE DESIGN
185. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 30,2020 TIME= 17:26:33 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	30-03-2020	29-05-2020	01-11-2023				Date	30-03-2020	29-05-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK/APL	RPK/APL	RPK/APL				Prepared By	RPK/APL	RPK/APL	RPK/APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 7 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-4**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 28-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART GROUND PORTAL FOUNDATIONS
JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
ENGINEER NAME RPK
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 8; 3 0 1.49012e-008 0; 4 0 1.49012e-008 8; 5 0 0.35 0;
6 0 0.35 8;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.8 ZD 0.5
5 PRIS YD 0.7 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 2 FIXED
*****************************************
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
JOINT LOAD
5 FX 0 FY -132.6 FZ -26.6
6 FX 0 FY -244.3 FZ 40.6
LOAD 2 LIVE LOAD
JOINT LOAD
5 FX 0 FY -83.6 FZ -15.1
6 FX 0 FY -84.3 FZ 15.2
LOAD 3 MATERIAL LOAD
JOINT LOAD
5 FX 0 FY -57.7 FZ -10.5
6 FX 0 FY -58.3 FZ 10.6
LOAD 4 WIND LOAD +Z
JOINT LOAD
5 FX 0 FY 376.8 FZ 71.9
```

```
6 FX 0 FY -376.8 FZ 71.9
LOAD 5 WIND LOAD -Z
JOINT LOAD
5 FX 0.02 FY -367.49 FZ -70.42
6 FX -0.02 FY 367.49 FZ -70.42
LOAD 6 WIND LOAD (+X)
JOINT LOAD
5 FX 7.42 FY 0 FZ 0
6 FX 7.42 FY 0 FZ 0
LOAD 7 WIND LOAD (-X)
JOINT LOAD
5 FX -7.42 FY 0 FZ 0
6 FX -7.42 FY 0 FZ 0
LOAD 8 TEMP LOAD 1
JOINT LOAD
5 FX 0 FY 0 FZ -78.91
6 FX 0 FY 0 FZ 78.91
LOAD 9 TEMP LOAD 2
JOINT LOAD
5 FX 0 FY 0 FZ 78.91
6 FX 0 FY 0 FZ -78.91
LOAD 10 EARTHQUAKE LOAD +X
JOINT LOAD
5 FX 0.05 FY -0.06 FZ -0.01
6 FX 0.05 FY -0.04 FZ 0.01
LOAD 11 EARTHQUAKE LOAD +Z
JOINT LOAD
5 FX 0 FY 17.28 FZ 2.65
6 FX 0 FY -17.28 FZ 2.65
*****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
```

```
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL( -Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL( +X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL( -X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL( +X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL( -X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL( +Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL( -Z)
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 ( DL + TL1) + 1.7  ( LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 ( DL + TL2) + 1.7  ( LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL( +Z) + 0.5  ( LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL( -Z) + 0.5  ( LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL( +X) + 0.5  ( LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL( -X) + 0.5  ( LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL( +X) + 0.5  ( LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL( -X) + 0.5  ( LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL( +Z) + 0.5  ( LL+ML)
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL( -Z) + 0.5  ( LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL( +Z)
1 0.9 4 1.6
LOAD COMB 413 0.9 DL + 1.6 WL( -Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL( +X)
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL( -X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL( +X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL( -X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL( +Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL( -Z)
```

```
1 0.9 11 -1.0
****************************************
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 1 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 30, 2020                *
*           Time=    17:27:46                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Conveyor CNV-3130-003\Civil\Documents\Staad\3130-CNV-003-TR4-Fdn.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 28-MAR-20
     4. JOB NAME MMGP-SAUDI ARABIA
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045/900707
     7. JOB REV 0
     8. JOB PART GROUND PORTAL FOUNDATIONS
     9. JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
    10. ENGINEER NAME RPK
    11. CHECKER NAME SBY
    12. APPROVED NAME PRI
    13. END JOB INFORMATION
    14. INPUT WIDTH 79
    15. UNIT METER KN
    16. JOINT COORDINATES
    17. 1 0 -3 0; 2 0 -3 8; 3 0 1.49012E-008 0; 4 0 1.49012E-008 8; 5 0 0.35 0
    18. 6 0 0.35 8
    19. MEMBER INCIDENCES
    20. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
    21. DEFINE MATERIAL START
    22. ISOTROPIC CONCRETE
    23. E 2.17185E+007
    24. POISSON 0.17
    25. DENSITY 23.5616
    26. ALPHA 1E-005
    27. DAMP 0.05
    28. TYPE CONCRETE
    29. STRENGTH FCU 27579
    30. END DEFINE MATERIAL
    31. MEMBER PROPERTY AMERICAN
    32. 1 TO 4 PRIS YD 0.8 ZD 0.5
    33. 5 PRIS YD 0.7 ZD 0.3
    34. CONSTANTS
    35. MATERIAL CONCRETE ALL
    36. SUPPORTS
    37. 1 2 FIXED
    38. ****************************************
```

```
39. LOAD 1 DEAD LOAD
40. SELFWEIGHT Y -1
41. JOINT LOAD
42. 5 FX 0 FY -132.6 FZ -26.6
43. 6 FX 0 FY -244.3 FZ 40.6
44. LOAD 2 LIVE LOAD
45. JOINT LOAD
46. 5 FX 0 FY -83.6 FZ -15.1
47. 6 FX 0 FY -84.3 FZ 15.2
48. LOAD 3 MATERIAL LOAD
49. JOINT LOAD
50. 5 FX 0 FY -57.7 FZ -10.5
51. 6 FX 0 FY -58.3 FZ 10.6
52. LOAD 4 WIND LOAD +Z
53. JOINT LOAD
54. 5 FX 0 FY 376.8 FZ 71.9
55. 6 FX 0 FY -376.8 FZ 71.9
56. LOAD 5 WIND LOAD -Z
57. JOINT LOAD
58. 5 FX 0.02 FY -367.49 FZ -70.42
59. 6 FX -0.02 FY 367.49 FZ -70.42
60. LOAD 6 WIND LOAD (+X)
61. JOINT LOAD
62. 5 FX 7.42 FY 0 FZ 0
63. 6 FX 7.42 FY 0 FZ 0
64. LOAD 7 WIND LOAD (-X)
65. JOINT LOAD
66. 5 FX -7.42 FY 0 FZ 0
67. 6 FX -7.42 FY 0 FZ 0
68. LOAD 8 TEMP LOAD 1
69. JOINT LOAD
70. 5 FX 0 FY 0 FZ -78.91
71. 6 FX 0 FY 0 FZ 78.91
72. LOAD 9 TEMP LOAD 2
73. JOINT LOAD
74. 5 FX 0 FY 0 FZ 78.91
75. 6 FX 0 FY 0 FZ -78.91
76. LOAD 10 EARTHQUAKE LOAD +X
77. JOINT LOAD
78. 5 FX 0.05 FY -0.06 FZ -0.01
79. 6 FX 0.05 FY -0.04 FZ 0.01
80. LOAD 11 EARTHQUAKE LOAD +Z
81. JOINT LOAD
82. 5 FX 0 FY 17.28 FZ 2.65
83. 6 FX 0 FY -17.28 FZ 2.65
84. ****************************************
85. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
86. LOAD COMB 301 1.0 DL
87. 1 1.0
88. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
89. 1 1.0 8 1.0 2 1.0 3 1.0
90. LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
91. 1 1.0 9 1.0 2 1.0 3 1.0
92. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
93. 1 1.0 4 0.75 2 0.75 3 0.75
94. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
```

```
 95. 1 1.0 5 0.75 2 0.75 3 0.75
 96. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
 97. 1 1.0 6 0.75 2 0.75 3 0.75
 98. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
 99. 1 1.0 7 0.75 2 0.75 3 0.75
100. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
101. 1 1.0 10 0.525 2 0.75 3 0.75
102. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
103. 1 1.0 10 -0.525 2 0.75 3 0.75
104. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
105. 1 1.0 11 0.525 2 0.75 3 0.75
106. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
107. 1 1.0 11 -0.525 2 0.75 3 0.75
108. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
109. 1 0.6 4 1.0
110. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
111. 1 0.6 5 1.0
112. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
113. 1 0.6 6 1.0
114. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
115. 1 0.6 7 1.0
116. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
117. 1 0.6 10 0.7
118. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
119. 1 0.6 10 -0.7
120. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
121. 1 0.6 11 0.7
122. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
123. 1 0.6 11 -0.7
124. ****************************************
125. *****LOAD COMBINATION FOR FOUNDATION DESIGN
126. LOAD COMB 401 1.4 DL
127. 1 1.4
128. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
129. 1 1.4 8 1.4 2 1.7 3 1.7
130. LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
131. 1 1.4 9 1.4 2 1.7 3 1.7
132. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
133. 1 1.2 4 1.6 2 0.5 3 0.5
134. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
135. 1 1.2 5 1.6 2 0.5 3 0.5
136. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
137. 1 1.2 6 1.6 2 0.5 3 0.5
138. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
139. 1 1.2 7 1.6 2 0.5 3 0.5
140. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
141. 1 1.2 10 1.0 2 0.5 3 0.5
142. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
143. 1 1.2 10 -1.0 2 0.5 3 0.5
144. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
145. 1 1.2 11 1.0 2 0.5 3 0.5
146. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
147. 1 1.2 11 -1.0 2 0.5 3 0.5
148. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
149. 1 0.9 4 1.6
150. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
```

```
151. 1 0.9 5 1.6
152. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
153. 1 0.9 6 1.6
154. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
155. 1 0.9 7 1.6
156. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
157. 1 0.9 10 1.0
158. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
159. 1 0.9 10 -1.0
160. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
161. 1 0.9 11 1.0
162. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
163. 1 0.9 11 -1.0
164. ****************************************
165. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**6**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 38 SO FAR.

```
166. LOAD LIST 301 TO 319
167. PRINT SUPPORT REACTION ALL
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	0.00	181.77	-0.35	-7.33	0.00	0.00
	302	0.00	323.04	-12.40	-14.92	0.00	0.00
	303	0.00	323.04	5.63	-3.78	0.00	0.00
	304	0.00	-11.92	-56.55	-121.35	0.00	0.00
	305	-0.01	580.03	50.19	101.34	0.00	0.05
	306	-5.57	287.72	-2.63	-8.85	0.00	18.64
	307	5.57	287.72	-2.63	-8.85	0.00	-18.64
	308	-0.03	287.75	-2.63	-8.85	0.00	0.09
	309	0.03	287.69	-2.63	-8.85	0.00	-0.09
	310	0.00	278.21	-4.02	-11.75	0.00	0.00
	311	0.00	297.23	-1.24	-5.94	0.00	0.00
	312	0.00	-290.45	-72.11	-154.40	0.00	0.00
	313	-0.02	498.80	70.21	142.51	0.00	0.06
	314	-7.42	109.06	-0.21	-4.40	0.00	24.86
	315	7.42	109.06	-0.21	-4.40	0.00	-24.86
	316	-0.04	109.10	-0.21	-4.40	0.00	0.12
	317	0.04	109.02	-0.21	-4.40	0.00	-0.12
	318	0.00	96.38	-2.07	-8.28	0.00	0.00
	319	0.00	121.74	1.64	-0.52	0.00	0.00
2	301	0.00	297.86	-13.65	-22.02	0.00	0.00
	302	0.00	440.49	-1.80	-14.84	0.00	0.00
	303	0.00	440.49	-19.83	-25.98	0.00	0.00
	304	0.00	704.47	-65.45	-133.32	0.00	0.00
	305	0.01	112.53	41.29	89.37	0.00	-0.05
	306	-5.57	404.83	-11.52	-20.81	0.00	18.64
	307	5.57	404.83	-11.52	-20.81	0.00	-18.64
	308	-0.03	404.85	-11.52	-20.81	0.00	0.09
	309	0.03	404.81	-11.52	-20.81	0.00	-0.09
	310	0.00	414.34	-12.91	-23.72	0.00	0.00
	311	0.00	395.32	-10.13	-17.91	0.00	0.00
	312	0.00	578.23	-80.09	-163.21	0.00	0.00
	313	0.02	-211.02	62.23	133.70	0.00	-0.06
	314	-7.42	178.72	-8.19	-13.21	0.00	24.86
	315	7.42	178.72	-8.19	-13.21	0.00	-24.86
	316	-0.04	178.74	-8.19	-13.21	0.00	0.12
	317	0.04	178.69	-8.19	-13.21	0.00	-0.12
	318	0.00	191.40	-10.04	-17.09	0.00	0.00
	319	0.00	166.04	-6.33	-9.33	0.00	0.00

********** END OF LATEST ANALYSIS RESULT **********

```
168. ****************************************
169. LOAD LIST 401 TO 419
170. START CONCRETE DESIGN
171. CODE ACI 2008
172. FC 35000 ALL
173. FYMAIN 420000 ALL
174. TRACK 1 ALL
175. CLB 0.05 ALL
176. CLT 0.05 ALL
177. CLS 0.05 ALL
178. MAXMAIN 25 ALL
179. MINMAIN 16 ALL
180. MINSEC 8 ALL
181. REINF 0 MEMB 1 TO 4
182. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 800.0 MMS, TIED
AREA OF STEEL REQUIRED = 4224.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 16 MM	1.206	412	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13783.17	11026.53	5191.57	1627.23	313.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
715.55	-2026.73	-528.48	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13783.17	11026.53	4894.76	976.39	199.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
429.11	-2026.73	-528.48	274.00	-0.17282

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 800.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
13469.31  10775.44   5189.38   1552.22     299.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  602.05  -1688.94    285.60      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
13469.31  10775.44   4903.45    934.93     190.7
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  363.40  -1688.94    285.60      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 800.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  4000.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
---------------------------------------------------------

 20 - 16 MM               1.005      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 256.00 MM

    COLUMN INTERACTION:  MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
13469.31  10775.44   5189.38   1552.22     299.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  602.05  -1688.94    580.64      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
13469.31  10775.44   4903.45    934.93     190.7
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  363.40  -1688.94    580.64     40.77    0.02340
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 800.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13469.31	10775.44	5189.38	1552.22	299.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
602.05	-1688.94	526.18	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13469.31	10775.44	4903.45	934.93	190.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
363.40	-1688.94	526.18	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

183. DESIGN BEAM 5

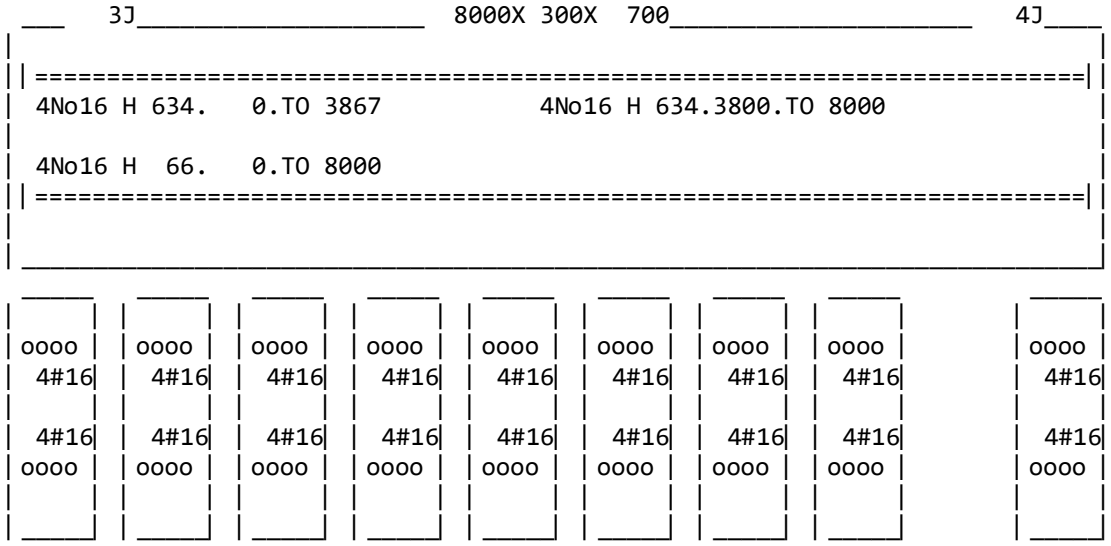
BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 8000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 700. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	4 - 16MM	0.	8000.	YES	YES
	CRITICAL POS MOMENT= 131.35 KN-MET AT 0.MM, LOAD 412					
	REQD STEEL= 667.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 586. MMS					
	Cracked Moment of Inertia Iz at above location = 167600.2 cm^4					
2	634.	4 - 16MM	0.	3867.	YES	NO
	CRITICAL NEG MOMENT= 162.03 KN-MET AT 0.MM, LOAD 405					
	REQD STEEL= 694.MM2, RHO=0.0036, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 586. MMS					
	Cracked Moment of Inertia Iz at above location = 167600.2 cm^4					
3	634.	4 - 16MM	3800.	8000.	NO	YES
	CRITICAL NEG MOMENT= 186.21 KN-MET AT 8000.MM, LOAD 404					
	REQD STEEL= 803.MM2, RHO=0.0042, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 586. MMS					
	Cracked Moment of Inertia Iz at above location = 166462.7 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 20.24 KNS Vc= 177.13 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 7.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 26.39 KNS Vc= 177.13 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 7.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

184. END CONCRETE DESIGN
185. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 30,2020 TIME= 17:27:47 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	30-03-2020	29-05-2020	01-11-2023				Date	30-03-2020	29-05-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK/APL	RPK/APL	RPK/APL				Prepared By	RPK/APL	RPK/APL	RPK/APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 8 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-5**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 28-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART GROUND PORTAL FOUNDATIONS
JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
ENGINEER NAME RPK
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION

INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 8; 3 0 1.49012e-008 0; 4 0 1.49012e-008 8; 5 0 0.35 0;
6 0 0.35 8;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.8 ZD 0.5
5 PRIS YD 0.7 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 2 FIXED
*****************************************
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
JOINT LOAD
5 FX 0 FY -169.9 FZ -28.6
6 FX 0 FY -144.3 FZ 23.4
LOAD 2 LIVE LOAD
JOINT LOAD
5 FX 0 FY -70.1 FZ -11.9
6 FX 0 FY -70.5 FZ 12
LOAD 3 MATERIAL LOAD
JOINT LOAD
5 FX 0 FY -46.8 FZ -7.9
6 FX 0 FY -46.6 FZ 7.9
LOAD 4 WIND LOAD +Z
JOINT LOAD
```

```
5 FX 0 FY 408.6 FZ 68.6
6 FX 0 FY -404.2 FZ 67.9
LOAD 5 WIND LOAD -Z
JOINT LOAD
5 FX 0 FY -383.91 FZ -65.35
6 FX 0 FY 383.91 FZ -65.35
LOAD 6 WIND LOAD (+X)
JOINT LOAD
5 FX 8.8 FY 0 FZ 0
6 FX 8.8 FY 0 FZ 0
LOAD 7 WIND LOAD (-X)
JOINT LOAD
5 FX -8.8 FY 0 FZ 0
6 FX -8.8 FY 0 FZ 0
LOAD 8 TEMP LOAD 1
JOINT LOAD
5 FX 0 FY 0 FZ -93.87
6 FX 0 FY 0 FZ 93.87
LOAD 9 TEMP LOAD 2
JOINT LOAD
5 FX 0 FY 0 FZ 93.87
6 FX 0 FY 0 FZ -93.87
LOAD 10 EARTHQUAKE LOAD +X
JOINT LOAD
5 FX 0.06 FY -0.5 FZ -0.08
6 FX 0.06 FY -0.5 FZ 0.08
LOAD 11 EARTHQUAKE LOAD +Z
JOINT LOAD
5 FX 0 FY 18.36 FZ 2.33
6 FX 0 FY -18.36 FZ 2.33
*****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
```

```
LOAD COMB 312 0.6 DL + 1.0 WL( +Z)
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL( -Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL( +X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL( -X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL( +X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL( -X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL( +Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL( -Z)
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 ( DL + TL1) + 1.7  ( LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 ( DL + TL2) + 1.7  ( LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL( +Z) + 0.5  ( LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL( -Z) + 0.5  ( LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL( +X) + 0.5  ( LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL( -X) + 0.5  ( LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL( +X) + 0.5  ( LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL( -X) + 0.5  ( LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL( +Z) + 0.5  ( LL+ML)
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL( -Z) + 0.5  ( LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL( +Z)
1 0.9 4 1.6
LOAD COMB 413 0.9 DL + 1.6 WL( -Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL( +X)
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL( -X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL( +X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL( -X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL( +Z)
1 0.9 11 1.0
```

```
LOAD COMB 419 0.9 DL + 1.0 EL( -Z)
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 1 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 30, 2020                *
*           Time=    17:30: 0                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Conveyor CNV-3130-003\Civil\Documents\Staad\3130-CNV-003-TR5-Fdn.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 28-MAR-20
     4. JOB NAME MMGP-SAUDI ARABIA
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045/900707
     7. JOB REV 0
     8. JOB PART GROUND PORTAL FOUNDATIONS
     9. JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
    10. ENGINEER NAME RPK
    11. CHECKER NAME SBY
    12. APPROVED NAME PRI
    13. END JOB INFORMATION
    15. INPUT WIDTH 79
    16. UNIT METER KN
    17. JOINT COORDINATES
    18. 1 0 -3 0; 2 0 -3 8; 3 0 1.49012E-008 0; 4 0 1.49012E-008 8; 5 0 0.35 0
    19. 6 0 0.35 8
    20. MEMBER INCIDENCES
    21. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
    22. DEFINE MATERIAL START
    23. ISOTROPIC CONCRETE
    24. E 2.17185E+007
    25. POISSON 0.17
    26. DENSITY 23.5616
    27. ALPHA 1E-005
    28. DAMP 0.05
    29. TYPE CONCRETE
    30. STRENGTH FCU 27579
    31. END DEFINE MATERIAL
    32. MEMBER PROPERTY AMERICAN
    33. 1 TO 4 PRIS YD 0.8 ZD 0.5
    34. 5 PRIS YD 0.7 ZD 0.3
    35. CONSTANTS
    36. MATERIAL CONCRETE ALL
    37. SUPPORTS
    38. 1 2 FIXED
    39. ****************************************
```

```
40. LOAD 1 DEAD LOAD
41. SELFWEIGHT Y -1
42. JOINT LOAD
43. 5 FX 0 FY -169.9 FZ -28.6
44. 6 FX 0 FY -144.3 FZ 23.4
45. LOAD 2 LIVE LOAD
46. JOINT LOAD
47. 5 FX 0 FY -70.1 FZ -11.9
48. 6 FX 0 FY -70.5 FZ 12
49. LOAD 3 MATERIAL LOAD
50. JOINT LOAD
51. 5 FX 0 FY -46.8 FZ -7.9
52. 6 FX 0 FY -46.6 FZ 7.9
53. LOAD 4 WIND LOAD +Z
54. JOINT LOAD
55. 5 FX 0 FY 408.6 FZ 68.6
56. 6 FX 0 FY -404.2 FZ 67.9
57. LOAD 5 WIND LOAD -Z
58. JOINT LOAD
59. 5 FX 0 FY -383.91 FZ -65.35
60. 6 FX 0 FY 383.91 FZ -65.35
61. LOAD 6 WIND LOAD (+X)
62. JOINT LOAD
63. 5 FX 8.8 FY 0 FZ 0
64. 6 FX 8.8 FY 0 FZ 0
65. LOAD 7 WIND LOAD (-X)
66. JOINT LOAD
67. 5 FX -8.8 FY 0 FZ 0
68. 6 FX -8.8 FY 0 FZ 0
69. LOAD 8 TEMP LOAD 1
70. JOINT LOAD
71. 5 FX 0 FY 0 FZ -93.87
72. 6 FX 0 FY 0 FZ 93.87
73. LOAD 9 TEMP LOAD 2
74. JOINT LOAD
75. 5 FX 0 FY 0 FZ 93.87
76. 6 FX 0 FY 0 FZ -93.87
77. LOAD 10 EARTHQUAKE LOAD +X
78. JOINT LOAD
79. 5 FX 0.06 FY -0.5 FZ -0.08
80. 6 FX 0.06 FY -0.5 FZ 0.08
81. LOAD 11 EARTHQUAKE LOAD +Z
82. JOINT LOAD
83. 5 FX 0 FY 18.36 FZ 2.33
84. 6 FX 0 FY -18.36 FZ 2.33
85. ************************************
86. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
87. LOAD COMB 301 1.0 DL
88. 1 1.0
89. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
90. 1 1.0 8 1.0 2 1.0 3 1.0
91. LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
92. 1 1.0 9 1.0 2 1.0 3 1.0
93. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
94. 1 1.0 4 0.75 2 0.75 3 0.75
95. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
```

```
 96. 1 1.0 5 0.75 2 0.75 3 0.75
 97. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
 98. 1 1.0 6 0.75 2 0.75 3 0.75
 99. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
100. 1 1.0 7 0.75 2 0.75 3 0.75
101. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
102. 1 1.0 10 0.525 2 0.75 3 0.75
103. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
104. 1 1.0 10 -0.525 2 0.75 3 0.75
105. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
106. 1 1.0 11 0.525 2 0.75 3 0.75
107. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
108. 1 1.0 11 -0.525 2 0.75 3 0.75
109. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
110. 1 0.6 4 1.0
111. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
112. 1 0.6 5 1.0
113. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
114. 1 0.6 6 1.0
115. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
116. 1 0.6 7 1.0
117. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
118. 1 0.6 10 0.7
119. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
120. 1 0.6 10 -0.7
121. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
122. 1 0.6 11 0.7
123. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
124. 1 0.6 11 -0.7
125. ****************************************
126. *****LOAD COMBINATION FOR FOUNDATION DESIGN
127. LOAD COMB 401 1.4 DL
128. 1 1.4
129. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
130. 1 1.4 8 1.4 2 1.7 3 1.7
131. LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
132. 1 1.4 9 1.4 2 1.7 3 1.7
133. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
134. 1 1.2 4 1.6 2 0.5 3 0.5
135. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
136. 1 1.2 5 1.6 2 0.5 3 0.5
137. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
138. 1 1.2 6 1.6 2 0.5 3 0.5
139. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
140. 1 1.2 7 1.6 2 0.5 3 0.5
141. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
142. 1 1.2 10 1.0 2 0.5 3 0.5
143. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
144. 1 1.2 10 -1.0 2 0.5 3 0.5
145. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
146. 1 1.2 11 1.0 2 0.5 3 0.5
147. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
148. 1 1.2 11 -1.0 2 0.5 3 0.5
149. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
150. 1 0.9 4 1.6
151. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
```

```
152. 1 0.9 5 1.6
153. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
154. 1 0.9 6 1.6
155. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
156. 1 0.9 7 1.6
157. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
158. 1 0.9 10 1.0
159. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
160. 1 0.9 10 -1.0
161. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
162. 1 0.9 11 1.0
163. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
164. 1 0.9 11 -1.0
165. ****************************************
166. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**6**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 38 SO FAR.

```
167. LOAD LIST 301 TO 319
168. PRINT SUPPORT REACTION ALL
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	0.00	222.09	10.12	13.30	0.00	0.00
	302	0.00	338.97	-2.93	5.17	0.00	0.00
	303	0.00	338.97	18.53	18.42	0.00	0.00
	304	0.00	-12.84	-42.78	-94.73	0.00	0.00
	305	0.00	613.14	57.39	114.54	0.00	0.00
	306	-6.60	309.75	8.38	12.17	0.00	22.11
	307	6.60	309.75	8.38	12.17	0.00	-22.11
	308	-0.03	310.01	8.37	12.17	0.00	0.11
	309	0.03	309.49	8.38	12.18	0.00	-0.11
	310	0.00	299.73	7.16	9.61	0.00	0.00
	311	0.00	319.77	9.60	14.74	0.00	0.00
	312	0.00	-296.87	-62.14	-134.56	0.00	0.00
	313	0.00	537.77	71.42	144.47	0.00	0.00
	314	-8.80	133.25	6.07	7.98	0.00	29.48
	315	8.80	133.25	6.07	7.98	0.00	-29.48
	316	-0.04	133.60	6.06	7.98	0.00	0.14
	317	0.04	132.90	6.08	7.99	0.00	-0.14
	318	0.00	119.89	4.44	4.56	0.00	0.00
	319	0.00	146.62	7.70	11.40	0.00	0.00
2	301	0.00	194.84	-4.92	-2.46	0.00	0.00
	302	0.00	311.96	8.03	5.46	0.00	0.00
	303	0.00	311.96	-13.43	-7.79	0.00	0.00
	304	0.00	601.97	-54.47	-108.43	0.00	0.00
	305	0.00	-20.71	45.76	100.88	0.00	0.00
	306	-6.60	282.68	-3.25	-1.49	0.00	22.11
	307	6.60	282.68	-3.25	-1.49	0.00	-22.11
	308	-0.03	282.94	-3.25	-1.48	0.00	0.11
	309	0.03	282.42	-3.26	-1.49	0.00	-0.11
	310	0.00	292.70	-4.48	-4.05	0.00	0.00
	311	0.00	272.66	-2.03	1.08	0.00	0.00
	312	0.00	542.62	-71.24	-144.06	0.00	0.00
	313	0.00	-287.61	62.40	135.01	0.00	0.00
	314	-8.80	116.91	-2.95	-1.48	0.00	29.48
	315	8.80	116.91	-2.95	-1.48	0.00	-29.48
	316	-0.04	117.26	-2.94	-1.47	0.00	0.14
	317	0.04	116.56	-2.96	-1.48	0.00	-0.14
	318	0.00	130.27	-4.58	-4.89	0.00	0.00
	319	0.00	103.54	-1.32	1.94	0.00	0.00

********** END OF LATEST ANALYSIS RESULT **********

```
169. ****************************************
170. LOAD LIST 401 TO 419
171. START CONCRETE DESIGN
172. CODE ACI 2008
173. FC 35000 ALL
174. FYMAIN 420000 ALL
175. TRACK 1 ALL
176. CLB 0.05 ALL
177. CLT 0.05 ALL
178. CLS 0.05 ALL
179. MAXMAIN 25 ALL
180. MINMAIN 16 ALL
181. MINSEC 8 ALL
182. REINF 0 MEMB 1 TO 4
183. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 800.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13469.31	10775.44	5189.38	1552.22	299.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
602.05	-1688.94	417.44	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13469.31	10775.44	4903.45	934.93	190.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
363.40	-1688.94	417.44	36.72	0.02932

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 800.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
13469.31  10775.44   5189.38   1552.22     299.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  602.05  -1688.94    365.94      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
13469.31  10775.44   4903.45    934.93     190.7
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  363.40  -1688.94    365.94      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 800.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  4000.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 20 - 16 MM               1.005      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 256.00 MM

    COLUMN INTERACTION:  MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
13469.31  10775.44   5189.38   1552.22     299.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  602.05  -1688.94    358.76      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
13469.31  10775.44   4903.45    934.93     190.7
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  363.40  -1688.94    358.76     26.47    0.02460
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 800.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 16 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13469.31	10775.44	5189.38	1552.22	299.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
602.05	-1688.94	310.80	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13469.31	10775.44	4903.45	934.93	190.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
363.40	-1688.94	310.80	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

184. DESIGN BEAM 5

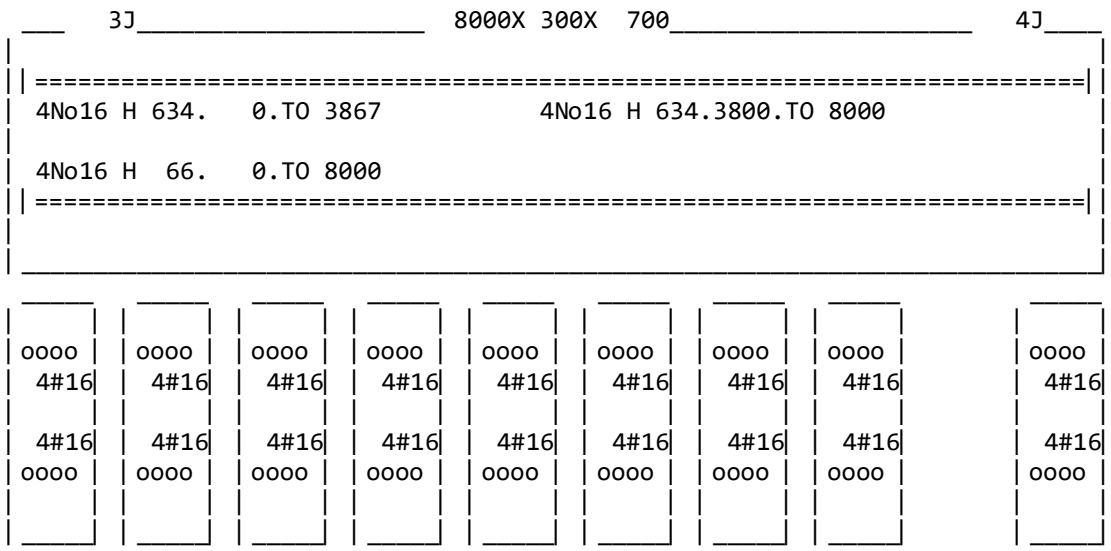
BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 8000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 700. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	4 - 16MM	0.	8000.	YES	YES
	CRITICAL POS MOMENT= 113.46 KN-MET AT 8000.MM, LOAD 413					
	REQD STEEL= 641.MM2, RHO=0.0034, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 586. MMS					
	Cracked Moment of Inertia Iz at above location = 167600.2 cm^4					
2	634.	4 - 16MM	0.	3867.	YES	NO
	CRITICAL NEG MOMENT= 165.11 KN-MET AT 0.MM, LOAD 405					
	REQD STEEL= 708.MM2, RHO=0.0037, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 586. MMS					
	Cracked Moment of Inertia Iz at above location = 167600.2 cm^4					
3	634.	4 - 16MM	3800.	8000.	NO	YES
	CRITICAL NEG MOMENT= 163.06 KN-MET AT 8000.MM, LOAD 404					
	REQD STEEL= 698.MM2, RHO=0.0037, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 586. MMS					
	Cracked Moment of Inertia Iz at above location = 167600.2 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 24.47 KNS Vc= 180.19 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 7.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 22.17 KNS Vc= 180.19 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 7.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

185. END CONCRETE DESIGN
186. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 30,2020 TIME= 17:30: 1 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	30-03-2020	29-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	30-03-2020	29-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK/APL	RPK/APL	RPK/APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

ATTACHMENT 9 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR VGTU**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 28-Mar-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART GROUND PORTAL FOUNDATIONS
JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
ENGINEER NAME RPK
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 4.5; 3 5 -3 0; 4 5 -3 4.5; 5 9 -3 0; 6 9 -3 4.5; 7 0 0.35 0;
8 0 0.35 4.5; 9 5 0.35 0; 10 5 0.35 4.5; 11 9 0.35 0; 12 9 0.35 4.5; 13 0 1 0;
14 0 1 4.5; 15 5 1 0; 16 5 1 4.5; 17 9 1 0; 18 9 1 4.5; 19 0 -3 8; 20 0 0.35 8;
21 0 1 8; 22 5 -3 8; 23 5 0.35 8; 24 5 1 8;
MEMBER INCIDENCES
1 1 7; 2 7 13; 3 2 8; 4 8 14; 5 3 9; 6 9 15; 7 4 10; 8 10 16; 9 5 11; 10 11 17;
11 6 12; 12 12 18; 13 7 8; 14 9 10; 15 11 12; 16 7 9; 17 9 11; 18 8 10;
19 10 12; 20 19 20; 21 20 21; 22 22 23; 23 23 24; 24 20 23; 25 8 20; 26 10 23;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
9 TO 12 PRIS YD 0.6 ZD 0.6
13 TO 15 25 26 PRIS YD 0.5 ZD 0.3
MEMBER PROPERTY AMERICAN
16 TO 19 24 PRIS YD 0.6 ZD 0.3
MEMBER PROPERTY AMERICAN
1 TO 8 20 TO 23 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 TO 6 19 22 FIXED
*****************************************
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
JOINT LOAD
17 FX -25.1 FY -10.2 FZ 2.7
18 FX -21.7 FY -148.3 FZ 5.9
15 FX -27.2 FY -238.5 FZ -3.2
16 FX -24.1 FY -213.7 FZ 14.9
13 FX 0 FY -229 FZ 3.8
```

```
14 FX -0.5 FY -328.5 FZ 4.8
24 FX -0.2 FY -36.9 FZ 0
21 FX -0.2 FY -23.2 FZ 0
LOAD 2 LIVE LOAD
JOINT LOAD
17 FX 2.5 FY -173.6 FZ -6.6
18 FX 2.9 FY -214 FZ 7.9
15 FX -2.2 FY -242.6 FZ -13.8
16 FX -2.7 FY -229.1 FZ 11.3
13 FX 0.1 FY -115.8 FZ 0.8
14 FX -0.4 FY -149.1 FZ 0.4
24 FX -0.1 FY -7.8 FZ 0
21 FX -0.1 FY -2.1 FZ 0
LOAD 3 MATERIAL LOAD
JOINT LOAD
17 FX -0.4 FY -21.6 FZ 0.2
18 FX 0.4 FY -27.2 FZ 0.3
15 FX -0.4 FY -1 FZ -0.2
16 FX 0.4 FY -0.7 FZ -0.1
13 FX -0.1 FY -35.5 FZ -0.2
14 FX 0.1 FY -30 FZ -0.2
24 FX 0 FY 0.1 FZ 0
21 FX 0 FY -0.1 FZ 0
LOAD 4 WIND LOAD +Z
JOINT LOAD
17 FX -2.4 FY 122.3 FZ 27.4
18 FX 2.3 FY -119 FZ 27.4
15 FX -2 FY 64.3 FZ 19
16 FX 1.9 FY -70.2 FZ 19.5
13 FX 0.1 FY 152.2 FZ 34
14 FX 0.1 FY -149.6 FZ 34
24 FX 0 FY -0.1 FZ 0.8
21 FX 0 FY 0.1 FZ 0.8
LOAD 5 WIND LOAD -Z
JOINT LOAD
17 FX 2.3 FY -121.92 FZ -27.47
18 FX -2.16 FY 118.8 FZ -27.42
15 FX 1.88 FY -64.38 FZ -18.97
16 FX -1.75 FY 69.89 FZ -19.48
13 FX -0.07 FY -150.11 FZ -33.57
14 FX -0.11 FY 147.68 FZ -33.58
24 FX -0.04 FY 0.12 FZ -0.8
21 FX -0.04 FY -0.07 FZ -0.78
LOAD 6 WIND LOAD (+X)
JOINT LOAD
17 FX 43.93 FY -172.16 FZ 0.87
18 FX 46.09 FY -132.07 FZ 1.26
15 FX 44.07 FY 133.38 FZ -3.05
16 FX 45.79 FY 82.81 FZ -7.28
13 FX 2.77 FY 38.78 FZ 4.12
14 FX 3.63 FY 49.25 FZ 4.09
24 FX 2.97 FY 0.44 FZ 0
21 FX 2.97 FY -0.44 FZ 0
LOAD 7 WIND LOAD (-X)
```

```
JOINT LOAD
17 FX -43.93 FY 172.16 FZ -0.87
18 FX -46.09 FY 132.07 FZ -1.26
15 FX -44.07 FY -133.38 FZ 3.05
16 FX -45.79 FY -82.81 FZ 7.28
13 FX -2.77 FY -38.78 FZ -4.12
14 FX -3.63 FY -49.25 FZ -4.09
24 FX -2.97 FY -0.44 FZ 0
21 FX -2.97 FY 0.44 FZ 0
LOAD 8 TEMP LOAD 1
JOINT LOAD
17 FX 0.74 FY 20.8 FZ 0.2
18 FX 3.2 FY -1.28 FZ 1.36
15 FX -2.51 FY -26.56 FZ -38.11
16 FX -1.31 FY -10.07 FZ 37.23
13 FX -0.44 FY 3.5 FZ -0.33
14 FX 0.39 FY 13.6 FZ -0.37
24 FX -0.03 FY 1.5 FZ 0.03
21 FX -0.03 FY -1.49 FZ -0.01
LOAD 9 TEMP LOAD 2
JOINT LOAD
17 FX 3.23 FY -36.99 FZ 0.29
18 FX -3.77 FY 0.05 FZ -0.84
15 FX 6.49 FY 38.35 FZ 37.68
16 FX 0.75 FY 12.77 FZ -38.15
13 FX 0.45 FY -17.81 FZ -2.9
14 FX -0.39 FY 3.63 FZ -2.86
24 FX 0.02 FY -1.47 FZ -0.03
21 FX 0.02 FY 1.46 FZ 0.01
LOAD 10 EARTHQUAKE LOAD +X
JOINT LOAD
17 FX 2.53 FY -11.75 FZ 0.09
18 FX 2.58 FY -8.51 FZ 0.12
15 FX 2.53 FY 8.16 FZ -0.19
16 FX 2.55 FY 3.52 FZ -0.38
13 FX 0 FY 2.89 FZ 0.18
14 FX 0.06 FY 4.39 FZ 0.18
24 FX 0.02 FY -0.27 FZ 0
21 FX 0.02 FY 0.27 FZ 0
LOAD 11 EARTHQUAKE LOAD +Z
JOINT LOAD
17 FX -0.21 FY 6.74 FZ 1.19
18 FX 0.2 FY -6.46 FZ 1.19
15 FX -0.18 FY 3.5 FZ 0.62
16 FX 0.17 FY -3.99 FZ 0.69
13 FX 0 FY 9.35 FZ 1.74
14 FX 0.01 FY -9.15 FZ 1.74
24 FX 0 FY -0.03 FZ 0
21 FX 0 FY 0.03 FZ 0
*****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
```

```
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL(+X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL(-X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL(+X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL(-X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
```

```
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL(+X)
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL(-X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL(+X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL(-X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION LIST ALL
*******************************************************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 19
FYMAIN 420000 MEMB 1 TO 19
TRACK 1 MEMB 1 TO 24
CLB 0.05 MEMB 1 TO 19
CLT 0.05 MEMB 1 TO 19
CLS 0.05 MEMB 1 TO 24
MAXMAIN 25 MEMB 1 TO 24
MINMAIN 16 MEMB 1 TO 24
MINSEC 8 MEMB 1 TO 24
REINF 0 MEMB 1 TO 12 20 TO 23
DESIGN COLUMN 1 TO 12 20 TO 23
DESIGN BEAM 13 TO 19 24
END CONCRETE DESIGN
FINISH
```

```
*****************************************************
*                                                   *
*           STAAD.Pro CONNECT Edition               *
*           Version  21.00.01.12                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    MAR 30, 2020                   *
*           Time=    17:31:16                       *
*                                                   *
*  Licensed to: MMH KOLKATTA                        *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE:  D:\F Drive\Maaden Gold\Conveyor CNV-3130-003\Civil\Documents\Staad\3130-CNV-003-VGTU-Fdn.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 28-MAR-20
     4. JOB NAME MMGP-SAUDI ARABIA
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045/900707
     7. JOB REV 0
     8. JOB PART GROUND PORTAL FOUNDATIONS
     9. JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
    10. ENGINEER NAME RPK
    11. CHECKER NAME SBY
    12. APPROVED NAME PRI
    13. END JOB INFORMATION
    14. INPUT WIDTH 79
    15. UNIT METER KN
    16. JOINT COORDINATES
    17. 1 0 -3 0; 2 0 -3 4.5; 3 5 -3 0; 4 5 -3 4.5; 5 9 -3 0; 6 9 -3 4.5; 7 0 0.35 0
    18. 8 0 0.35 4.5; 9 5 0.35 0; 10 5 0.35 4.5; 11 9 0.35 0; 12 9 0.35 4.5; 13 0 1 0
    19. 14 0 1 4.5; 15 5 1 0; 16 5 1 4.5; 17 9 1 0; 18 9 1 4.5; 19 0 -3 8; 20 0 0.35 8
    20. 21 0 1 8; 22 5 -3 8; 23 5 0.35 8; 24 5 1 8
    21. MEMBER INCIDENCES
    22. 1 1 7; 2 7 13; 3 2 8; 4 8 14; 5 3 9; 6 9 15; 7 4 10; 8 10 16; 9 5 11; 10 11 17
    23. 11 6 12; 12 12 18; 13 7 8; 14 9 10; 15 11 12; 16 7 9; 17 9 11; 18 8 10
    24. 19 10 12; 20 19 20; 21 20 21; 22 22 23; 23 23 24; 24 20 23; 25 8 20; 26 10 23
    25. DEFINE MATERIAL START
    26. ISOTROPIC CONCRETE
    27. E 2.17185E+007
    28. POISSON 0.17
    29. DENSITY 23.5616
    30. ALPHA 1E-005
    31. DAMP 0.05
    32. TYPE CONCRETE
    33. STRENGTH FCU 27579
    34. END DEFINE MATERIAL
    35. MEMBER PROPERTY AMERICAN
    36. 9 TO 12 PRIS YD 0.6 ZD 0.6
    37. 13 TO 15 25 26 PRIS YD 0.5 ZD 0.3
    38. MEMBER PROPERTY AMERICAN
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39. 16 TO 19 24 PRIS YD 0.6 ZD 0.3
40. MEMBER PROPERTY AMERICAN
41. 1 TO 8 20 TO 23 PRIS YD 0.6 ZD 0.6
42. CONSTANTS
43. MATERIAL CONCRETE ALL
44. SUPPORTS
45. 1 TO 6 19 22 FIXED
46. ****************************************
47. LOAD 1 DEAD LOAD
48. SELFWEIGHT Y -1
49. JOINT LOAD
50. 17 FX -25.1 FY -10.2 FZ 2.7
51. 18 FX -21.7 FY -148.3 FZ 5.9
52. 15 FX -27.2 FY -238.5 FZ -3.2
53. 16 FX -24.1 FY -213.7 FZ 14.9
54. 13 FX 0 FY -229 FZ 3.8
55. 14 FX -0.5 FY -328.5 FZ 4.8
56. 24 FX -0.2 FY -36.9 FZ 0
57. 21 FX -0.2 FY -23.2 FZ 0
58. LOAD 2 LIVE LOAD
59. JOINT LOAD
60. 17 FX 2.5 FY -173.6 FZ -6.6
61. 18 FX 2.9 FY -214 FZ 7.9
62. 15 FX -2.2 FY -242.6 FZ -13.8
63. 16 FX -2.7 FY -229.1 FZ 11.3
64. 13 FX 0.1 FY -115.8 FZ 0.8
65. 14 FX -0.4 FY -149.1 FZ 0.4
66. 24 FX -0.1 FY -7.8 FZ 0
67. 21 FX -0.1 FY -2.1 FZ 0
68. LOAD 3 MATERIAL LOAD
69. JOINT LOAD
70. 17 FX -0.4 FY -21.6 FZ 0.2
71. 18 FX 0.4 FY -27.2 FZ 0.3
72. 15 FX -0.4 FY -1 FZ -0.2
73. 16 FX 0.4 FY -0.7 FZ -0.1
74. 13 FX -0.1 FY -35.5 FZ -0.2
75. 14 FX 0.1 FY -30 FZ -0.2
76. 24 FX 0 FY 0.1 FZ 0
77. 21 FX 0 FY -0.1 FZ 0
78. LOAD 4 WIND LOAD +Z
79. JOINT LOAD
80. 17 FX -2.4 FY 122.3 FZ 27.4
81. 18 FX 2.3 FY -119 FZ 27.4
82. 15 FX -2 FY 64.3 FZ 19
83. 16 FX 1.9 FY -70.2 FZ 19.5
84. 13 FX 0.1 FY 152.2 FZ 34
85. 14 FX 0.1 FY -149.6 FZ 34
86. 24 FX 0 FY -0.1 FZ 0.8
87. 21 FX 0 FY 0.1 FZ 0.8
88. LOAD 5 WIND LOAD -Z
89. JOINT LOAD
90. 17 FX 2.3 FY -121.92 FZ -27.47
91. 18 FX -2.16 FY 118.8 FZ -27.42
92. 15 FX 1.88 FY -64.38 FZ -18.97
93. 16 FX -1.75 FY 69.89 FZ -19.48
94. 13 FX -0.07 FY -150.11 FZ -33.57
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 95. 14 FX -0.11 FY 147.68 FZ -33.58
 96. 24 FX -0.04 FY 0.12 FZ -0.8
 97. 21 FX -0.04 FY -0.07 FZ -0.78
 98. LOAD 6 WIND LOAD (+X)
 99. JOINT LOAD
100. 17 FX 43.93 FY -172.16 FZ 0.87
101. 18 FX 46.09 FY -132.07 FZ 1.26
102. 15 FX 44.07 FY 133.38 FZ -3.05
103. 16 FX 45.79 FY 82.81 FZ -7.28
104. 13 FX 2.77 FY 38.78 FZ 4.12
105. 14 FX 3.63 FY 49.25 FZ 4.09
106. 24 FX 2.97 FY 0.44 FZ 0
107. 21 FX 2.97 FY -0.44 FZ 0
108. LOAD 7 WIND LOAD (-X)
109. JOINT LOAD
110. 17 FX -43.93 FY 172.16 FZ -0.87
111. 18 FX -46.09 FY 132.07 FZ -1.26
112. 15 FX -44.07 FY -133.38 FZ 3.05
113. 16 FX -45.79 FY -82.81 FZ 7.28
114. 13 FX -2.77 FY -38.78 FZ -4.12
115. 14 FX -3.63 FY -49.25 FZ -4.09
116. 24 FX -2.97 FY -0.44 FZ 0
117. 21 FX -2.97 FY 0.44 FZ 0
118. LOAD 8 TEMP LOAD 1
119. JOINT LOAD
120. 17 FX 0.74 FY 20.8 FZ 0.2
121. 18 FX 3.2 FY -1.28 FZ 1.36
122. 15 FX -2.51 FY -26.56 FZ -38.11
123. 16 FX -1.31 FY -10.07 FZ 37.23
124. 13 FX -0.44 FY 3.5 FZ -0.33
125. 14 FX 0.39 FY 13.6 FZ -0.37
126. 24 FX -0.03 FY 1.5 FZ 0.03
127. 21 FX -0.03 FY -1.49 FZ -0.01
128. LOAD 9 TEMP LOAD 2
129. JOINT LOAD
130. 17 FX 3.23 FY -36.99 FZ 0.29
131. 18 FX -3.77 FY 0.05 FZ -0.84
132. 15 FX 6.49 FY 38.35 FZ 37.68
133. 16 FX 0.75 FY 12.77 FZ -38.15
134. 13 FX 0.45 FY -17.81 FZ -2.9
135. 14 FX -0.39 FY 3.63 FZ -2.86
136. 24 FX 0.02 FY -1.47 FZ -0.03
137. 21 FX 0.02 FY 1.46 FZ 0.01
138. LOAD 10 EARTHQUAKE LOAD +X
139. JOINT LOAD
140. 17 FX 2.53 FY -11.75 FZ 0.09
141. 18 FX 2.58 FY -8.51 FZ 0.12
142. 15 FX 2.53 FY 8.16 FZ -0.19
143. 16 FX 2.55 FY 3.52 FZ -0.38
144. 13 FX 0 FY 2.89 FZ 0.18
145. 14 FX 0.06 FY 4.39 FZ 0.18
146. 24 FX 0.02 FY -0.27 FZ 0
147. 21 FX 0.02 FY 0.27 FZ 0
148. LOAD 11 EARTHQUAKE LOAD +Z
149. JOINT LOAD
150. 17 FX -0.21 FY 6.74 FZ 1.19
```

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151. 18 FX 0.2 FY -6.46 FZ 1.19
152. 15 FX -0.18 FY 3.5 FZ 0.62
153. 16 FX 0.17 FY -3.99 FZ 0.69
154. 13 FX 0 FY 9.35 FZ 1.74
155. 14 FX 0.01 FY -9.15 FZ 1.74
156. 24 FX 0 FY -0.03 FZ 0
157. 21 FX 0 FY 0.03 FZ 0
158. ****************************************
159. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
160. LOAD COMB 301 1.0 DL
161. 1 1.0
162. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
163. 1 1.0 8 1.0 2 1.0 3 1.0
164. LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
165. 1 1.0 9 1.0 2 1.0 3 1.0
166. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
167. 1 1.0 4 0.75 2 0.75 3 0.75
168. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
169. 1 1.0 5 0.75 2 0.75 3 0.75
170. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
171. 1 1.0 6 0.75 2 0.75 3 0.75
172. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
173. 1 1.0 7 0.75 2 0.75 3 0.75
174. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
175. 1 1.0 10 0.525 2 0.75 3 0.75
176. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
177. 1 1.0 10 -0.525 2 0.75 3 0.75
178. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
179. 1 1.0 11 0.525 2 0.75 3 0.75
180. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
181. 1 1.0 11 -0.525 2 0.75 3 0.75
182. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
183. 1 0.6 4 1.0
184. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
185. 1 0.6 5 1.0
186. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
187. 1 0.6 6 1.0
188. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
189. 1 0.6 7 1.0
190. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
191. 1 0.6 10 0.7
192. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
193. 1 0.6 10 -0.7
194. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
195. 1 0.6 11 0.7
196. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
197. 1 0.6 11 -0.7
198. ****************************************
199. *****LOAD COMBINATION FOR FOUNDATION DESIGN
200. LOAD COMB 401 1.4 DL
201. 1 1.4
202. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
203. 1 1.4 8 1.4 2 1.7 3 1.7
204. LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
205. 1 1.4 9 1.4 2 1.7 3 1.7
206. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
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```
207. 1 1.2 4 1.6 2 0.5 3 0.5
208. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
209. 1 1.2 5 1.6 2 0.5 3 0.5
210. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
211. 1 1.2 6 1.6 2 0.5 3 0.5
212. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
213. 1 1.2 7 1.6 2 0.5 3 0.5
214. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
215. 1 1.2 10 1.0 2 0.5 3 0.5
216. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
217. 1 1.2 10 -1.0 2 0.5 3 0.5
218. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
219. 1 1.2 11 1.0 2 0.5 3 0.5
220. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
221. 1 1.2 11 -1.0 2 0.5 3 0.5
222. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
223. 1 0.9 4 1.6
224. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
225. 1 0.9 5 1.6
226. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
227. 1 0.9 6 1.6
228. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
229. 1 0.9 7 1.6
230. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
231. 1 0.9 10 1.0
232. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
233. 1 0.9 10 -1.0
234. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
235. 1 0.9 11 1.0
236. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
237. 1 0.9 11 -1.0
238. ****************************************
239. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**24**	**NUMBER OF MEMBERS**	**26**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**8**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 96
TOTAL LOAD COMBINATION CASES = 38 SO FAR.

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240. LOAD LIST 301 TO 319
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241. PRINT SUPPORT REACTION LIST ALL

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	20.63	287.57	0.40	-3.72	-0.11	-42.78
	302	21.85	436.66	0.67	-2.75	-0.72	-44.94
	303	17.69	456.50	2.21	1.06	0.42	-35.91
	304	22.05	280.48	-11.19	-40.83	-1.20	-45.28
	305	19.94	521.00	12.30	33.76	0.58	-41.18
	306	-2.02	361.33	-0.49	-6.83	-0.82	8.95
	307	43.97	441.78	1.75	0.22	0.23	-95.31
	308	20.06	399.61	0.60	-3.41	-0.31	-41.13
	309	21.89	403.50	0.66	-3.20	-0.28	-45.23
	310	21.04	396.42	0.22	-4.61	-0.34	-43.32
	311	20.90	406.69	1.04	-2.00	-0.24	-43.04
	312	13.82	11.11	-15.51	-52.26	-1.27	-28.46
	313	11.01	331.80	15.80	47.19	1.10	-23.01
	314	-18.28	118.91	-1.26	-6.93	-0.77	43.84
	315	43.04	226.18	1.74	2.46	0.63	-95.17
	316	11.16	169.95	0.20	-2.37	-0.09	-22.93
	317	13.60	175.14	0.28	-2.10	-0.04	-28.40
	318	12.47	165.70	-0.30	-3.97	-0.13	-25.85
	319	12.29	179.39	0.78	-0.49	-0.00	-25.48
2	301	17.27	390.70	-4.58	-8.78	-1.59	-35.66
	302	16.72	555.28	-4.28	-7.78	-1.67	-34.02
	303	17.83	565.84	-2.96	-4.24	-1.36	-36.84
	304	16.27	631.92	-24.11	-53.59	-2.18	-33.54
	305	18.06	418.09	14.37	35.49	-1.12	-37.05
	306	-5.10	477.67	-6.33	-12.68	0.12	15.18
	307	39.50	570.98	-3.18	-4.87	-3.40	-85.89
	308	16.34	521.67	-4.80	-8.88	-1.56	-33.43
	309	18.05	526.98	-4.71	-8.67	-1.72	-37.28
	310	17.14	528.95	-5.43	-10.34	-1.67	-35.24
	311	17.25	519.70	-4.08	-7.21	-1.61	-35.47
	312	9.13	377.88	-28.56	-65.02	-1.67	-18.98
	313	11.52	92.78	22.75	53.75	-0.26	-23.65
	314	-19.37	172.21	-4.84	-10.47	1.39	45.98
	315	40.09	296.63	-0.65	-0.07	-3.30	-88.77
	316	9.22	230.88	-2.80	-5.41	-0.85	-18.84
	317	11.50	237.96	-2.69	-5.13	-1.06	-23.96
	318	10.29	240.59	-3.64	-7.35	-0.99	-21.25
	319	10.44	228.26	-1.85	-3.19	-0.92	-21.55
3	301	18.98	303.08	-2.03	-6.85	0.14	-42.12
	302	19.04	573.53	-10.65	-10.70	0.39	-43.09
	303	15.30	507.72	4.06	-0.02	0.60	-34.32
	304	19.64	431.99	-11.30	-31.75	1.00	-43.89
	305	17.85	539.59	5.00	19.35	-0.54	-40.02
	306	-8.30	381.15	-1.60	-2.14	0.17	15.64
	307	45.75	590.37	-4.68	-10.21	0.32	-99.44
	308	17.68	481.30	-3.08	-6.02	0.25	-39.66

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	309	19.77	490.22	-3.20	-6.33	0.24	-44.14
	310	18.79	483.79	-3.35	-6.83	0.26	-42.04
	311	18.66	487.74	-2.94	-5.52	0.23	-41.76
	312	12.62	110.15	-12.09	-38.22	1.09	-27.92
	313	10.23	253.62	9.63	29.92	-0.96	-22.76
	314	-24.64	42.37	0.83	1.27	-0.02	51.45
	315	47.42	321.33	-3.27	-9.49	0.19	-101.99
	316	10.00	175.90	-1.14	-3.91	0.09	-22.29
	317	12.78	187.80	-1.30	-4.31	0.08	-28.26
	318	11.48	179.22	-1.49	-4.98	0.10	-25.46
	319	11.30	184.48	-0.94	-3.24	0.07	-25.09
4	301	15.69	285.56	-3.47	-8.68	-1.18	-34.95
	302	13.89	520.19	7.67	5.12	-0.93	-31.94
	303	16.35	503.28	-7.54	-10.96	-0.75	-36.26
	304	14.41	506.53	-14.14	-35.26	-0.30	-32.46
	305	15.92	405.99	12.05	25.48	-1.82	-35.78
	306	-11.11	388.49	0.75	-0.57	0.69	21.81
	307	41.50	523.81	-2.81	-9.14	-2.79	-90.17
	308	14.21	454.08	-0.96	-4.69	-0.97	-32.07
	309	16.18	458.23	-1.10	-5.02	-1.13	-36.30
	310	15.14	458.18	-1.36	-5.64	-1.03	-34.07
	311	15.25	454.13	-0.70	-4.08	-1.07	-34.30
	312	8.37	238.50	-19.56	-45.74	0.30	-18.67
	313	10.38	104.45	15.35	35.24	-1.73	-23.10
	314	-25.66	81.12	0.29	0.51	1.62	53.68
	315	44.49	261.55	-4.46	-10.92	-3.03	-95.62
	316	8.10	168.56	-1.99	-4.99	-0.60	-18.15
	317	10.73	174.10	-2.18	-5.43	-0.81	-23.79
	318	9.34	174.03	-2.53	-6.25	-0.68	-20.82
	319	9.48	168.64	-1.64	-4.17	-0.73	-21.13
5	301	12.23	45.86	-2.03	-9.17	-0.05	-35.45
	302	13.11	219.99	-5.28	-14.35	0.63	-37.12
	303	9.82	279.95	-3.94	-11.88	0.34	-28.90
	304	12.74	89.76	-22.77	-63.85	1.72	-37.02
	305	12.08	294.63	15.23	40.58	-1.31	-34.28
	306	-5.99	336.87	-4.44	-13.23	0.68	13.34
	307	30.78	47.81	-3.12	-10.11	-0.26	-84.55
	308	11.69	199.09	-3.83	-11.79	0.23	-33.70
	309	13.10	185.58	-3.73	-11.55	0.19	-37.51
	310	12.43	188.46	-4.35	-13.24	0.26	-35.71
	311	12.36	196.22	-3.21	-10.10	0.16	-35.50
	312	7.80	-109.26	-26.53	-75.07	1.98	-23.15
	313	6.92	163.90	24.13	64.17	-2.06	-19.51
	314	-17.17	220.22	-2.10	-7.58	0.60	43.99
	315	31.85	-165.19	-0.34	-3.42	-0.66	-86.53
	316	6.40	36.52	-1.28	-5.66	0.00	-18.73

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	317	8.28	18.51	-1.15	-5.34	-0.06	-23.81
	318	7.38	22.34	-1.98	-7.59	0.04	-21.41
	319	7.29	32.69	-0.46	-3.41	-0.09	-21.13
6	301	10.33	189.06	-5.80	-13.17	0.12	-29.73
	302	10.54	432.83	-4.76	-14.36	0.37	-28.60
	303	10.94	430.06	-5.02	-13.47	0.82	-30.96
	304	10.33	470.38	-24.08	-65.62	1.91	-28.45
	305	10.81	270.46	14.24	39.15	-1.26	-30.75
	306	-7.18	485.74	-5.53	-14.79	0.52	17.94
	307	28.36	254.92	-4.35	-11.77	0.15	-77.25
	308	9.93	375.44	-4.98	-13.39	0.34	-27.86
	309	11.26	365.22	-4.89	-13.16	0.32	-31.46
	310	10.57	374.06	-5.51	-14.85	0.38	-29.57
	311	10.62	366.60	-4.37	-11.71	0.28	-29.74
	312	5.85	246.83	-29.01	-77.69	2.17	-16.23
	313	6.49	-19.73	22.09	62.00	-2.05	-19.29
	314	-17.50	267.32	-4.27	-9.92	0.32	45.62
	315	29.89	-40.44	-2.69	-5.89	-0.18	-81.30
	316	5.31	120.25	-3.54	-8.06	0.09	-15.44
	317	7.08	106.63	-3.42	-7.75	0.05	-20.24
	318	6.16	118.41	-4.25	-10.00	0.14	-17.72
	319	6.23	108.46	-2.72	-5.81	0.00	-17.96
19	301	5.12	78.08	-5.44	-9.48	-1.58	-7.88
	302	5.12	82.42	-5.64	-9.00	-1.52	-7.87
	303	5.32	78.63	-3.96	-5.15	-1.59	-8.34
	304	4.96	91.85	-24.86	-53.30	-2.00	-7.52
	305	5.36	69.13	12.92	33.35	-1.22	-8.41
	306	0.48	78.79	-7.42	-13.49	0.40	3.40
	307	9.82	82.06	-4.28	-5.92	-3.61	-19.30
	308	5.04	80.23	-5.89	-9.81	-1.52	-7.68
	309	5.27	80.61	-5.81	-9.60	-1.70	-8.22
	310	5.14	80.81	-6.53	-11.24	-1.63	-7.92
	311	5.17	80.04	-5.18	-8.17	-1.59	-7.98
	312	2.81	62.08	-28.60	-63.81	-1.47	-4.16
	313	3.35	31.79	21.77	51.72	-0.43	-5.35
	314	-3.16	44.67	-5.35	-10.74	1.73	10.40
	315	9.30	49.03	-1.17	-0.64	-3.62	-19.86
	316	2.92	46.60	-3.31	-5.83	-0.83	-4.37
	317	3.22	47.10	-3.21	-5.55	-1.06	-5.09
	318	3.05	47.37	-4.16	-7.73	-0.97	-4.69
	319	3.09	46.33	-2.36	-3.64	-0.92	-4.77
22	301	-1.26	89.80	-5.95	-11.03	-1.77	-1.25
	302	-1.28	98.93	-6.44	-9.64	-1.76	-1.21
	303	-1.05	97.87	-4.74	-7.74	-1.81	-1.71
	304	-1.42	104.40	-18.49	-39.09	-2.40	-0.90
	305	-1.03	88.41	6.70	19.53	-1.25	-1.76

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	306	-5.95	97.27	-3.69	-5.17	0.37	10.09
	307	3.49	95.56	-8.07	-14.33	-4.01	-12.72
	308	-1.35	96.56	-5.80	-9.58	-1.73	-1.04
	309	-1.12	96.26	-5.96	-9.92	-1.92	-1.58
	310	-1.25	96.64	-6.21	-10.51	-1.85	-1.28
	311	-1.22	96.18	-5.55	-8.99	-1.79	-1.34
	312	-1.00	64.54	-20.38	-45.74	-1.83	-0.20
	313	-0.49	43.22	13.21	32.42	-0.30	-1.35
	314	-7.05	55.02	-0.65	-0.51	1.86	14.46
	315	5.54	52.74	-6.49	-12.73	-3.98	-15.96
	316	-0.91	54.08	-3.46	-6.39	-0.93	-0.39
	317	-0.60	53.68	-3.68	-6.85	-1.19	-1.11
	318	-0.77	54.19	-4.01	-7.63	-1.10	-0.71
	319	-0.74	53.58	-3.13	-5.61	-1.02	-0.79

********** END OF LATEST ANALYSIS RESULT **********

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242. ********************************************************************************
243. LOAD LIST 401 TO 419
244. START CONCRETE DESIGN
245. CODE ACI 2008
246. FC 35000 MEMB 1 TO 19
247. FYMAIN 420000 MEMB 1 TO 19
248. TRACK 1 MEMB 1 TO 24
249. CLB 0.05 MEMB 1 TO 19
250. CLT 0.05 MEMB 1 TO 19
251. CLS 0.05 MEMB 1 TO 24
252. MAXMAIN 25 MEMB 1 TO 24
253. MINMAIN 16 MEMB 1 TO 24
254. MINSEC 8 MEMB 1 TO 24
255. REINF 0 MEMB 1 TO 12 20 TO 23
256. DESIGN COLUMN 1 TO 12 20 TO 23
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COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	558.19	56.75	0.03035

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	558.19	10.89	0.00582

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COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

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 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    493.23      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    493.23      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION:  MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    780.31     47.82    0.01829
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    780.31     14.09    0.00539
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     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	707.54	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	707.54	0.00	0.00000

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COLUMN NO. 5 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	591.59	46.26	0.02334

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

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 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    591.59      0.10    0.00005
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COLUMN NO. 6 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION:  MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    513.69      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    513.69      0.00    0.00000
 ---------------------------------------------------------
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COLUMN NO. 7 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM
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BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

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12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	553.84	37.91	0.02043

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	553.84	6.37	0.00343

==========

COLUMN NO. 8 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

---

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	460.28	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	460.28	0.00	0.00000

==========

COLUMN NO. 9 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	37.57	11.91	0.09458

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	37.57	5.10	0.04051

==========

COLUMN NO. 10 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37     21.97      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37     21.97      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.    11  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION:  MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    346.01     10.48    0.00904
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    346.01     13.51    0.01165
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.    12  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	319.42	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	319.42	0.00	0.00000

==========

COLUMN NO. 20 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
9910.37	7928.30	3822.91	883.66	231.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
399.46	-1559.56	106.98	19.96	0.05570

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
 9910.37   7928.30   3822.91    883.66     231.2
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  399.46  -1559.56    106.98     18.80    0.05246
 ---------------------------------------------------------
```

==========

COLUMN NO. 21 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 413.7  FC -  27.6 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION:  MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
 9910.37   7928.30   3822.91    883.66     231.2
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  399.46  -1559.56     49.97      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
 9910.37   7928.30   3822.91    883.66     231.2
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  399.46  -1559.56     49.97      0.00    0.00000
 ---------------------------------------------------------
```

==========

COLUMN NO. 22 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 413.7  FC -  27.6 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM
```

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
9910.37	7928.30	3822.91	883.66	231.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
399.46	-1559.56	132.22	11.77	0.02657

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
9910.37	7928.30	3822.91	883.66	231.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
399.46	-1559.56	132.22	19.18	0.04330

==========

COLUMN NO. 23 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
9910.37	7928.30	3822.91	883.66	231.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
399.46	-1559.56	79.48	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
9910.37	7928.30	3822.91	883.66	231.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
399.46	-1559.56	79.48	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

257. DESIGN BEAM 13 TO 19 24

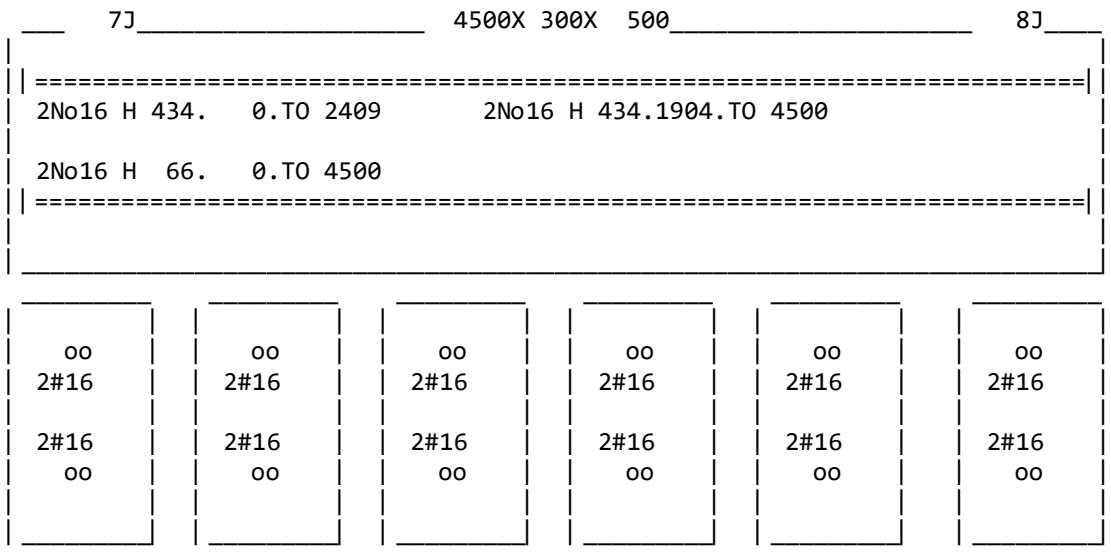
BEAM NO. 13 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	4500.	YES	YES
	CRITICAL POS MOMENT= 36.13 KN-MET AT 0.MM, LOAD 412					
	REQD STEEL= 297.MM2, RHO=0.0023, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
2	434.	2 - 16MM	0.	2409.	YES	NO
	CRITICAL NEG MOMENT= 41.50 KN-MET AT 0.MM, LOAD 405					
	REQD STEEL= 341.MM2, RHO=0.0026, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
3	434.	2 - 16MM	1904.	4500.	NO	YES
	CRITICAL NEG MOMENT= 42.75 KN-MET AT 4500.MM, LOAD 404					
	REQD STEEL= 352.MM2, RHO=0.0027, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					

B E A M N O. 13 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.38 KNS Vc= 225.14 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.59 KNS Vc= 126.41 KNS Vs= 0.00 KNS
Tu= 0.17 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 14 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	4500.	YES	YES

CRITICAL POS MOMENT= 25.33 KN-MET AT 0.MM, LOAD 412
REQD STEEL= 207.MM2, RHO=0.0016, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

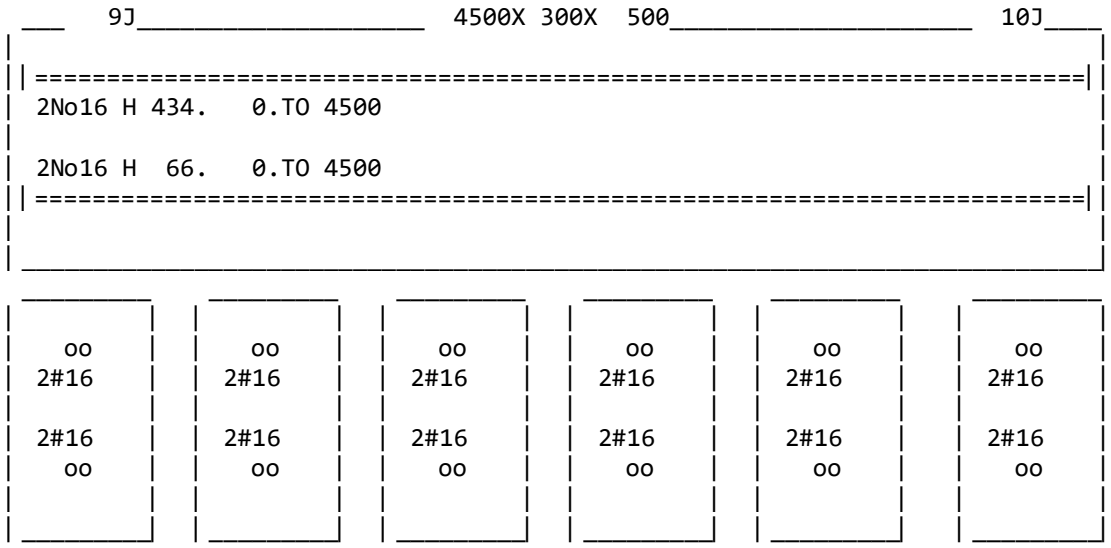
2	434.	2 - 16MM	0.	4500.	YES	YES

CRITICAL NEG MOMENT= 35.14 KN-MET AT 4500.MM, LOAD 404
REQD STEEL= 288.MM2, RHO=0.0022, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

B E A M N O. 14 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 6.44 KNS Vc= 126.19 KNS Vs= 0.00 KNS
Tu= 0.19 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.53 KNS Vc= 126.19 KNS Vs= 0.00 KNS
Tu= 0.19 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 15 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4500.	YES	YES
	CRITICAL POS MOMENT= 49.84 KN-MET AT 0.MM, LOAD 412					
	REQD STEEL= 411.MM2, RHO=0.0032, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
2	434.	3 - 16MM	0.	2409.	YES	NO

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     51.82 KN-MET  AT     0.MM, LOAD  405|
|   REQD STEEL=   428.MM2, RHO=0.0033, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

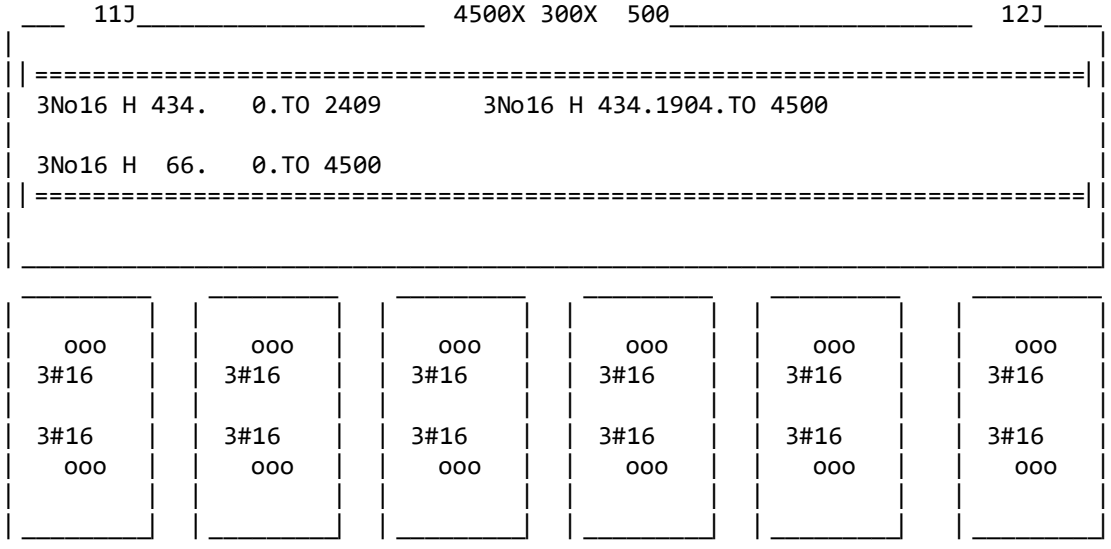
  Cracked Moment of Inertia Iz at above location =    58050.0 cm^4

  3        434.     3 - 16MM      1904.         4500.        NO   YES
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     63.15 KN-MET  AT  4500.MM, LOAD  404|
|   REQD STEEL=   457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    58050.0 cm^4
```

B E A M N O. 15 D E S I G N R E S U L T S - SHEAR

```
  AT START SUPPORT - Vu=    6.14 KNS  Vc=  132.49 KNS  Vs=    0.00 KNS
  Tu=    0.29 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                    STIRRUPS ARE NOT REQUIRED.
  AT END   SUPPORT - Vu=   11.83 KNS  Vc=  125.86 KNS  Vs=    0.00 KNS
  Tu=    0.29 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                    STIRRUPS ARE NOT REQUIRED.
```



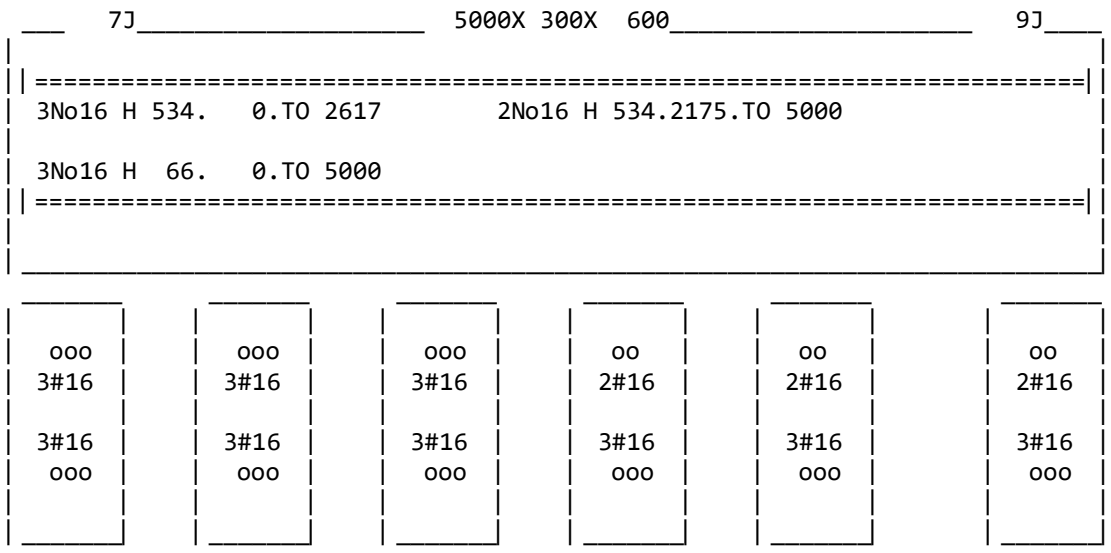
BEAM NO. 16 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	5000.	YES	YES
	CRITICAL POS MOMENT= 66.64 KN-MET AT 5000.MM, LOAD 407					
	REQD STEEL= 446.MM2, RHO=0.0028, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
2	534.	3 - 16MM	0.	2617.	YES	NO
	CRITICAL NEG MOMENT= 87.89 KN-MET AT 0.MM, LOAD 407					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
3	534.	2 - 16MM	2175.	5000.	NO	YES
	CRITICAL NEG MOMENT= 45.06 KN-MET AT 5000.MM, LOAD 414					
	REQD STEEL= 300.MM2, RHO=0.0019, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 63898.6 cm^4					

B E A M N O. 16 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 21.81 KNS Vc= 154.61 KNS Vs= 0.00 KNS
Tu= 0.13 KN-MET Tc= 6.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 1.54 KNS Vc= 150.24 KNS Vs= 0.00 KNS
Tu= 0.13 KN-MET Tc= 6.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 17 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES

CRITICAL POS MOMENT= 98.12 KN-MET AT 4000.MM, LOAD 407
REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

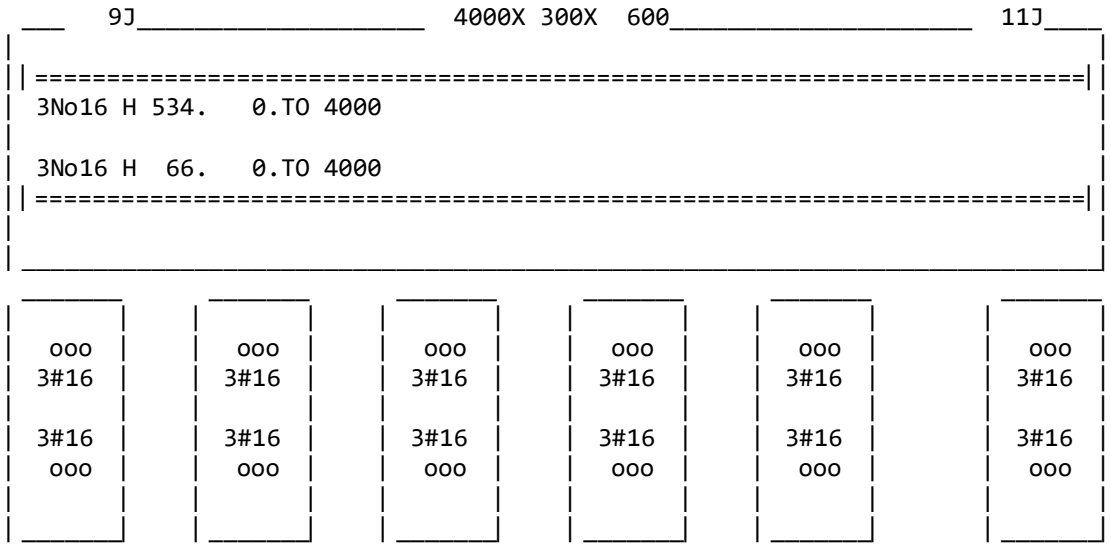
2	534.	3 - 16MM	0.	4000.	YES	YES

CRITICAL NEG MOMENT= 96.72 KN-MET AT 0.MM, LOAD 407
REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

B E A M N O. 17 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 26.45 KNS Vc= 155.83 KNS Vs= 0.00 KNS
Tu= 0.40 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 9.04 KNS Vc= 151.64 KNS Vs= 0.00 KNS
Tu= 0.40 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 18 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	5000.	YES	YES
	CRITICAL POS MOMENT= 58.25 KN-MET AT 5000.MM, LOAD 407					
	REQD STEEL= 389.MM2, RHO=0.0024, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 63898.6 cm^4					
2	534.	3 - 16MM	0.	2617.	YES	NO

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     79.58 KN-MET  AT     0.MM, LOAD  407|
|   REQD STEEL=   534.MM2, RHO=0.0033, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                        |
|-------------------------------------------------------------|

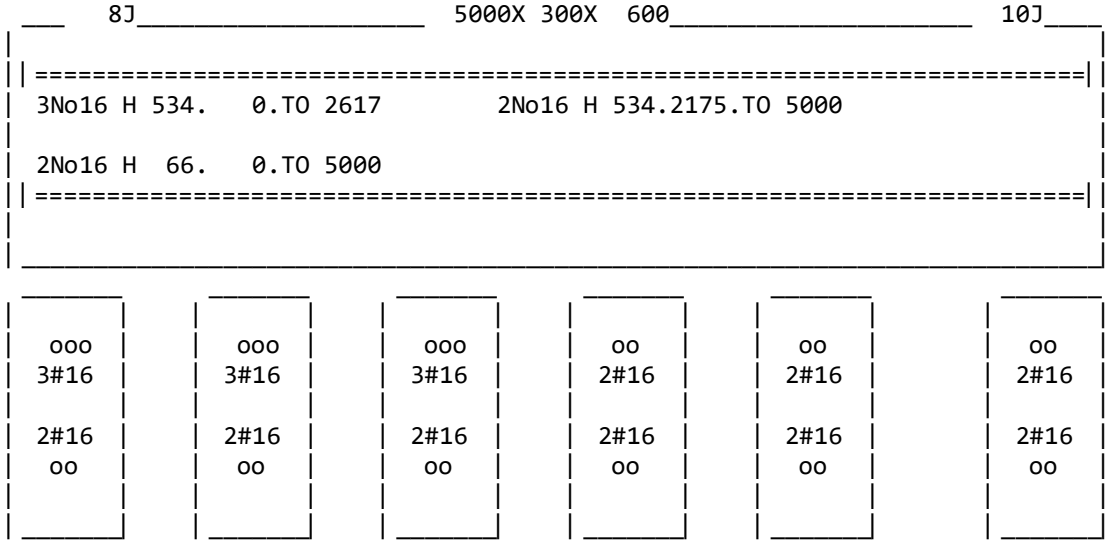
  Cracked Moment of Inertia Iz at above location =    90761.5 cm^4

  3        534.     2 - 16MM      2175.         5000.        NO   YES
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     46.47 KN-MET  AT  5000.MM, LOAD  414|
|   REQD STEEL=   309.MM2, RHO=0.0019, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                        |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    63898.6 cm^4
```

B E A M N O. 18 D E S I G N R E S U L T S - SHEAR

```
  AT START SUPPORT - Vu=   19.33 KNS  Vc=  154.64 KNS  Vs=    0.00 KNS
  Tu=    0.24 KN-MET  Tc=    6.9 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                    STIRRUPS ARE NOT REQUIRED.
  AT END   SUPPORT - Vu=    4.02 KNS  Vc=  152.67 KNS  Vs=    0.00 KNS
  Tu=    0.24 KN-MET  Tc=    6.9 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                    STIRRUPS ARE NOT REQUIRED.
```



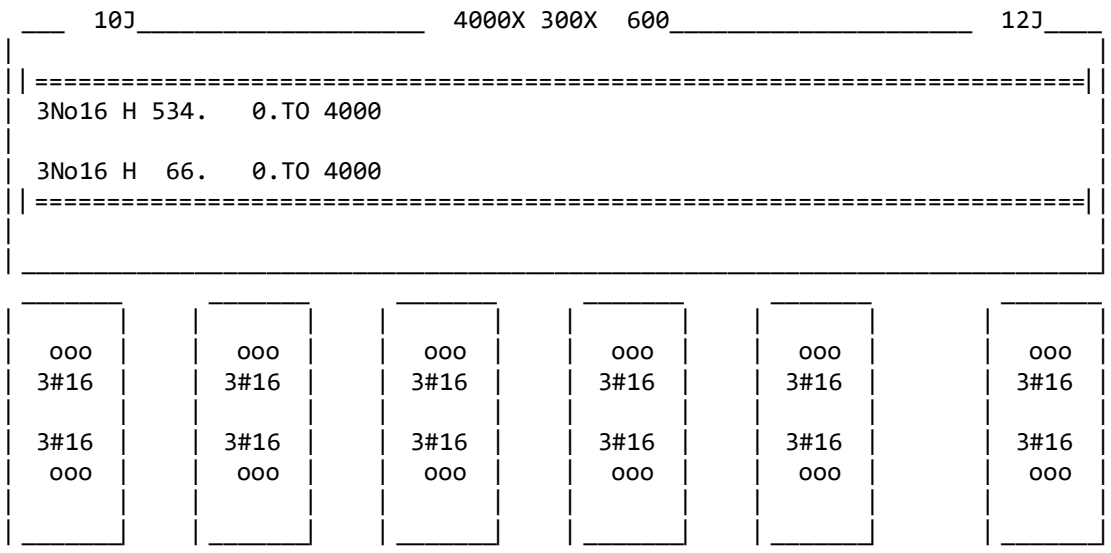
BEAM NO. 19 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL POS MOMENT= 96.06 KN-MET AT 4000.MM, LOAD 407					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
2	534.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL NEG MOMENT= 95.63 KN-MET AT 0.MM, LOAD 407					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					

B E A M N O. 19 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 24.99 KNS Vc= 155.87 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 7.58 KNS Vc= 151.51 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 24 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

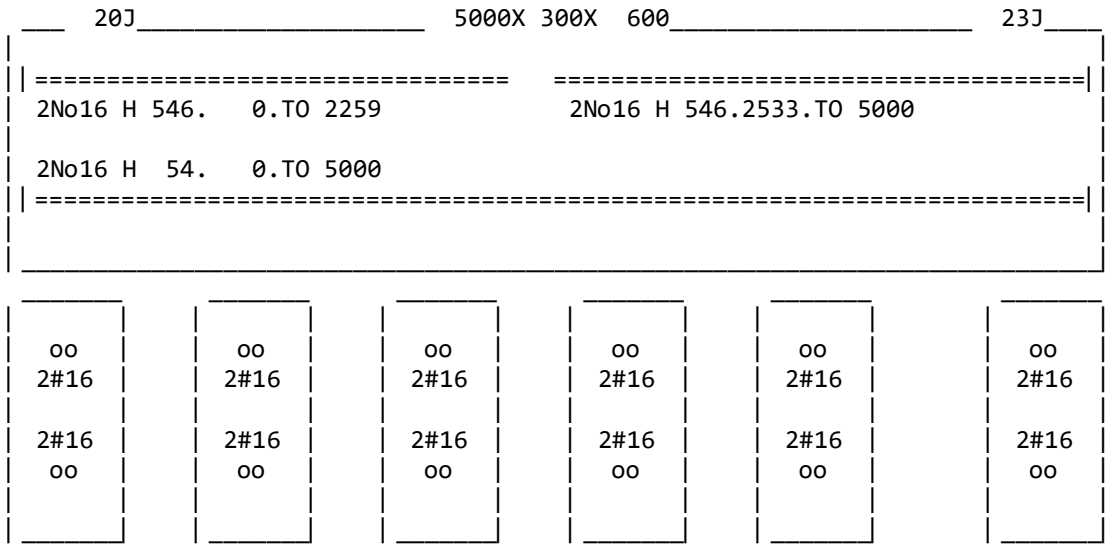
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	54.	2 - 16MM	0.	5000.	YES	YES
	CRITICAL POS MOMENT= 11.80 KN-MET AT 3750.MM, LOAD 407					
	REQD STEEL= 77.MM2, RHO=0.0005, RHOMX=0.0214 RHOMN=0.0033					
	MAX/MIN/ACTUAL BAR SPACING= 266./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 455. MMS					
	Cracked Moment of Inertia Iz at above location = 74322.3 cm^4					
2	546.	2 - 16MM	0.	2259.	YES	NO
	CRITICAL NEG MOMENT= 26.73 KN-MET AT 0.MM, LOAD 407					
	REQD STEEL= 176.MM2, RHO=0.0011, RHOMX=0.0214 RHOMN=0.0033					
	MAX/MIN/ACTUAL BAR SPACING= 266./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 455. MMS					
	Cracked Moment of Inertia Iz at above location = 74322.3 cm^4					
3	546.	2 - 16MM	2533.	5000.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     20.08 KN-MET  AT  5000.MM, LOAD  406|
|   REQD STEEL=   132.MM2, RHO=0.0008, RHOMX=0.0214 RHOMN=0.0033 |
|   MAX/MIN/ACTUAL BAR SPACING=   266./   41./  168. MMS        |
|   REQD. DEVELOPMENT LENGTH =   455. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 74322.3 cm^4

B E A M N O. 24 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.12 KNS Vc= 137.50 KNS Vs= 0.00 KNS
Tu= 0.09 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.08 KNS Vc= 250.87 KNS Vs= 0.00 KNS
Tu= 0.09 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

258. END CONCRETE DESIGN
259. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 30, 2020 TIME= 17: 31: 17 ****

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* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
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